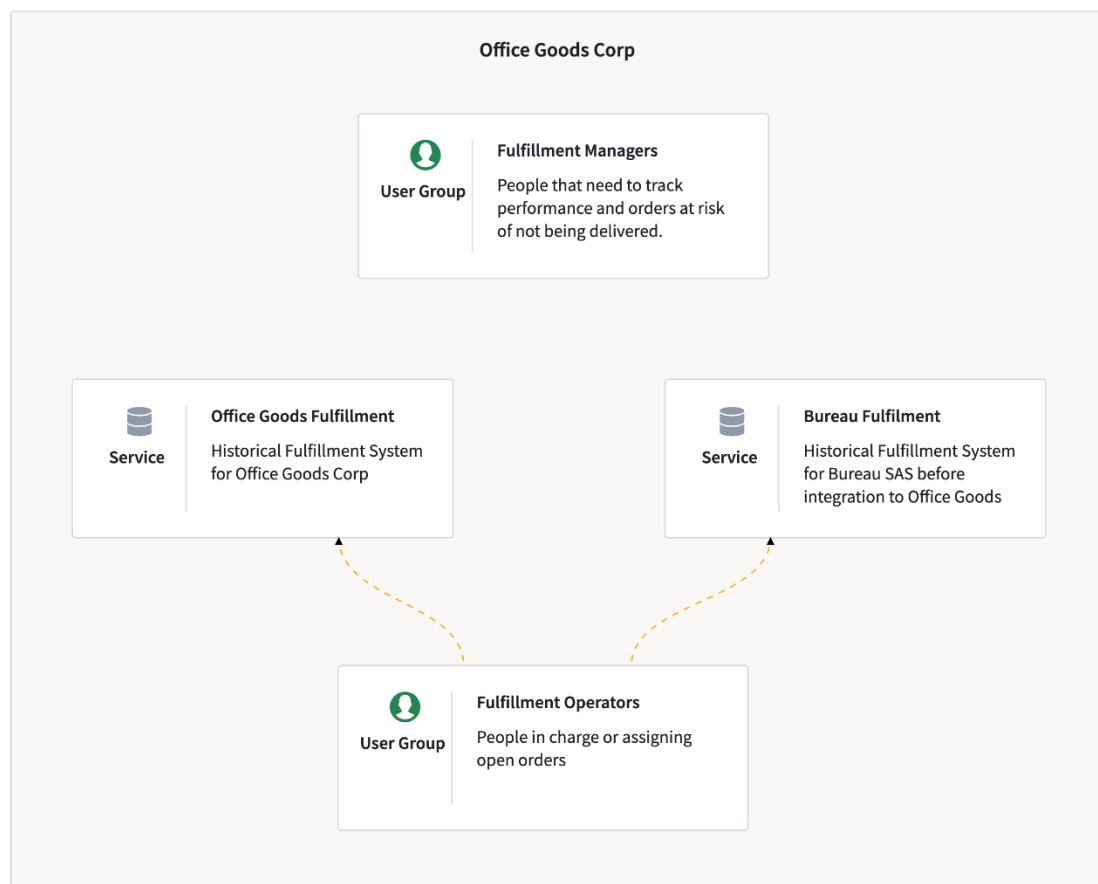


Speedrun: Your First End-to-End Workflow

The Use Case

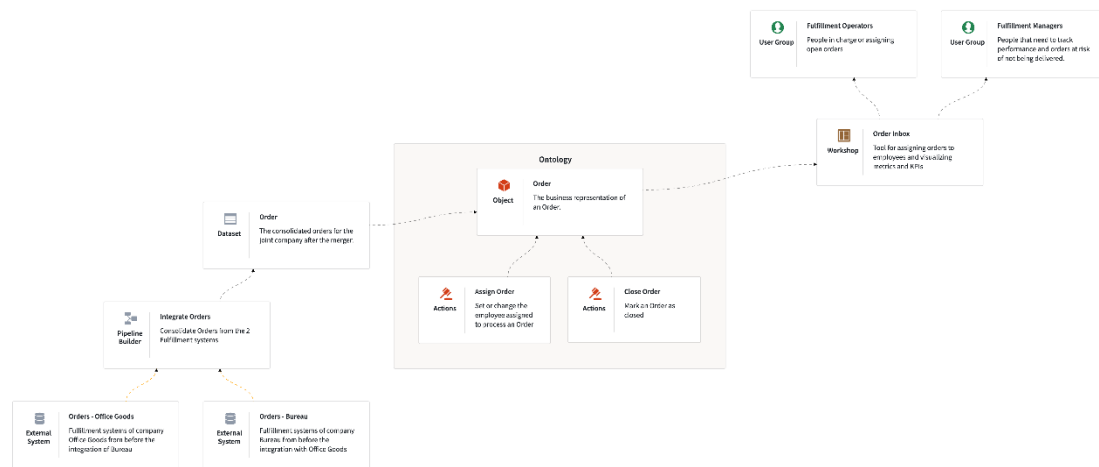
You work for **Office Goods Corp**, a paper and office goods company. Recently, your company acquired one of its competitors, **Bureau SAS**. The IT unification project won't start for another year, which means that your colleagues are stuck tracking and managing orders on two different systems.

They've survived a couple months by manually stitching spreadsheets together, but with demand growing, things have started to fall through the cracks. Incorrect metrics, unfulfilled orders, and unhappy customers. The fulfillment department has reached out to you for help, to see what can be done quickly ahead of the unification project.



After understanding the problem a little better, you have decided to create a simple operational tool for the fulfillment department that connects to both IT systems. It will create a single location for managers to assign orders, show what orders haven't been assigned, and what is at risk of not being delivered. Along the way, this tool should end debates about which spreadsheet is the most up-to-date!

You've sketched together the below diagram showing what you're going to build and how it's going to work.



Setting up your Project and Folder

Introduction

All of the resources you create in Foundry need to live inside a Project. For production use cases, these resources likely need to be shared across your organization and it is recommended you create a Project for each stage of the workflow.

During this learning course, you will be creating resources that will be specific to your learning experience. Depending on how your environment is set up, you will want to use a naming convention to easily distinguish your resources from those of other users. We will use the convention <username> throughout this course to represent your user login id. Wherever you see <username> please replace with your own user login id.

We advise using a single Foundry Project to store all your training materials for all courses on this site. You should then create a dedicated Folder within your training Project for each learning course. If your organization prefers a different Project set-up convention, you may need to adapt some instructions. Contact your platform support team for more details.

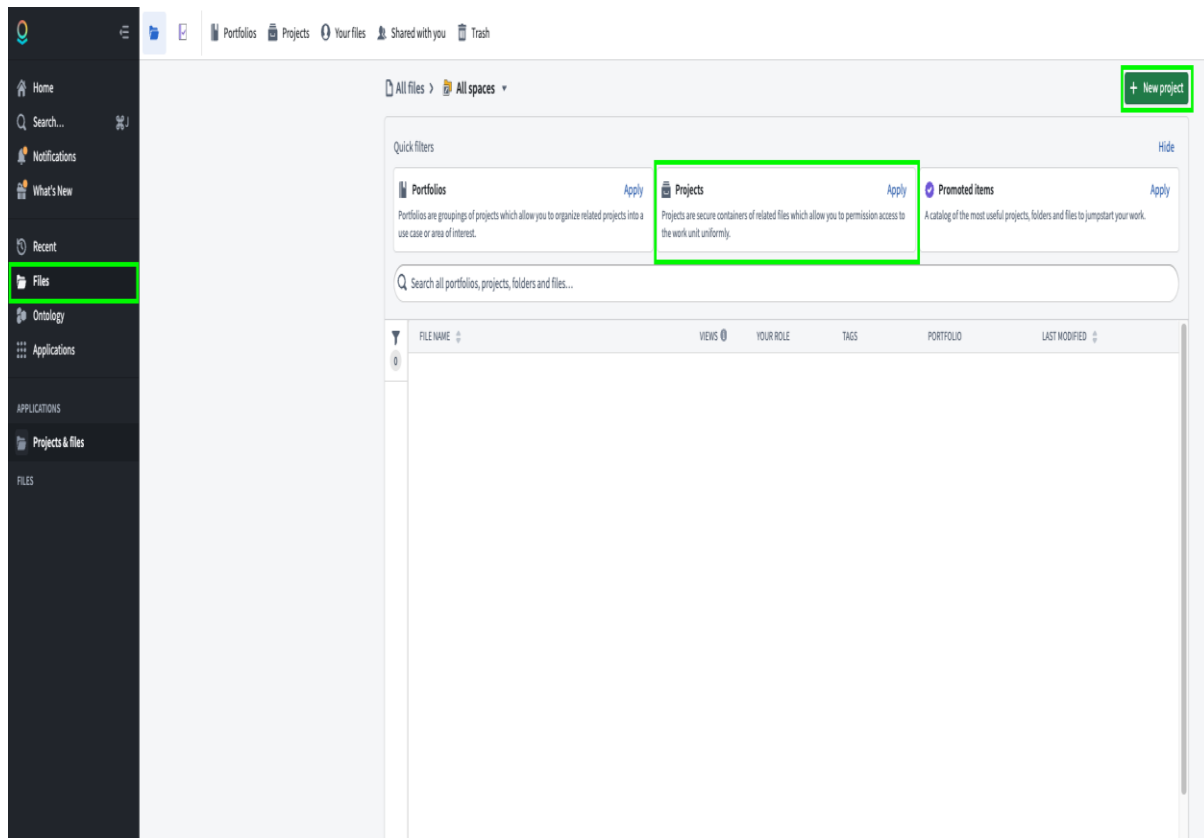
Create a Foundry Learning Project

Note if you already have your own Learning Project on your Foundry instance you can skip this step and move on to the next step to create your course-specific folder.

Step 1: Create a Project

1. Click on *Files* in the left sidebar
2. Click on *Projects* in the top bar
3. Click on *+ New Project* in the top right

Note: If you do not have permissions to create a project, please select a project you feel comfortable using and continue with the next instructions to create a folder. Do not use "Your files" nor your home folder as certain tools cannot be accessed from your home folder.



4. Select your organization's space from the dropdown

5. Select a Template for your project (Your specific Foundry enrollment may have different templates to what is shown in the screenshot below)

Create new project [Close]

Space: Palantir [Dropdown]

Default Template →

testing

[Manage project templates](#) [External Link]

6. Give your project a name, such as: Learning (<username>)

7. Give your project a description that would help another user understand what the project contains and is used for

8. If visible, do not select a Portfolio, as this is optional

9. In the 'Advanced' dropdown, this typically defaults to granting others the **Editor** role. To keep your training uneditable, you can change this to **Viewer**

10. Click on **Create project**

 **Create new project** ×

Name

Learning <username>

Project description (optional)

Personal Learning Project for Palantir Learn courses.

Advanced

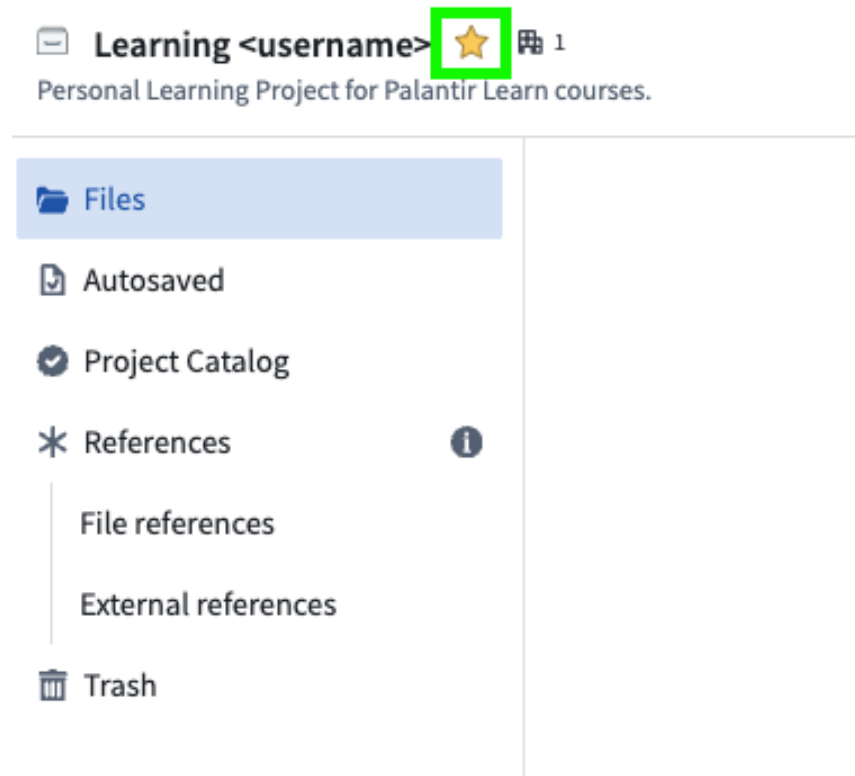
The project will be created in  **Palantir**. Everyone from  **Palantir** will be able to see its existence . They will need a role on the project to see files within it. ▼

[Back](#) ✓ Create project

Create a Course-Specific Training Folder

Step 1: Create a Folder

1. In the top left, click on the star to favorite your project. This will allow you to find it quickly later on.
2. Click on *New* in the top-right corner
3. Select *Folder*



4. Name the new folder after the current learning course - "Speedrun: Your First E2E Workflow"

NAME ^		L
Files	Speedrun: Your First E2E Workflow	1
Autosaved		
Project Catalog		
References		
File references		
External references		
Trash		

Ingestion: Getting Data Into Foundry

Introduction

Foundry includes a powerful **Data Connection** application. It comes with connectors for over 200 systems and databases and a flexible plugin architecture for the rest. It also has powerful DevOps tools for managing recurring batch and streaming ingestions. However, for the sake of this tutorial, we will manually upload data.

You should have access to three files below. Download these to your device by clicking the blue links below.

1. `orders_office_goods.csv`: All the orders from Office Goods Corp., the fictional company you work for
2. `orders_bureau_transactional_system.csv`: All the orders from Bureau SAS, the fictional company your company bought
3. `consolidated_customers.csv`: A consolidated list of customers across both fictional companies

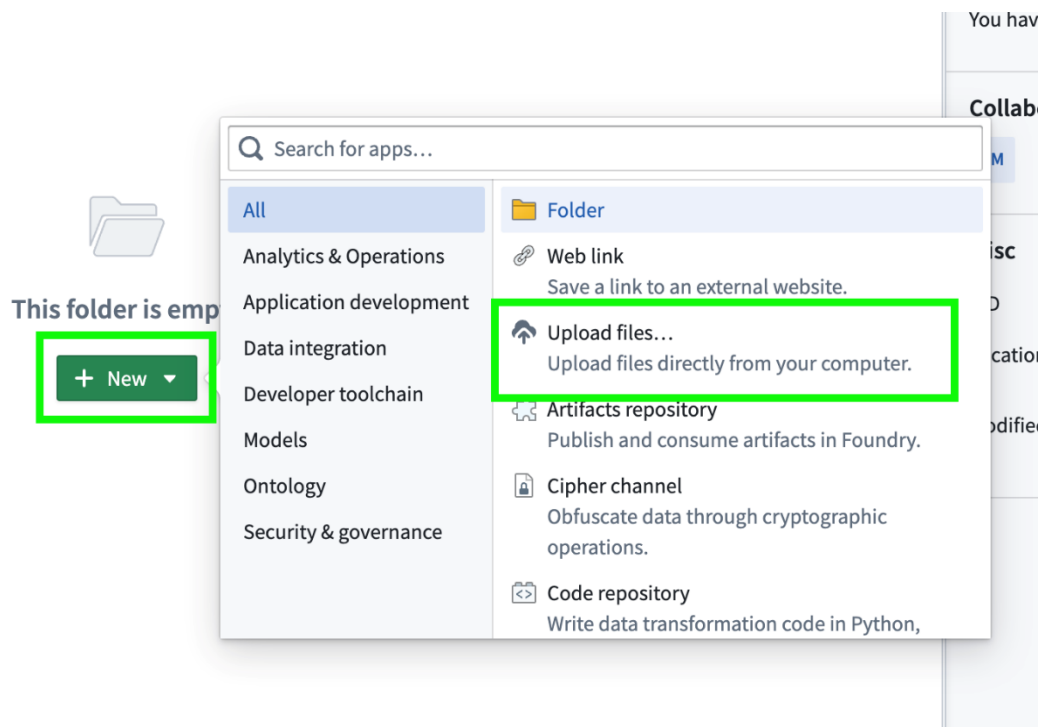
Downloads

- [orders_office_goods.csv](#)
- [orders_bureau_transactional_system.csv](#)
- [consolidated_customers.csv](#)

Manual Upload

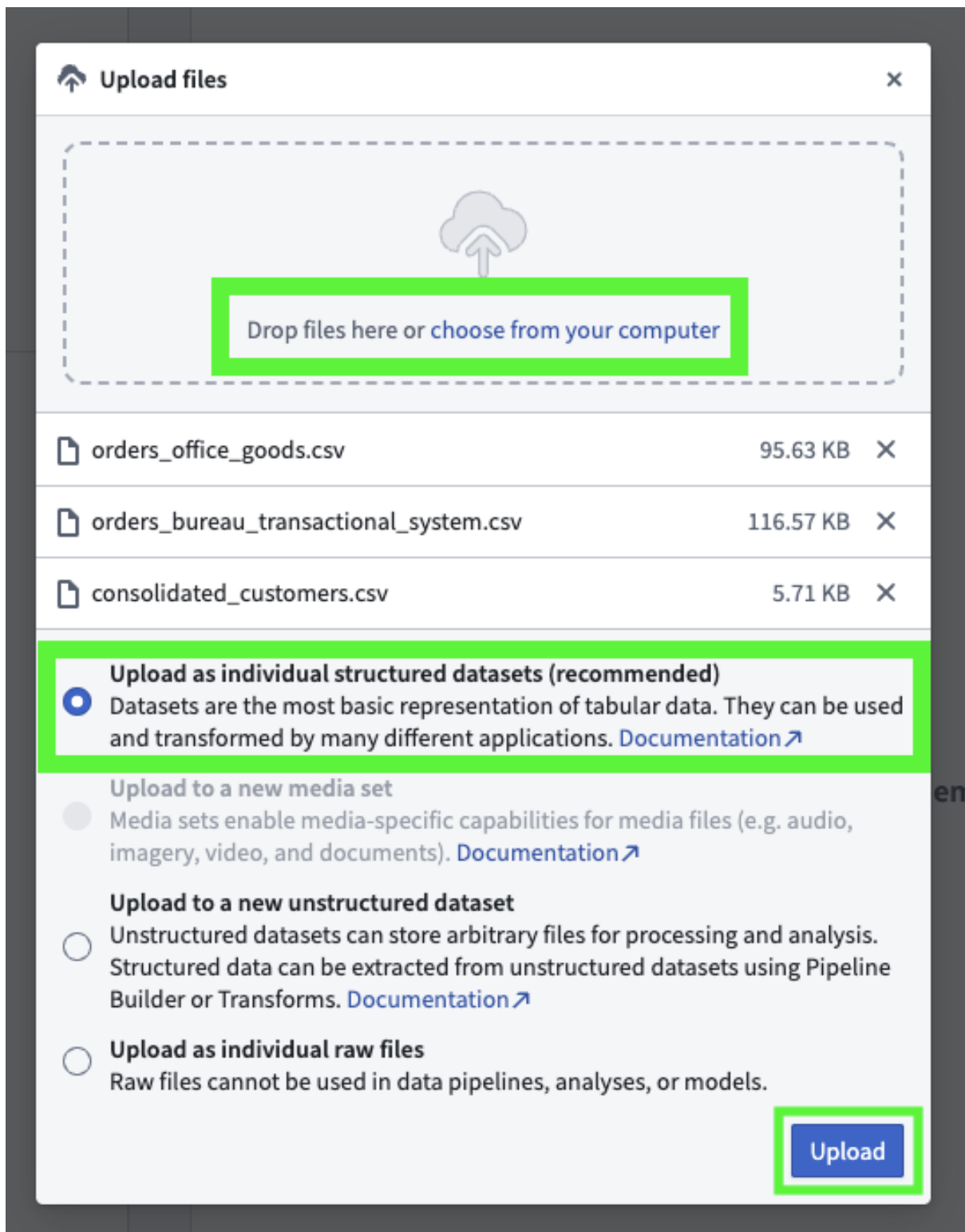
Step 1: Upload Data

1. Open your Learning Project and click into your folder for this course
2. Click on *New*
3. Click on *Upload files...*



4. Select the file by *clicking on choose from your computer*. Do this for all three files
5. Toggle *Upload as individual structured datasets (recommended)*
6. Click on *Upload*

Tip: You can also import by dragging files directly to your folder.



Conclusion

You should now have three datasets in Foundry! Although we manually uploaded them in this training, the following steps would be the same as if we had ingested them directly from systems via Data Connection.

Files	NAME ^	LAST UPDATED	TAGS
Autosaved	consolidated_customers	Thu, Oct 10, 2024, 3:50:35 PM	
Project Catalog	orders_bureau_transactional_system	Thu, Oct 10, 2024, 3:50:35 PM	
References	orders_office_goods	Thu, Oct 10, 2024, 3:50:36 PM	
File references			
External references			
Trash			

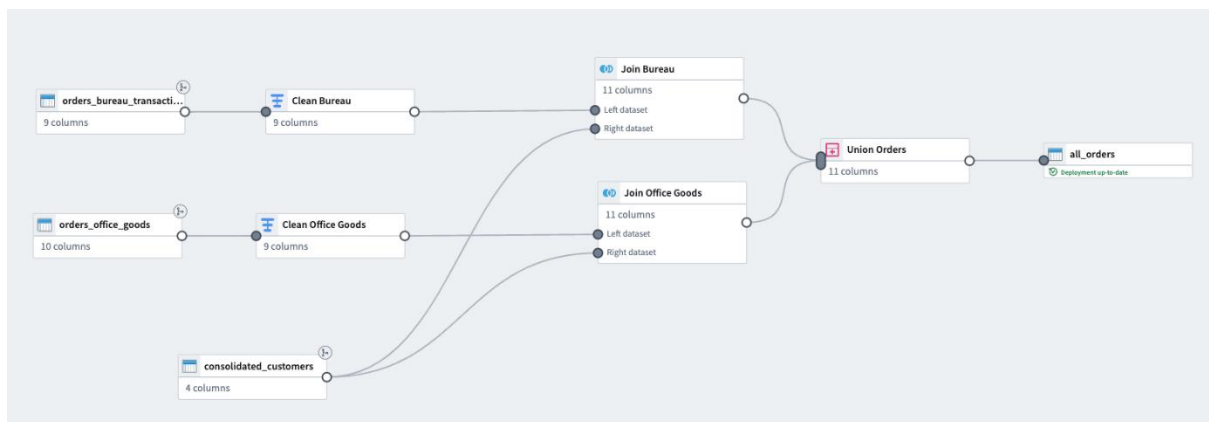
Transforming the Data

Introduction

In the next steps, we are going to clean the datasets, join them with the master customer list, and union them so that we have a single dataset containing all orders.

Foundry has several ways to do this. In this tutorial, we will use **Pipeline Builder**, which is a flexible low-code/no-code app for rapidly prototyping and building production-grade pipelines.

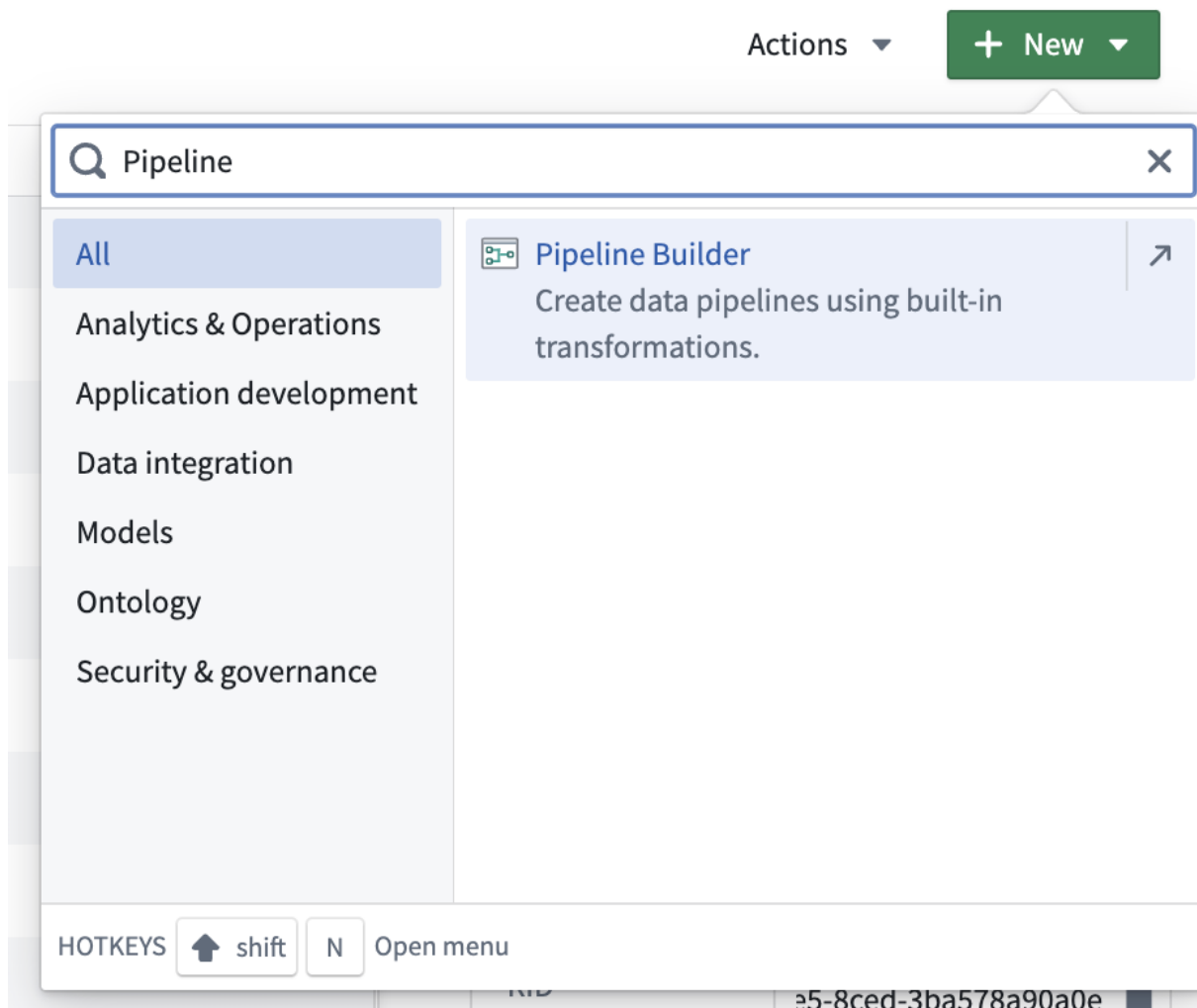
By the end of this section, you will have created the following pipeline and outputs:



Create a new Pipeline and add data

Step 1: Create a Pipeline

1. Navigate to your folder for this training course
2. Click **+ New** in the top right
3. Search and select *Pipeline Builder*



4. Give your pipeline a name like Orders Pipeline
5. Click on *Batch Pipeline* and *Standard*. We do not need the high-frequency updates of a streaming pipeline, or the fast executions of a Faster pipeline for this use case
6. Click on *Create Pipeline*

Configuration

**Batch pipeline**
Builds and transforms entire datasets on each deploy. For data that is ingested periodically.

**Streaming pipeline**
Transforms data continuously as new data is made available. For high frequency ingestion.

Select batch compute

**Standard**

Default

Build your pipelines using advanced expressions. Backed by Spark.

**Faster**

New

Speed up your pipelines and save compute. Backed by DataFusion.
[Learn More](#)

**External**

Beta

Build your pipelines using external compute platforms. [Learn More](#)

Compare types

← Back

Create pipeline

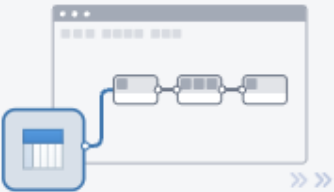
Step 2: Import Data

1. On the next screen, click *Add Foundry data*

Welcome to Pipeline Builder

Get started by adding datasets, then define transform logic to derive target outputs.

 Take a tour



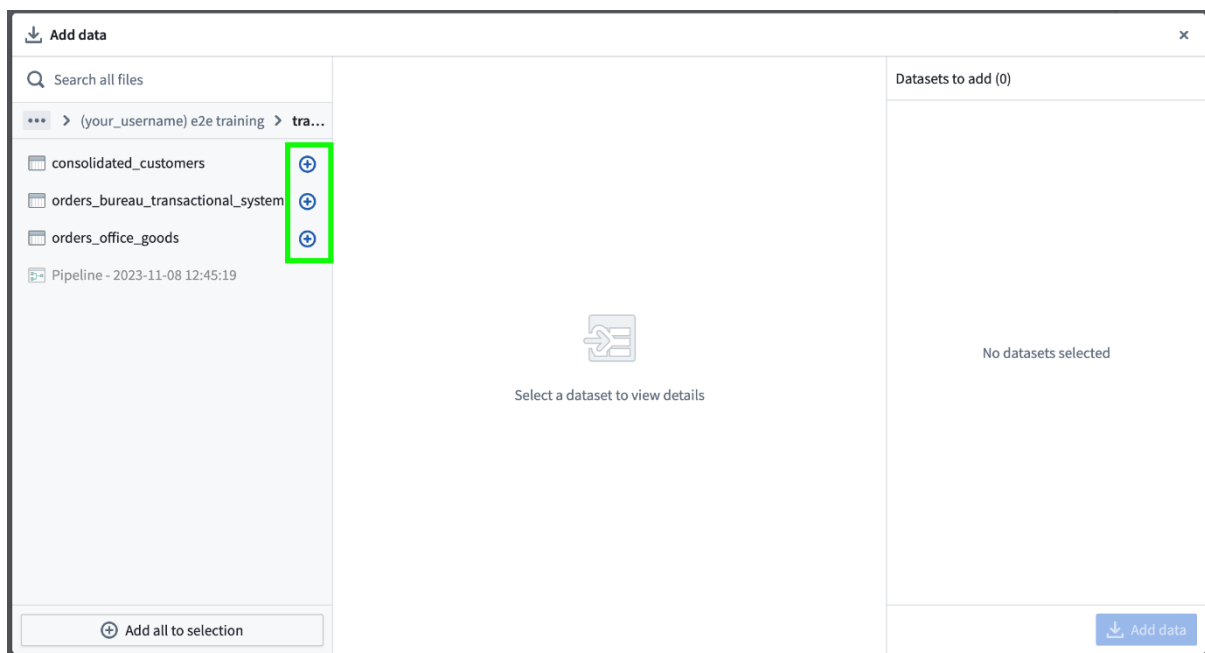
**Add Foundry data**
Recommended if you have already ingested data into Foundry.

**Add data to Foundry**
Import data from outside Foundry and start using it now

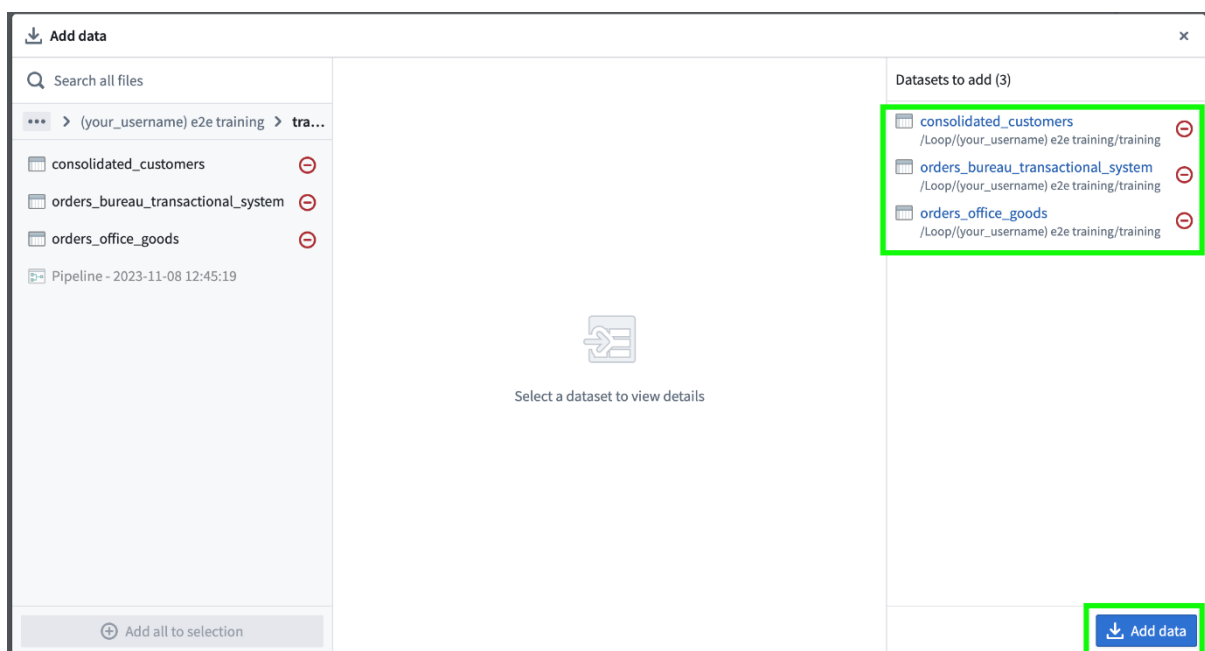
**Upload from your computer**
Recommended if you have sample data available locally.

**Manually enter data**
Recommended if you do not have data available to import.

2. Add each of the three datasets previously imported by using + icon



3. Click *Add Data*

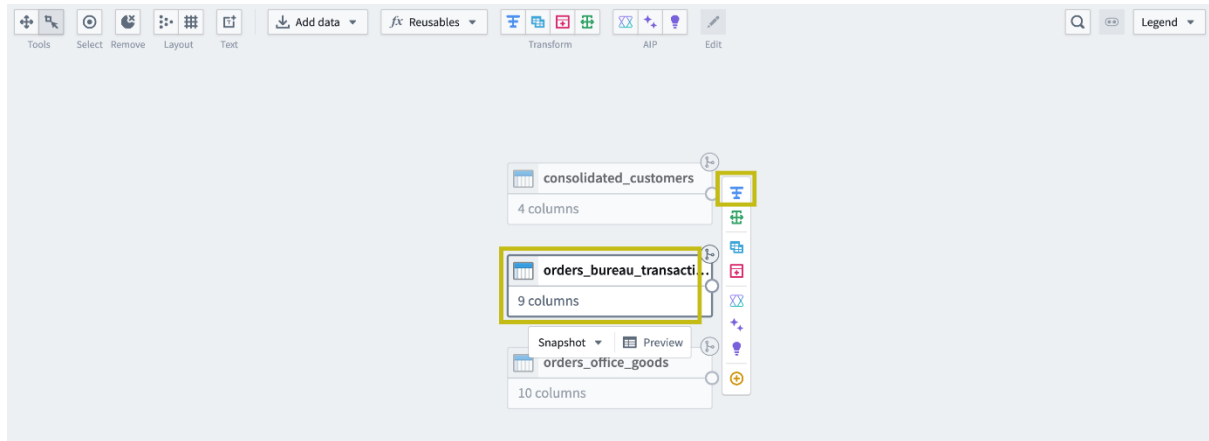


Clean Bureau SAS

Step 1: Open Transform

Your screen should now show three rectangular nodes representing each of your three imported datasets.

1. Click on *order_bureau_transactional*
2. Click on *Transform* (the three blue lines icon)

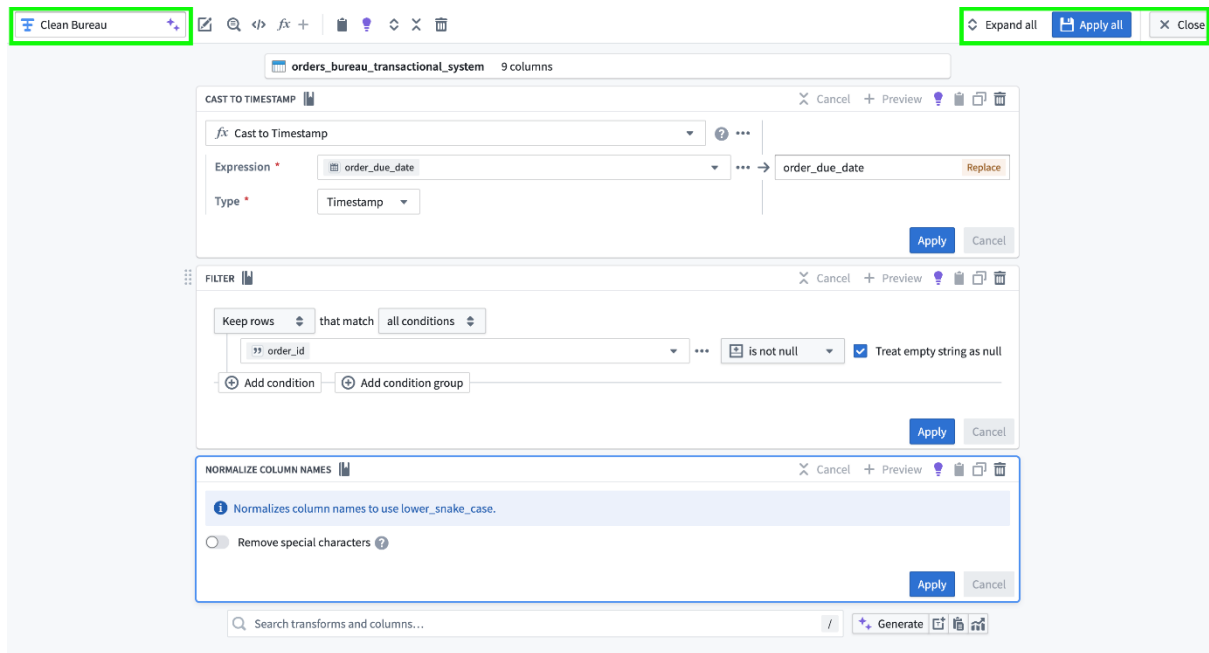


Step 2: Clean Data

In this step, we'll start to clean the Bureau transaction data. Refer to the screenshot and green boxes below for help navigating the transform window.

The cleaning tasks below standardizes date formats into timestamps, removes orders without a proper primary key identifier (`order_id`), and normalizes the format of our column names.

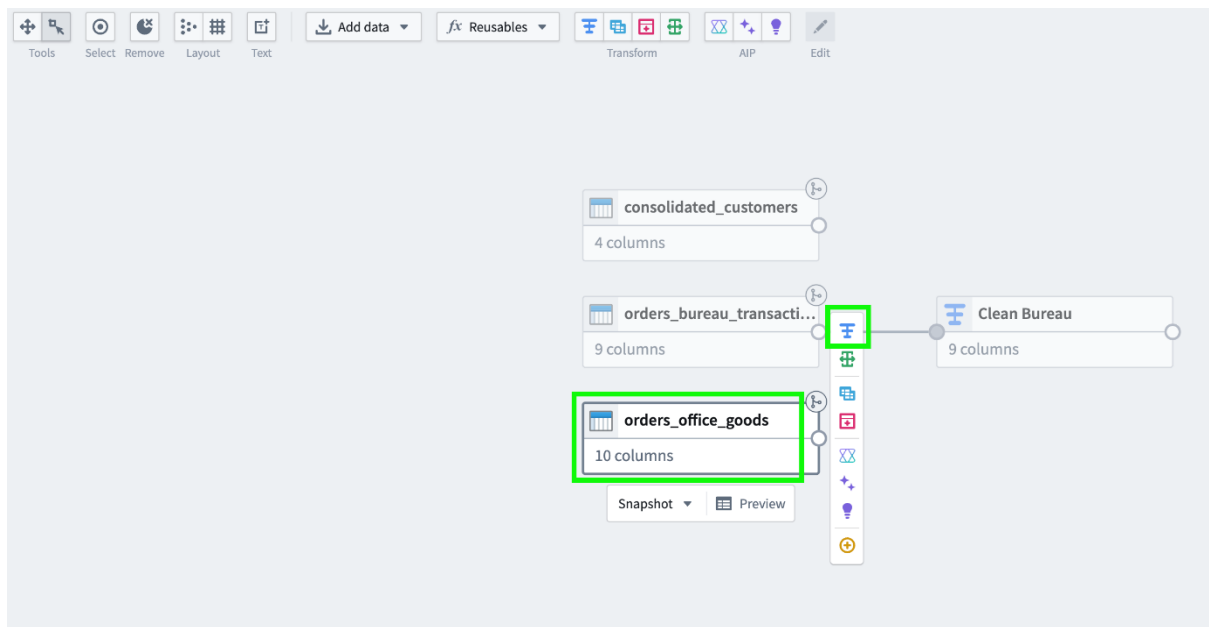
1. In the top left, rename the transform to Clean Bureau
2. Search Cast in the *Search transforms and column...* search bar
 1. Cast `order_due_date` from Date to Timestamp
 2. Click *Apply*
3. Search Filter Rows
 1. Filter rows on the column `order_id` set to *is not null* and select the box for *Treat empty string as null*
 2. Click *Apply*
4. Search Normalize column names
 1. Leave the default settings
 2. Click *Apply*
5. Click on *Apply all changes* in the top right (if everything is already applied, this will be greyed out, and you can proceed to the next step)
6. Click *Close* in the top right



Clean Office Goods Corp

Step 1: Open Transform

1. Click on *orders_office_goods*
2. Click on *Transform*

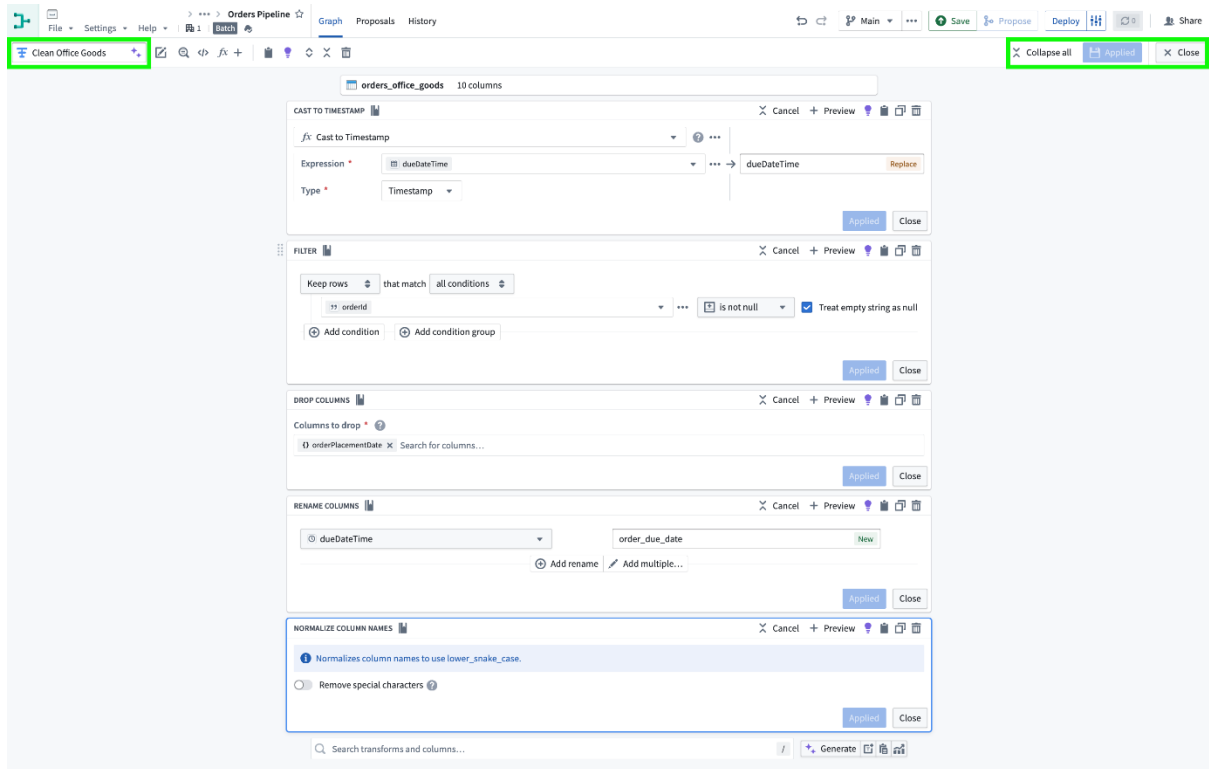


Step 2: Clean Data

In these steps, we'll start to clean the Office Goods data. Refer to the screenshot and green boxes below for help navigating the transform window.

Similar to the cleaning steps we took on the Bureau data, the steps below standardize date formats into timestamps, remove orders without a proper primary key identifier (*orderId*), remove an unnecessary column, rename a column, and normalize the format of our column names.

1. In the top left, rename the transform to Clean Office Goods
2. Search Cast
 1. Cast *dueDateTime* from Date to Timestamp
 2. Click *Apply*
3. Search Filter Rows
 1. Filter *orderId* to *is not null*. Select the box for *Treat empty string as null*
 2. Click *Apply*
4. Search Drop columns
 1. Type in and select the *orderPlacementDate* column
 2. Click *Apply*
5. Search Rename columns
 1. Select the *dueDateTime* column and rename to *order_due_date*
 2. Click *Apply*
6. Search Normalize column names
 1. Click *Apply*
7. Click *Apply all changes* (if everything is already applied, this will be greyed out, and you can proceed to the next step)
8. Click *Close*

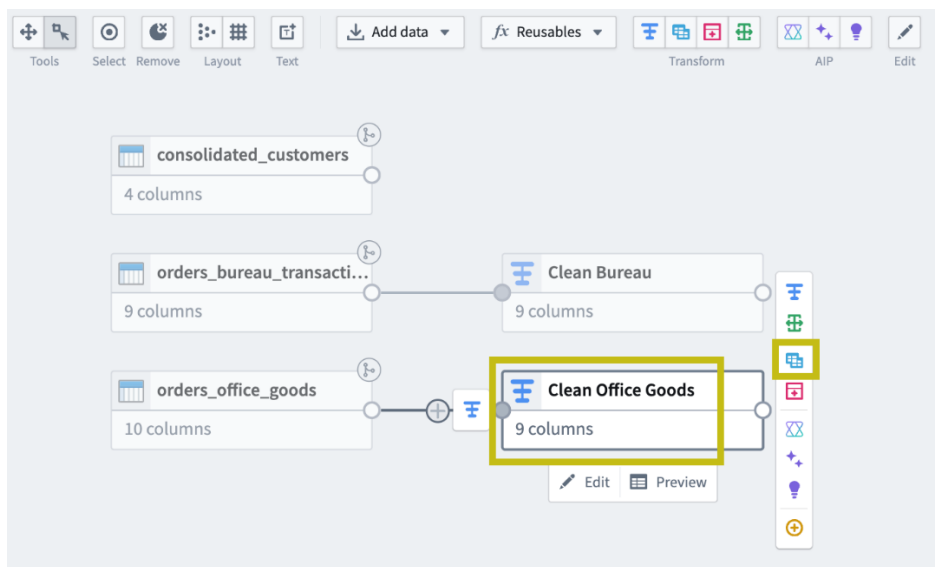


Join Clean Office Goods with Customer List

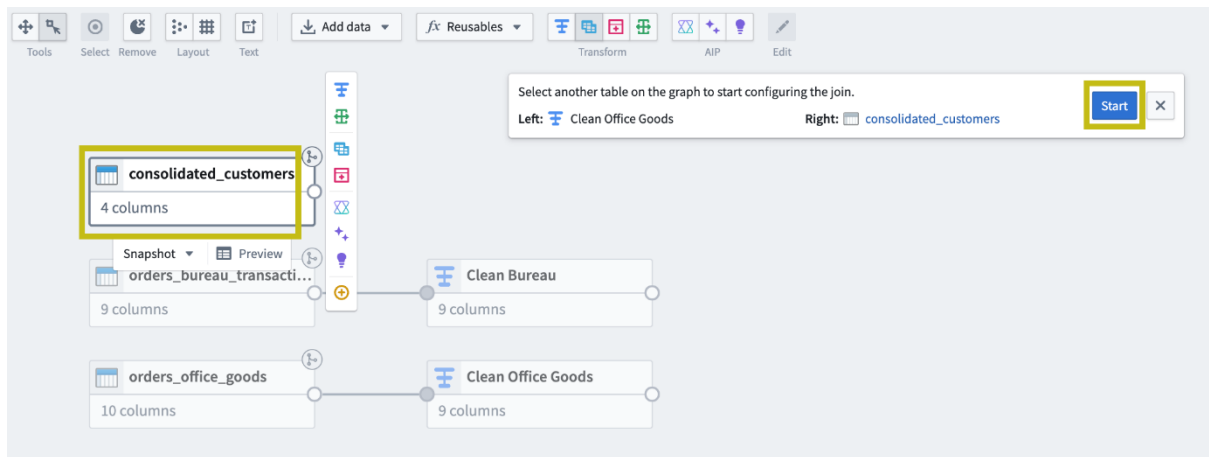
Step 1: Select Datasets to Join

In the following steps, we will join the Office Goods and Customers data together in order to pull customer-specific information into our Office Goods dataset. If you have not yet, ahead of these steps is a good time to explore the data that is in each of these datasets.

1. Navigate to the graph view
2. Click on *Clean Office Goods*
3. Click on *Join* (the two rectangle icon)



4. Click on *consolidated_customers*
5. Click on *Start*



Step 2: Select Join Properties

1. In the top left, rename join to Join Office Goods
2. Set your match condition as: *customer_id is equal to officegoods_customer_id*
3. On the right hand side, deselect *officegoods_customer_id* and *bureau_customer_id*
4. Click on *Apply*
5. Click on *Close*

Join Office Goods

Description

Applied

Close

Join type

Left join

Joins two datasets together, keeping all rows from the left table and only rows which satisfy the provided condition from the right table.

Input tables

Clean Office Goods

Swap

consolidated_customers

Match condition

rows that match all conditions

Basic Advanced

customer_id

is equal to

officegoods_customer_id

Add match condition

Select columns

Left: Clean Office Goods

Right: consolidated_customers

☒ Auto-select all left columns

Search for columns...

Select types

9 of 9 columns selected

Selected Not selected

Select all Deselect all

order_due_date

Timestamp

✓

order_id

String

✓

customer_id

String

✓

status

String

✓

assignee

String

✓

quantity

Integer

✓

Prefix right columns

☐ Auto-select all right columns

Search for columns...

Select types

2 of 4 columns selected

Selected Not selected

Select all Deselect all

officegoods_customer_id

String

bureau_customer_id

String

consolidated_customer_id

String

✓

customer_name

String

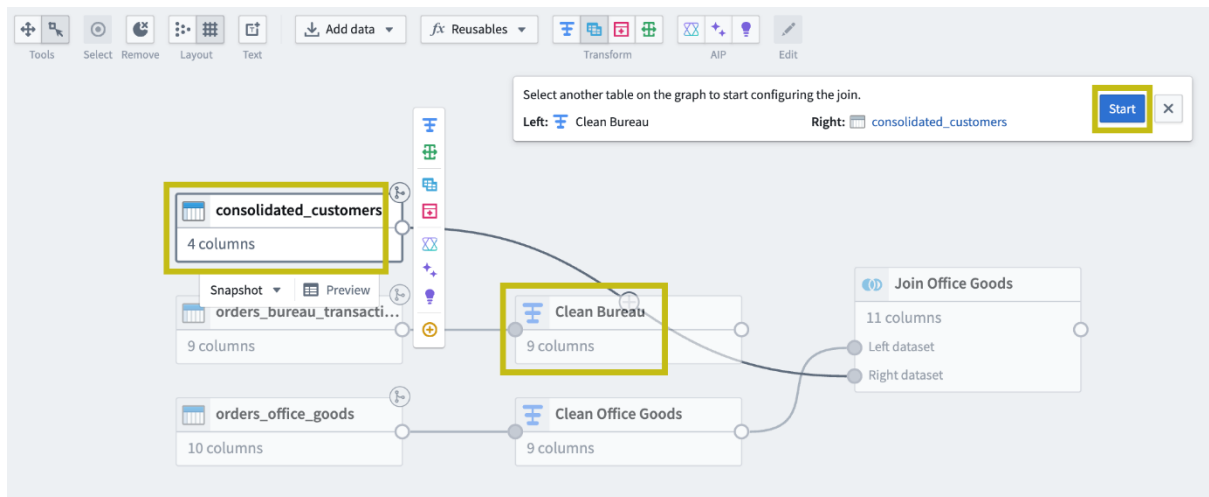
✓

Join Clean Bureau with Customer List

Step 1: Select Datasets to Join

Similar to the previous step, here we will join the Bureau orders dataset and the Customers data together in order to pull customer-specific information into our Bureau orders dataset.

1. From the graph, click on *Clean Bureau*
2. Click on *Join*
3. Click on *consolidated_customers*
4. Click *Start*



Step 2: Select the Join Properties

1. In the top left, rename the join node to *Join Bureau*
2. Set the match condition to *customer_id is equal to bureau_customer_id*
3. On the right hand side, deselect *officegoods_customer_id* and *bureau_customer_id*
4. Click on *Apply*
5. Click on *Close*

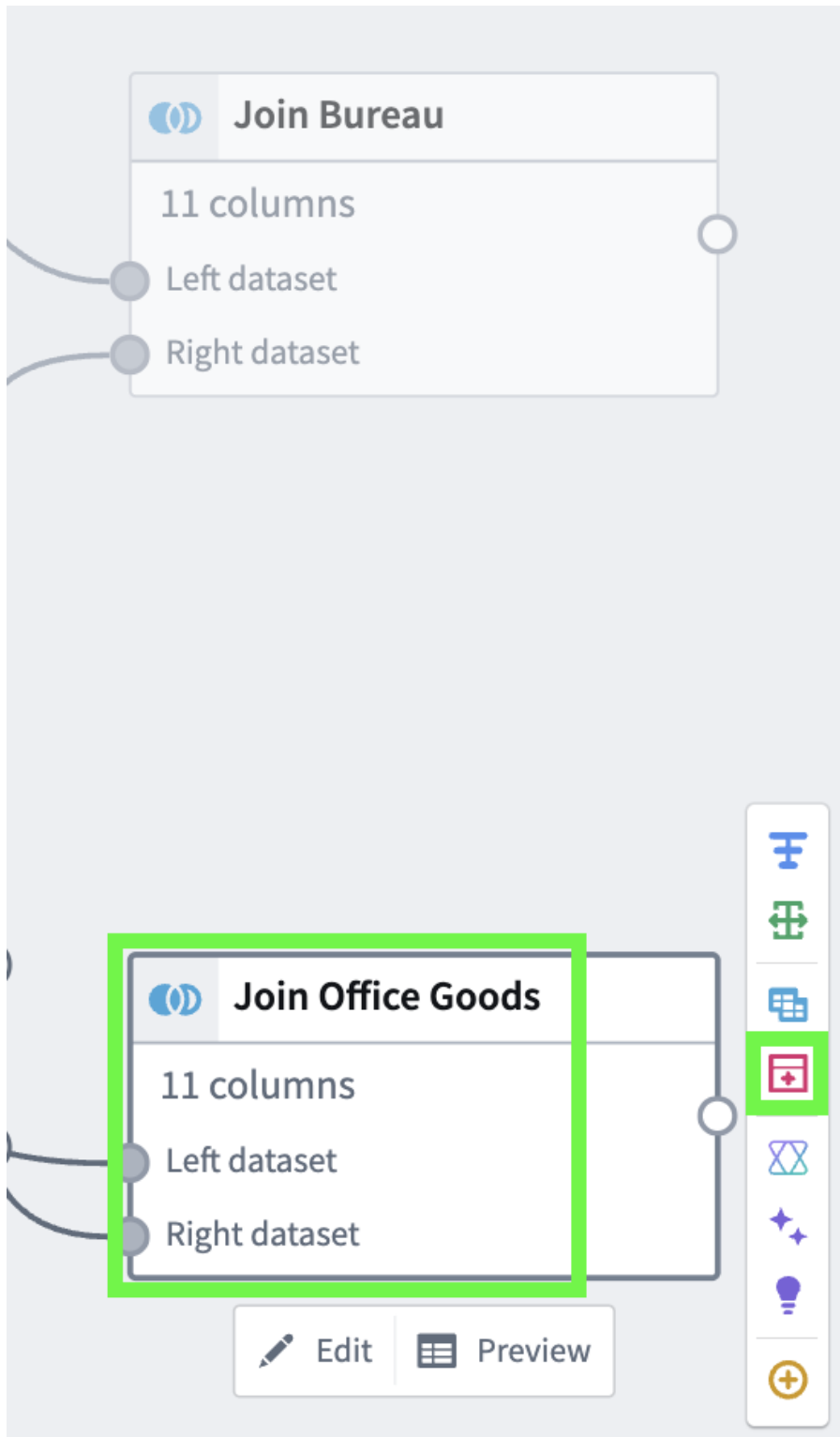
Union Datasets

Step 1: Select Datasets

In the following steps, we will union the two orders datasets into one master dataset. This will provide us with one source-of-truth for orders and their related customer information. Note that in order to union datasets together they need to have the exact same schema

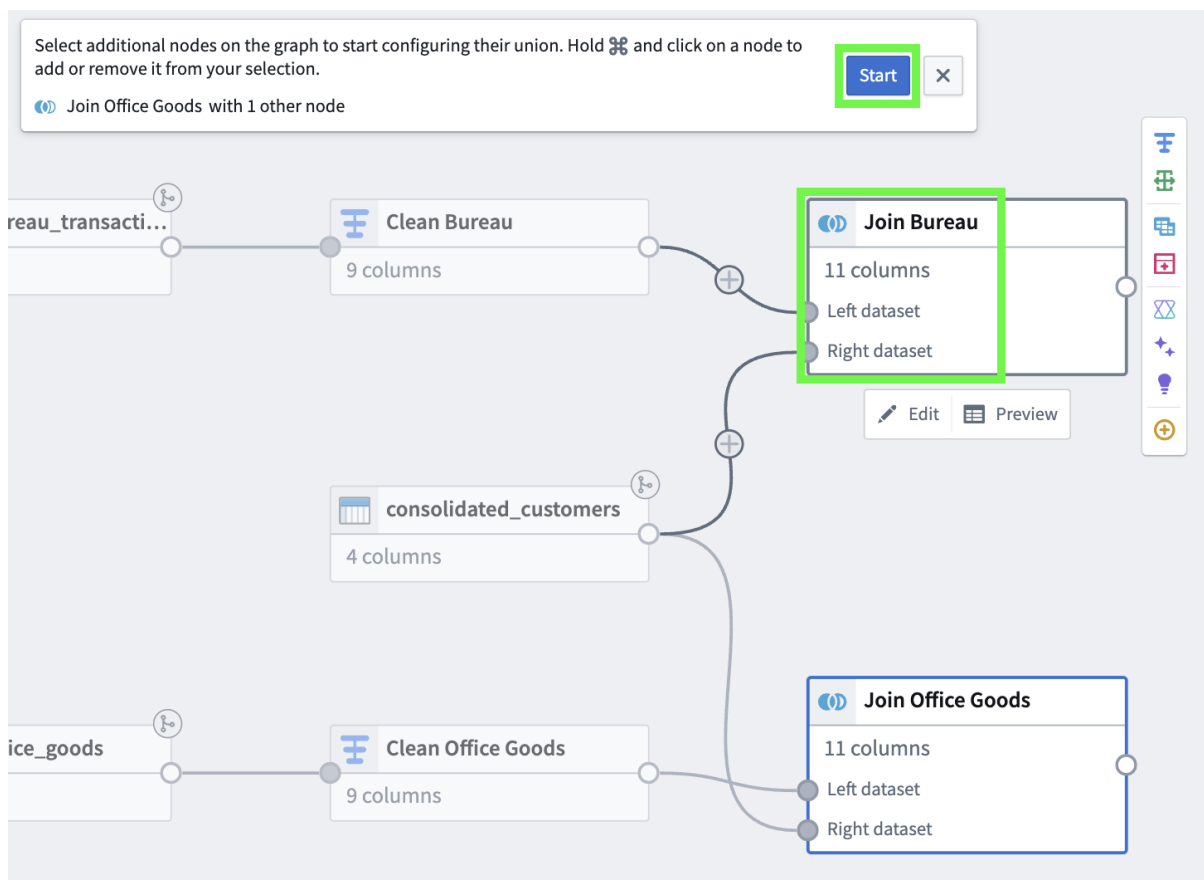
(column types). Our previous cleaning steps have allowed us to normalize the datasets and union them.

1. Click on *Join Office Goods*
2. Click on *Union* (the red square icon)



3. Click on *Join Bureau*

4. Click on *Start*



Step 2: Select Union

1. In the top left, rename to Union Orders
2. Click on *Create union*
 - *If you get a schema error here, double check that order_id column name in both datasets match. One might be called orderId, instead of order_id. You can resolve this by renaming the column in a previous transform node.*
3. Click *Close*

Union Orders

Description

Apply

Close

Inputs

Join Office Goods

Join Bureau

Output

Union Orders

Union type

Union by name

Create union transform to preview the output

Create union

4. Double click on the *Union Orders* node to explore the data. There should be 11 columns, resembling the screenshot below

5. Click on *Close*

Orders Pipeline

Batch

Graph

Proposals

History

Save

Propose

Deploy

Share

ion

Applied

Close

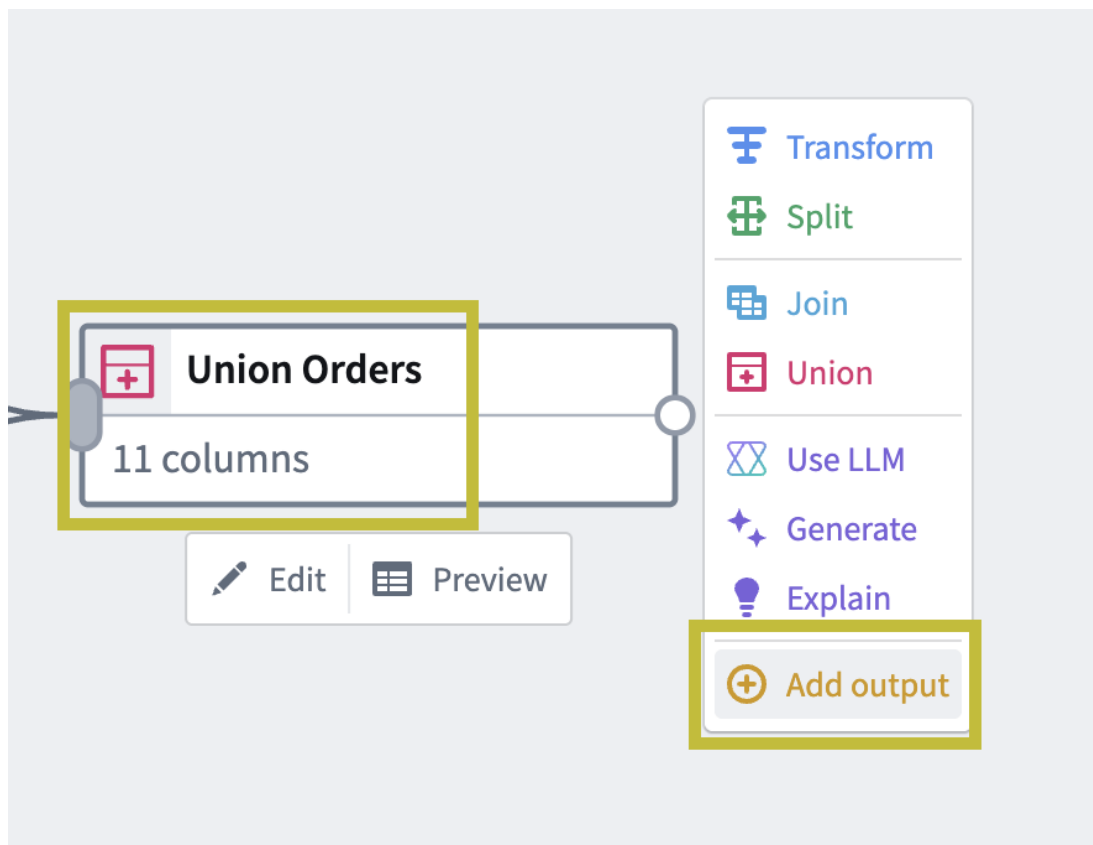
Union Orders		Previewing 500 rows		11 columns		Search columns...	
Search 11 columns...		No inputs sampled					
order_due_date	order_id	customer_id	status	assignee	quantity	item_name	unit_price
2023-06-10T00:00:00.000Z	A8144497-e8b47bbe-99	A94201908	closed	Kristen Mohr			
2023-07-11T00:00:00.000Z	A8144497-2876d968-98	A94201908	closed	Kristen Mohr			
2023-07-24T00:00:00.000Z	A8144497-fb778d8b-9a	A94201908	closed	Kristen Mohr			
2023-07-26T00:00:00.000Z	A8144497-c206b101-f8	A94201908	closed	Kristen Mohr			
2023-06-15T00:00:00.000Z	A90934179-bb6408d1-d	A94201908	assigned	Alfredo Bins			
2023-07-27T00:00:00.000Z	A90934179-42d95fcf-a	A94201908	assigned	Alfredo Bins			
2023-07-16T00:00:00.000Z	A31854563-15357af5-0	A94201908	closed	Kristen Mohr			
2023-06-11T00:00:00.000Z	A31854563-e8b47bbe-9	A94201908	closed	Kristen Mohr			
2023-07-19T00:00:00.000Z	A31854563-bb6408d1-d	A94201908	closed	Kristen Mohr			
2023-06-06T00:00:00.000Z	A31854563-fb778d8b-9	A94201908	closed	Kristen Mohr			
2023-06-28T00:00:00.000Z	A31854563-2876d968-9	A94201908	closed	Kristen Mohr			
2023-06-16T00:00:00.000Z	A31854563-42d95fcf-a	A94201908	closed	Kristen Mohr			
2023-06-02T00:00:00.000Z	A17351508-fd748d48-6	A94201908	closed	Alfredo Bins			
2023-06-29T00:00:00.000Z	A17351508-15357af5-0	A94201908	closed	Alfredo Bins			
2023-07-22T00:00:00.000Z	A17351508-c206b101-f	A94201908	closed	Alfredo Bins			
2023-07-13T00:00:00.000Z	A17351508-e8b47bbe-9	A94201908	closed	Alfredo Bins			
2023-07-02T00:00:00.000Z	A69256104-c206b101-f	A94201908	closed	Alfredo Bins			
2023-07-30T00:00:00.000Z	A69256104-42d95fcf-a	A94201908	closed	Alfredo Bins			
2023-06-22T00:00:00.000Z	A28496095-6d3cbc43-8	A94201908	assigned	John Dooley			
2023-06-02T00:00:00.000Z	A28496095-42d95fcf-a	A94201908	assigned	John Dooley			
2023-07-26T00:00:00.000Z	A28496095-e8b47bbe-9	A94201908	assigned	John Dooley			
2023-07-28T00:00:00.000Z	A28496095-bb6408d1-d	A94201908	assigned	John Dooley			
2023-07-19T00:00:00.000Z	A28496095-c206b101-f	A94201908	assigned	John Dooley			
2023-06-22T00:00:00.000Z	A28496095-fd748d48-6	A94201908	assigned	John Dooley			
2023-07-09T00:00:00.000Z	A28496095-15357af5-0	A94201908	assigned	John Dooley			
2023-07-25T00:00:00.000Z	A28496095-fc220883-6	A94201908	assigned	John Dooley			
2023-07-14T00:00:00.000Z	A19183848-bb6408d1-d	A94201908	closed	Latoya Gulgowski			
2023-06-13T00:00:00.000Z	A19183848-fc220883-6	A94201908	closed	Latoya Gulgowski			
2023-07-20T00:00:00.000Z	A42657692-2876d968-9	A94201908	assigned	Taylor Hilll			
2023-07-24T00:00:00.000Z	A42657692-fd748d48-6	A94201908	assigned	Taylor Hilll			
2023-07-10T00:00:00.000Z	A42657692-c206b101-f	A94201908	assigned	Taylor Hilll			
2023-06-27T00:00:00.000Z	A42657692-fc220883-6	A94201908	assigned	Taylor Hilll			
2023-06-01T00:00:00.000Z	A78902267-c206b101-f	A94201908	closed	Erick Quigley			

Set a Dataset Output

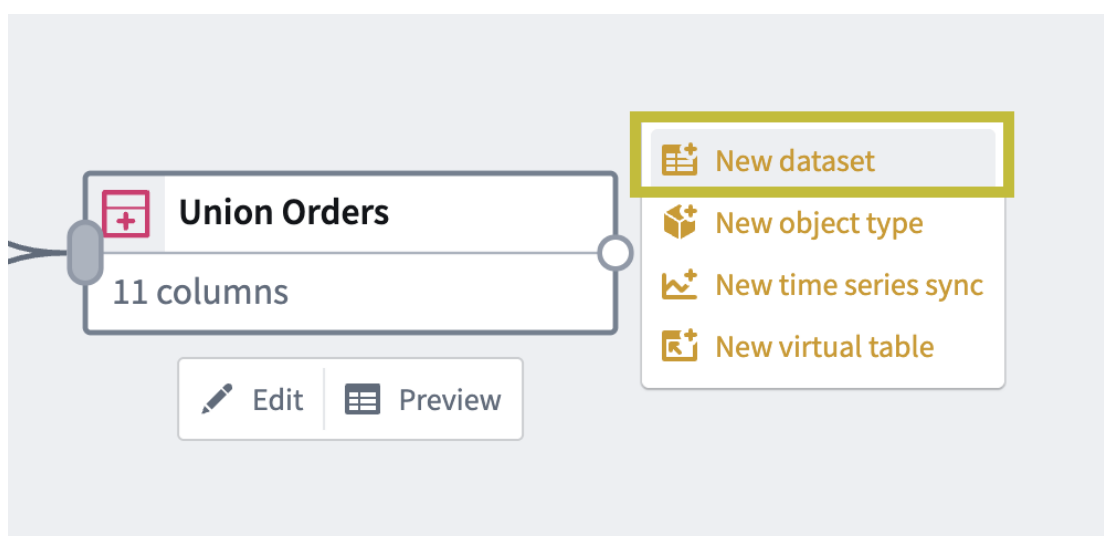
Step 1: Add Output

In the following steps, we will create a dataset where our transformed data can be written to and accessed by other applications in Foundry. Note that the dataset previews we've seen at each step of the pipeline are not actually saved in the platform outside of this pipeline - we need to manually define a dataset and deploy it to be accessible outside the pipeline.

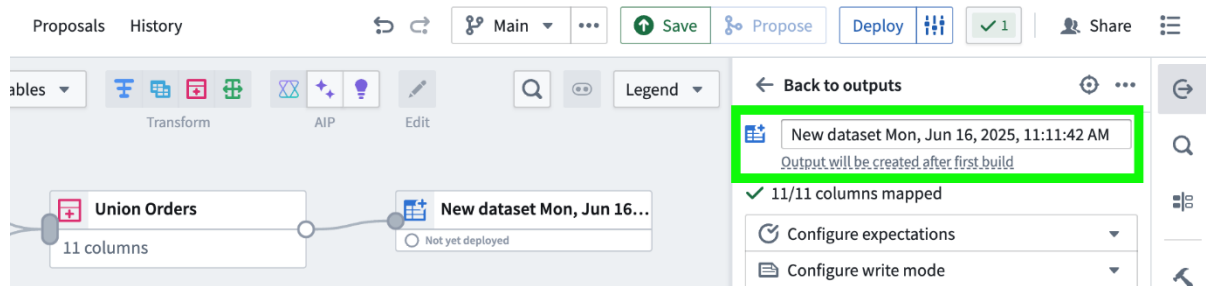
1. Click on *Union Orders*
2. Click on *Add output*



3. Click on *New dataset*



4. Rename *New dataset [date]...* to *all_orders*



For this example, we will keep all columns to remain in our deployed dataset, but note you could deselect columns here to be removed

Finalize and Run Pipeline

Step 1: Deploy Pipeline

In the step below, we'll deploy the pipeline and output which writes the new *all_orders* dataset to Foundry. This creates a dataset file in Foundry we can use in other applications in the platform.

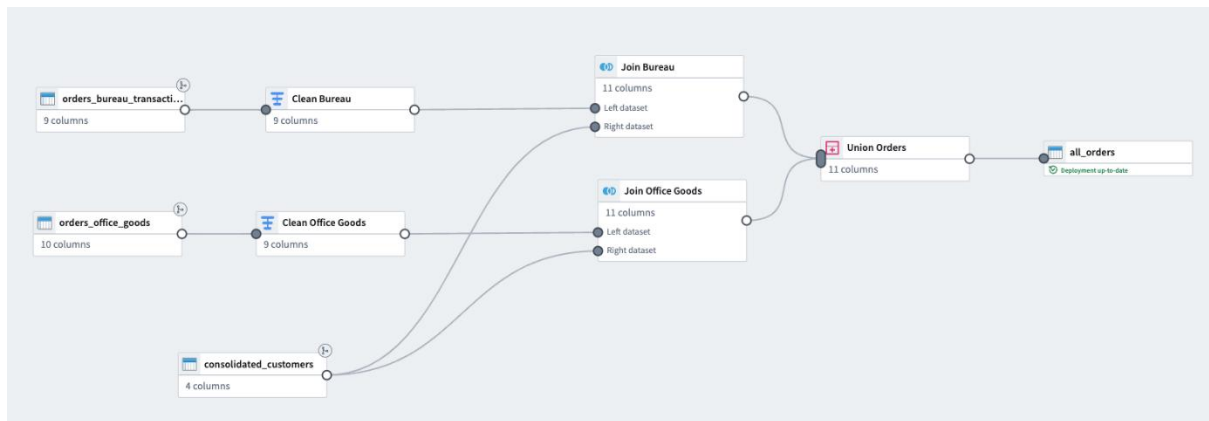
1. Navigate to the far right of your screen
2. Click on *Save*
3. Click on *Deploy*
4. Click on *Deploy pipeline* (this may take 1-2 minutes to complete)

The screenshot displays the Databricks Pipeline Builder interface. At the top, a navigation bar includes buttons for 'Main', 'Saved', 'Propose', 'Deploy', and 'Share'. The 'Saved' and 'Deploy' buttons are highlighted with green boxes. Below the navigation bar, the main workspace shows a pipeline diagram with a single job named 'all_orders'. To the right, a sidebar titled 'Deploy this pipeline' provides configuration options. It includes a 'Last deployment' status of 'None', 'Deploy settings' with 'Errors 0', a 'Compute profile' set to 'Default' (with an option for 'Extra small - warm pool'), and a 'Select outputs to build' section. In the 'Select outputs to build' section, 'All' and 'Single jobs' are selected, and under 'Single jobs', the 'all_orders' job is also selected. At the bottom of the sidebar, there are two buttons: 'View changes' and 'Deploy pipeline', with the 'Deploy pipeline' button highlighted by a green box.

Conclusion

You've just built your first pipeline!

There are some advanced topics we didn't cover like branching, custom code, promoting code to prod, pipelining best practices, and scheduling data refreshes; however, you're now familiar with the basic of Pipeline Builder and have built a functioning data pipeline.



Creating the Ontology

Introduction

We now need to make the *all_orders* dataset expressive and queryable, which is done through the **Ontology**.

The Ontology is a relational “digital twin” of your business comprised of **Objects**.

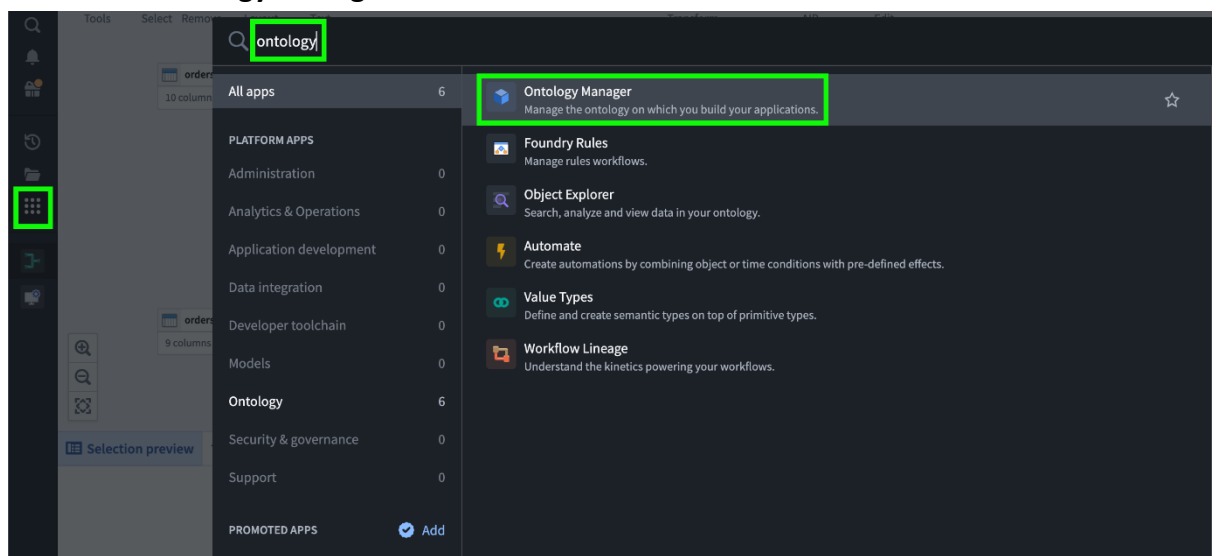
The Ontology is a compounding asset: Once you build the Objects for one use case, you can reuse them for other use cases and avoid reinventing the wheel. You can even create API bindings so that your applications outside of Foundry can interact with the Ontology in a first-class way.

We will define the Order object by using the **Ontology Manager App**.

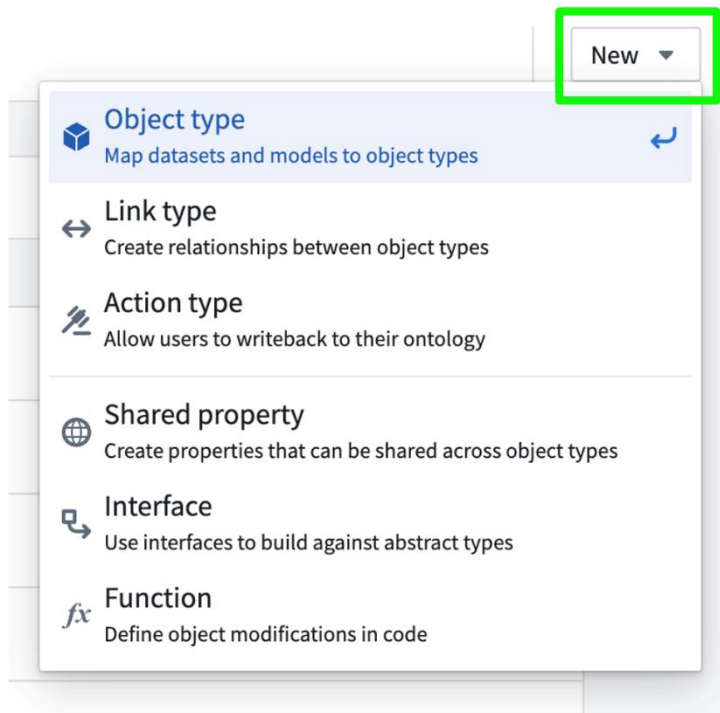
Create your Object

Step 1: Navigate to Ontology Manager

1. From the left-hand sidebar, select Applications
2. Search for **Ontology manager**



3. Click on *New* in the top right corner. If you're not able to select the *New* button proceed to steps 1b and 1c below.
4. Choose *Object Type*

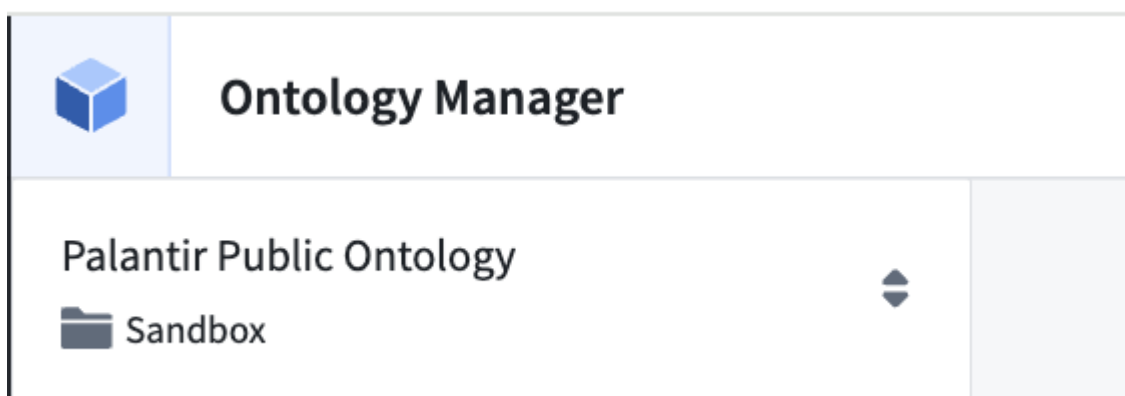


The following steps are only relevant for users where the *New* button is not selectable. You can skip to 'Step 2' below if you are able to select the *New* button.

Step 1b: Select a new Ontology

Your organization may have multiple Ontologies with differing permissions. By default, users may view an Ontology where they do not have create permissions.

1. In Ontology Manager, navigate to the left-side of the screen and note if there is a drop-down toggle to select Ontologies. If you do not have this toggle, proceed to Step 1c below.



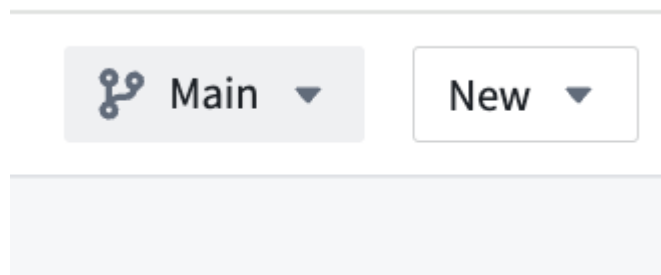
2. Toggle to an Ontology where the *New* button described in the steps above becomes selectable. You may need to contact your organization's Foundry administrator to find out which Ontology is preferred for training purposes.
3. Click *New*
4. Move on to Step 2 below.

Step 1c: Create an Ontology Branch

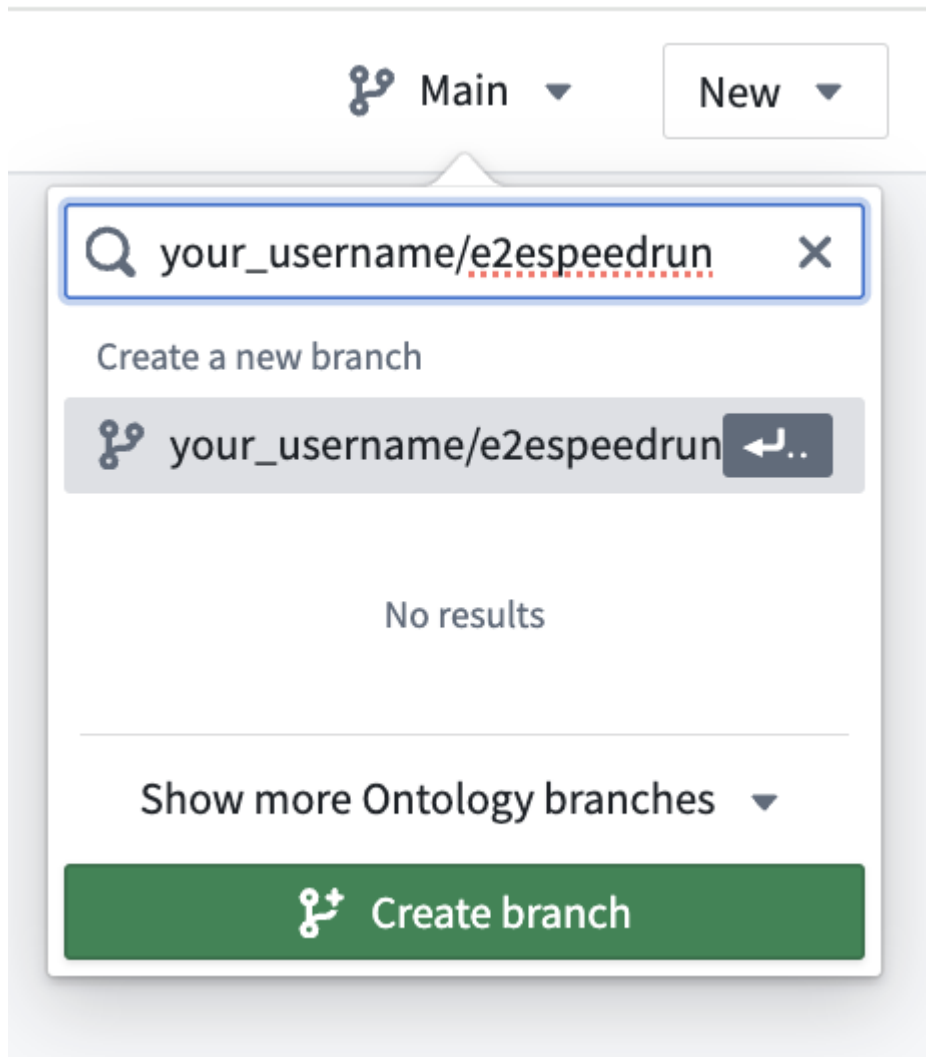
Note that some Organizations limit edits to the production Ontology via Ontology Proposals. If the 'New' button in the screenshot above is not clickable, you may need to create a custom branch of the Ontology for the following steps and seek approval from your Organization's Administrator for merging in the changes to the Main branch of the Ontology.

If you can not directly edit the Main Ontology, you can continue by creating a custom branch of the Ontology with the steps below. At the end of the 'Creating the Ontology' module, you will need to seek approval for merging in your changes to the Main Ontology before continuing. Further instructions for this can be found in the [Foundry documentation](#).

1. Click the *Main* button in the top right of your screen next to the *New* button



2. Type the name for your custom branch like [username]/e2espeedrun
3. Click *Create branch*



4. If you have the *Enable indexing* toggle available on your Foundry instance, select the toggle on and proceed to the next steps. Do not populate the *Optional* section.
5. If you do not have the toggle available, you will need to have your branch merged into the Main Ontology branch in order to proceed with the following steps. Contact your organization's Foundry Platform Administrator or Palantir representative for assistance.
6. Click *Create Branch*

no description

Create branch

×

This will create a Branch with your Ontology changes and a draft Proposal to review those changes. Open the Branch to view and add edits. Once all edits are approved, release the Proposal.

Branch name

username/e2espeedrun

×

👁

Enable indexing on this branch

Enable indexing if you need to preview schema changes for modified entities. This will use additional storage and compute. Only supported by Object Storage V2.

Dataset branch name (Optional)

Set this if you want to preview data or schema changes for modified entities from a branch other than master.

Cancel

🔗

Create branch

Step 2: Define Object Properties

1. In the *Create a new object type* window, select *Use existing datasource*
2. Click *Select datasource*, then search for and select your `all_orders` dataset you deployed in the previous steps
3. Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

5 Save location

STEP 1

Object type backing

Datasource

Continue without datasource

Create, edit, and delete object instances entirely through actions, with no dataset backing this object type

Use existing datasource

Back this object type with a dataset, restricted view, or other datasource you already have in Foundry

✓

Select a datasource to back this object type

all_orders

/Palantir/[REDACTED]/Speedrun: Your First E2E Workflow

✎

Replace

i

Instance permission

▼

Next

- In the Name field, input [username] Order. The *Plural name* field should auto-populate
- Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

5 Save location

STEP 2

Configure object type metadata

Icon

Name

Plural name

[username] Order

×

[username] Orders

×

Description

Enter optional description...

Groups (optional)

No group selected

+ Add to group

Back

Next

6. Select *Order Id* as the primary key of the new Object; this serves as each Object's unique identifier
7. Select *Item Name* as the title of the new Object; this serves as the name of the Object that will be displayed across the platform

Note that all columns of the backing dataset were mapped to Object properties when we selected the "all_orders" dataset to back the Order Object Type. On this tab, you could remove or add additional properties to be mapped. For the purpose of this training, we will keep all properties that were auto-mapped as-is.

8. Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

5 Save location

Apply interface

Build against abstract types. Once an object type implements an interface it will inherit its properties. [Learn more](#)

+ Add

SourceProperty+ Add property

> order_due_date

Order Due Date

> order_id

Order Id

Primary key

> customer_id

Customer Id

> status

Status

> assignee

Assignee

> quantity

Quantity

Primary key ?

Title ?

Order Id

Item Name

Back

Next

The "Actions" tab allows you to set the type of commonly-used Actions that can be taken upon the new Object Type, and what users are allowed to make these Actions. You can create these Actions now, or manually create them in Ontology Manager later. We will create an Action manually and cover the topic more in-depth further along in this training.

9. Leave all of the Actions types deselected

10. Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

5 Save location

STEP 4

Generate action types

Select action types to generate ?

☐

Create [username] Order

Set Order Due Date, Order Id, Customer Id, and 8 more properties

☐

Modify [username] Order

Modify Order Due Date, Customer Id, Status, and 7 more properties

☐

Delete [username] Order

Allows deleting object instances and all of their properties

Choose who can submit this action type

☐

Organization

☐

Group

☐

User

Back

Next

11. Click the *Create* button

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

5 Save location

STEP 5

Save location

Select a location to save this object type to

Browse

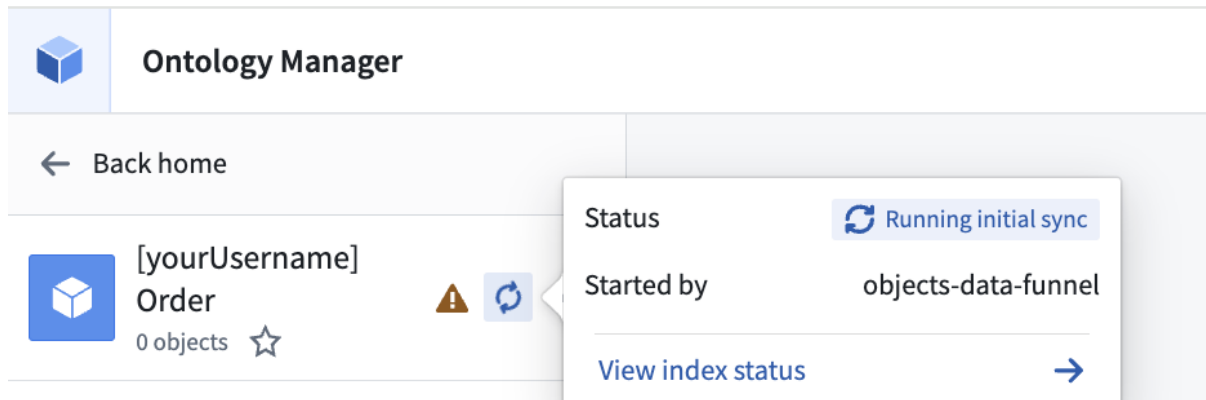
Permissions

Back

Create

Note that it may take a few minutes for your Object Type to index, which essentially creates all the individual Objects from your backing dataset. You won't be able to select your Objects in Workshop in the next step until the indexing is complete.

You can check the status of Object indexing on the Object overview page in the top left. Refreshing the page should pull up the current status of the indexing, and you should see the number of Objects increase above zero. Once the indexing is complete, you can move on to the next steps.



Conclusion

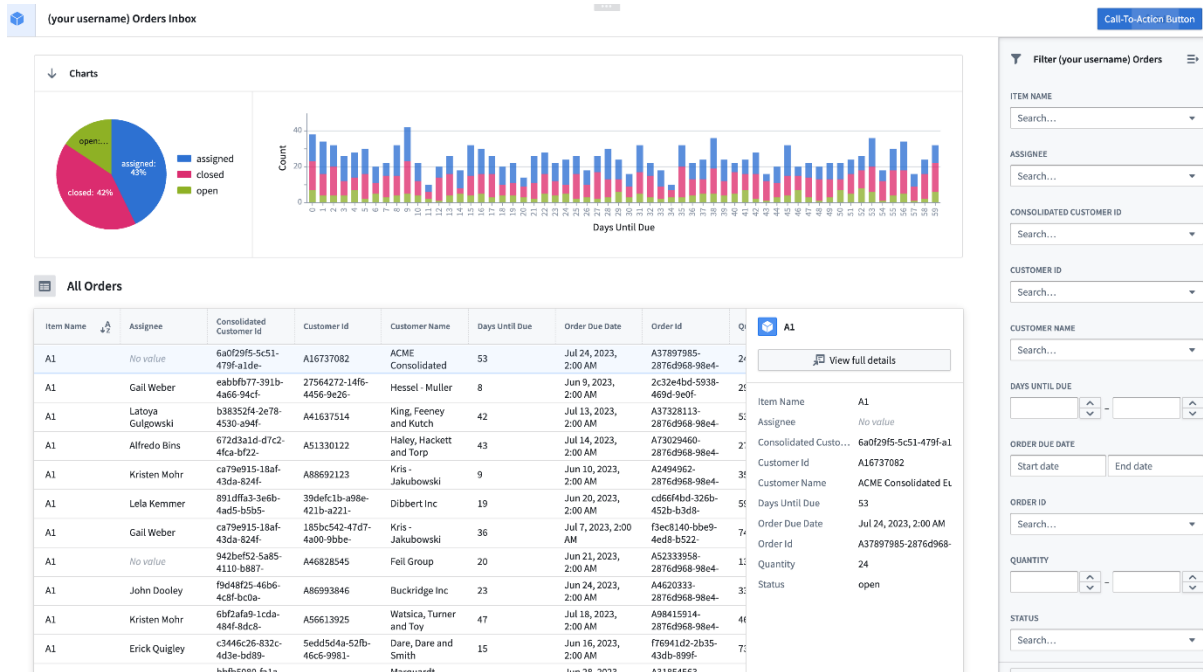
We've now created the Order object. We've skipped over more advanced things like configuring conditional formatting, defining API bindings, and applying security policies.

Building the Workshop Application

Introduction

The next step is to create a tool that solves the Fulfillment Department's problems. **Workshop** is a tool in Foundry for rapidly prototyping and creating powerful applications that leverage the Ontology.

By the end of this section, you will have created a classic dashboard with basic KPIs, charts, and filtering capabilities. Later on, we will make this interactive and collaborative by allowing you to modify who is assigned to an order.

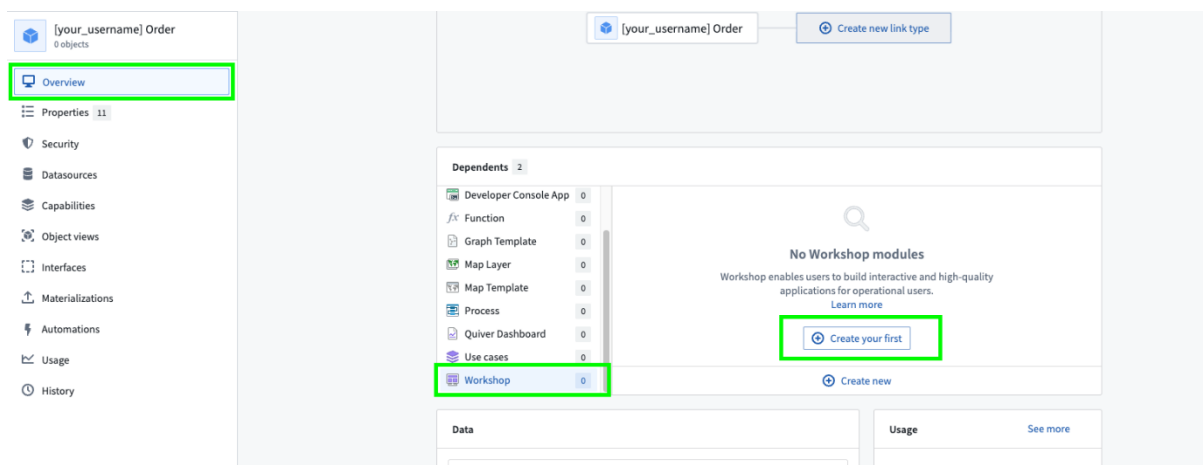


Create the Workshop Module

Step 1: Create Workshop Module

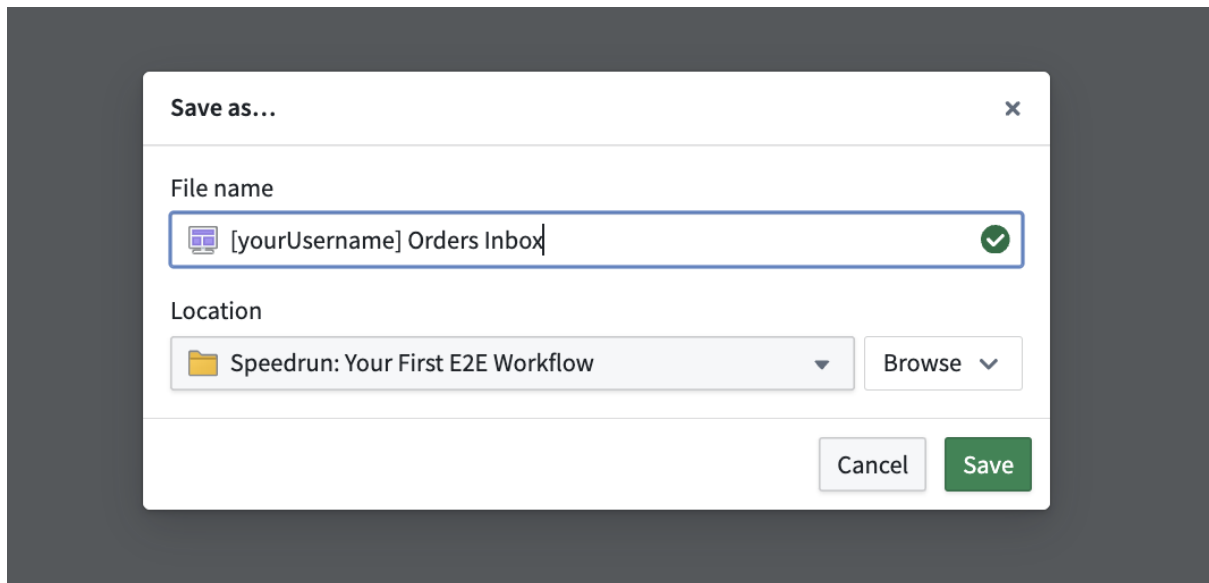
Ensure you've saved your previous changes to the Ontology before moving on to these steps.

1. Open your [username] Orders object in Ontology Manager
2. Click on *Overview*
3. Scroll down to *Dependents*
4. Select *Workshop*
5. Click on *Create your first*



1. Under *File name* name your Workshop App [username] Orders Inbox

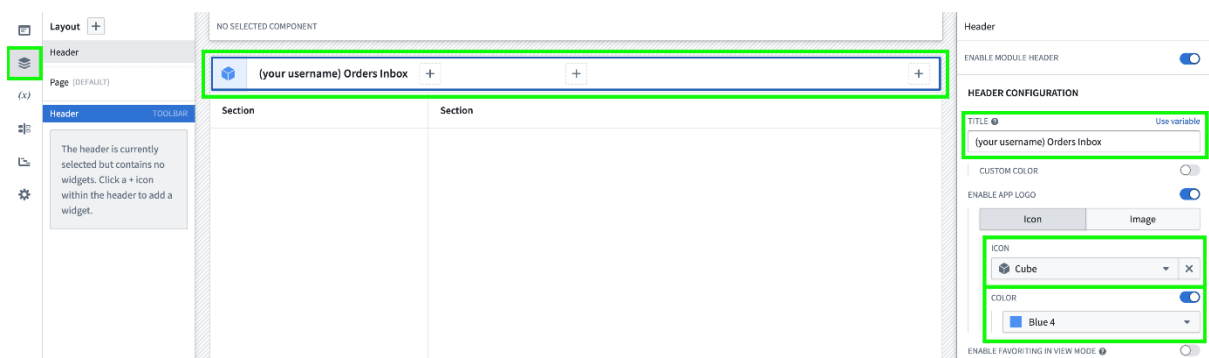
2. Select your course-specific training folder
3. Click *Save*



Add Workshop Title

Step 1: Rename Header

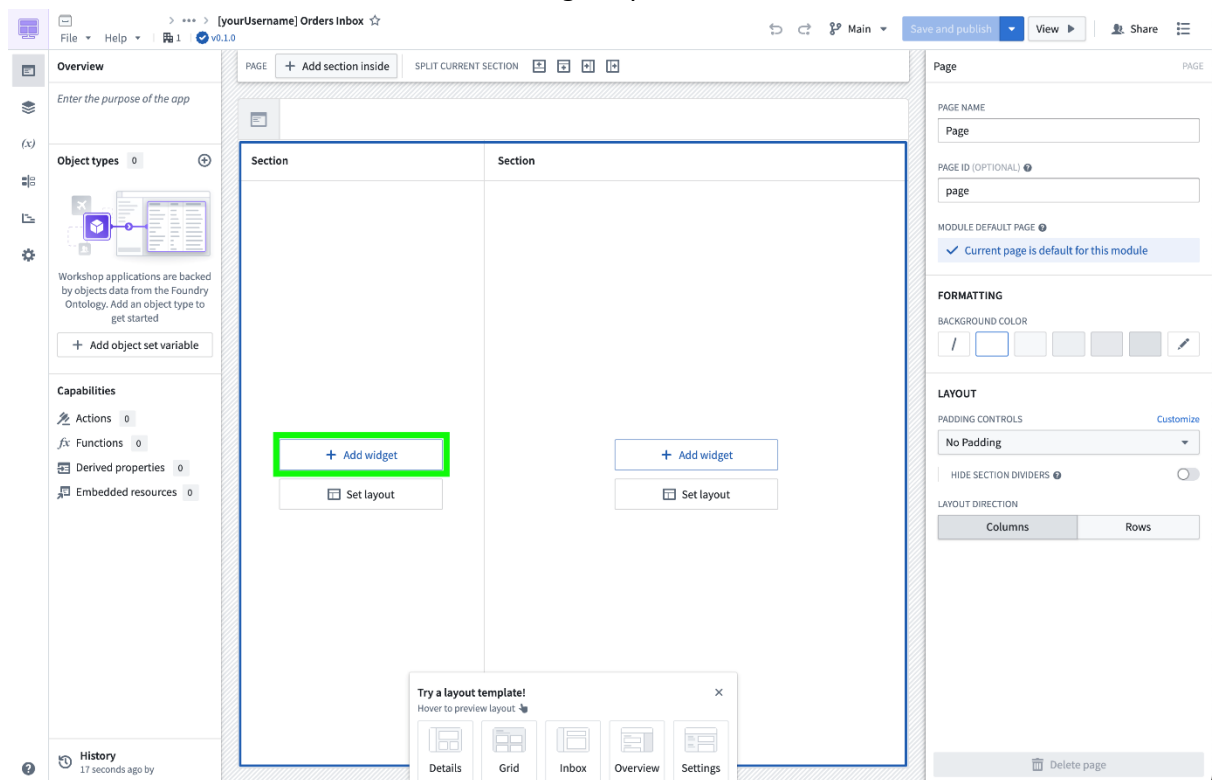
1. Select the *Section* at the top of the Workshop
 - Alternatively, open the *Layout* tab in the left, and select *Header*
2. In the edit-window on the right, rename the *Title* to *(your username) Orders Inbox*
3. Under *Icon*, search for and select *Cube*
4. Set *Color* to *Blue 4*



Create an Object Table

Step 1: Add Object Table Widget

1. In the Section on the left, click the *Add widget* option

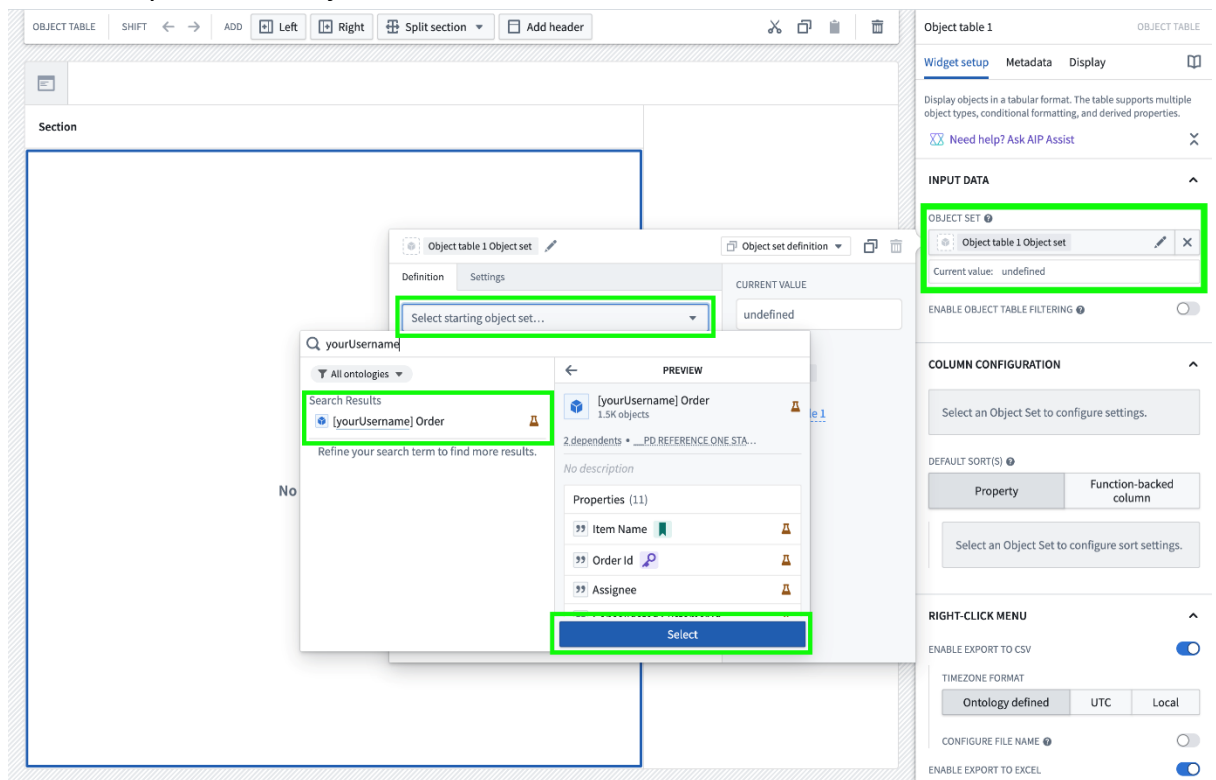


2. Search for the *Object table* widget and click it to add it

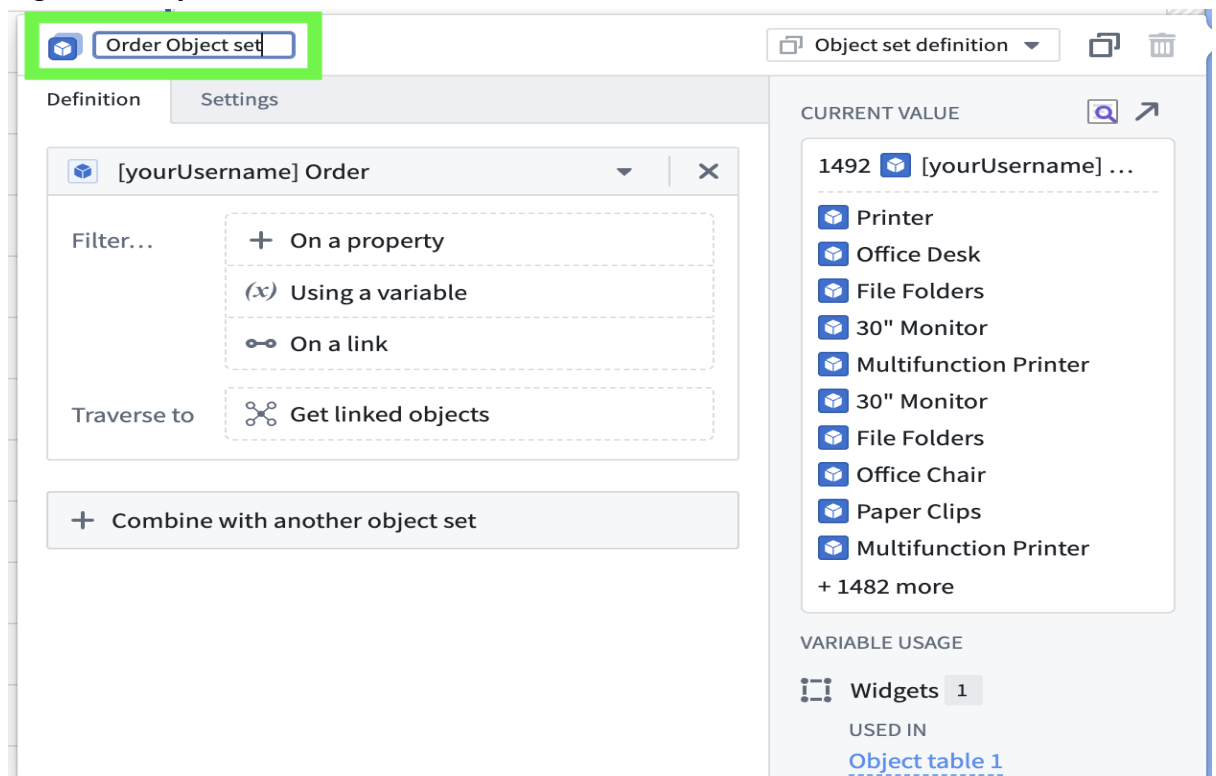
Step 2: Add the object set

1. A new edit window should have opened on the far-right of the screen. Locate the *Input Data* section and under *Object Set* click the *Select object set variable...* dropdown.
2. You will need to create a new variable to hold your *Order* object set. Select *New object set variable*
3. Click *Select starting object set...*

4. Search for your *Order* object set and *Select* it



5. You can also rename the variable that holds your *Order* object set to help keep track of your variables. Click the *pen* icon next to the variable name and enter a new name e.g. *Order Object set*



Step 3: Add properties to the Object Table

1. Under the *Column Configuration* section, select *Add all properties*
2. You can reorganize the order of the columns in the object table by dragging the properties up and down in the edit window using the 6 dots to the left of the property name.
3. To sort objects by *Item name*, scroll down in the edit window to find the *default sort* and click *select a property to sort by* and choose *Item name*

Object table 1

Widget setup Metadata Display

COLUMNS

- Title
- Assignee
- Consolidated Customer Id
- Customer Id
- Customer Name
- Days Until Due
- Order Due Date
- Order Id
- Quantity
- Status
- Unit Price

+ Add column

ENABLE INLINE EDITING

DEFAULT SORT(S)

Property Function-backed column

Item Name

Select a property to sort by

4. Rename the *Title* property to *Item Name* by clicking the arrow next to the *Title* property in the edit window and entering in the new column name

(your username) Orders Inbox

Section

Section

COLUMN CONFIGURATION

Order

COLUMNS

COLUMN NAME Use variable

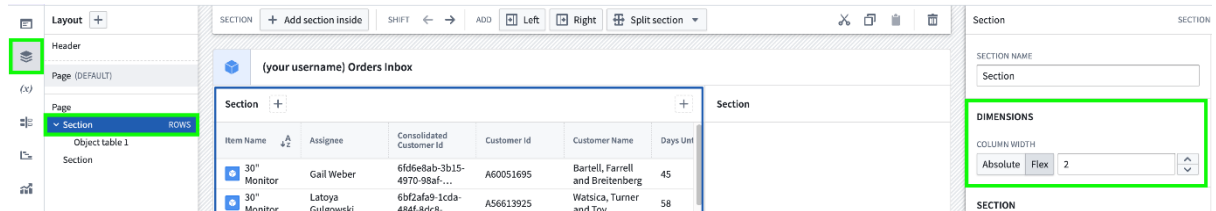
Title

Item Name

Item Name

Step 4: Configure the section dimensions

1. Click the *Layout* icon on the far left
2. Select the *Section* containing the *Object table*
3. Under *Dimensions*, change the *Column Width* to *Flex* and change the number to 2



Create Filter

Step 1: Add Filter widget

1. In the section on the right, click *Add widget*

Consolidated Customer

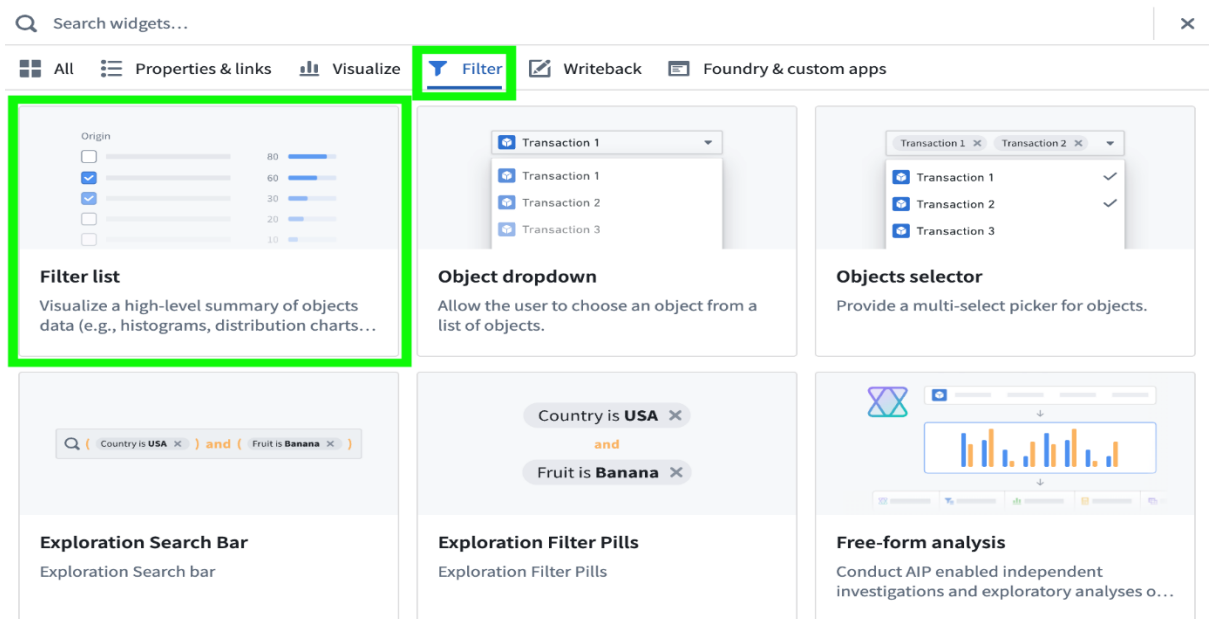
666768011ee-b905f5a9fd491c-8a33c474b54f65-a2f6fd6e8a14970-98891dffa34ad5-b56bf2afa9484f-8dc3dba68c4aa4-a907c73394c03-a0f9c5e02143d7-9ebbf508433f-b5e3c474b54f65-a2f5f5a9fd491c-8a3eabbfb74a66-94c90cef63e4953-a3c0beca041c7-82c672d3a14fca-bf26a0f29f5

Section

+ Add widget

Set layout

2. Search for *Filter list*, or click the *Filter* tab under the search bar, and add it by clicking it



Step 2: Add the object set

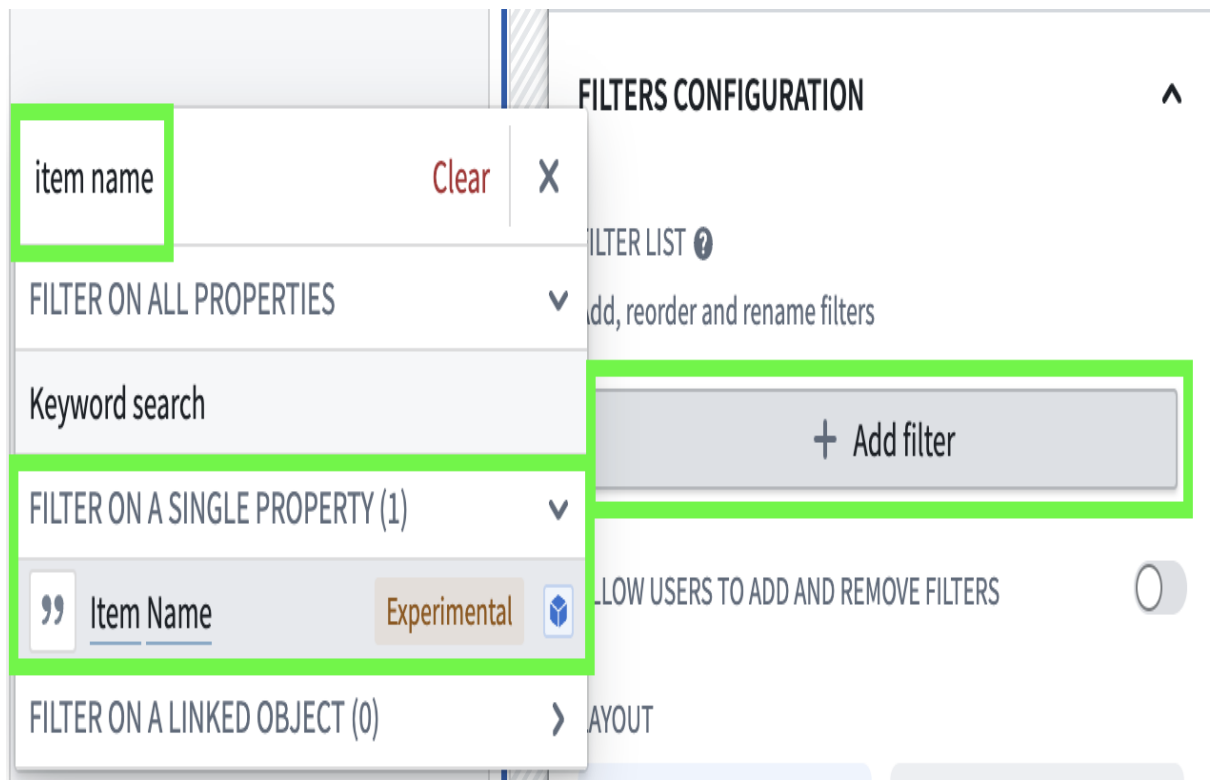
1. This time, you can select the variable you created when adding the object set to the *object table*

The screenshot shows the configuration interface for a 'Filter list' widget. On the left, a large preview area displays the widget's visual representation: a funnel icon with an 'x' and the text 'Filter list' and 'Select this widget to edit configuration.' Below this is a 'Section' header. On the right, the configuration panel is titled 'Filter list 1' and includes tabs for 'Widget setup', 'Metadata', and 'Display'. The 'Widget setup' tab is active, showing instructions: 'Display and update a filter variable that can be used to dynamically filter downstream object set variables.' Below this is a 'Need help? Ask AIP Assist' link. The 'INPUT DATA' section is highlighted with a green border and contains an 'OBJECT SET' dropdown menu. The selected value is 'Order Object set', and the 'Current value' is '1492 [yourUsername] Orders'. The 'FILTERS CONFIGURATION' section includes a 'FILTER LIST' header, instructions to 'Add, reorder and rename filters', an 'Add filter' button, a toggle for 'ALLOW USERS TO ADD AND REMOVE FILTERS', and two layout options: 'Vertical' and 'Pills'. The 'OUTPUT DATA' section shows a 'FILTER OUTPUT' dropdown menu with the selected value 'Filter list 1 Filter output' and a 'Current value' of 'No filters'.

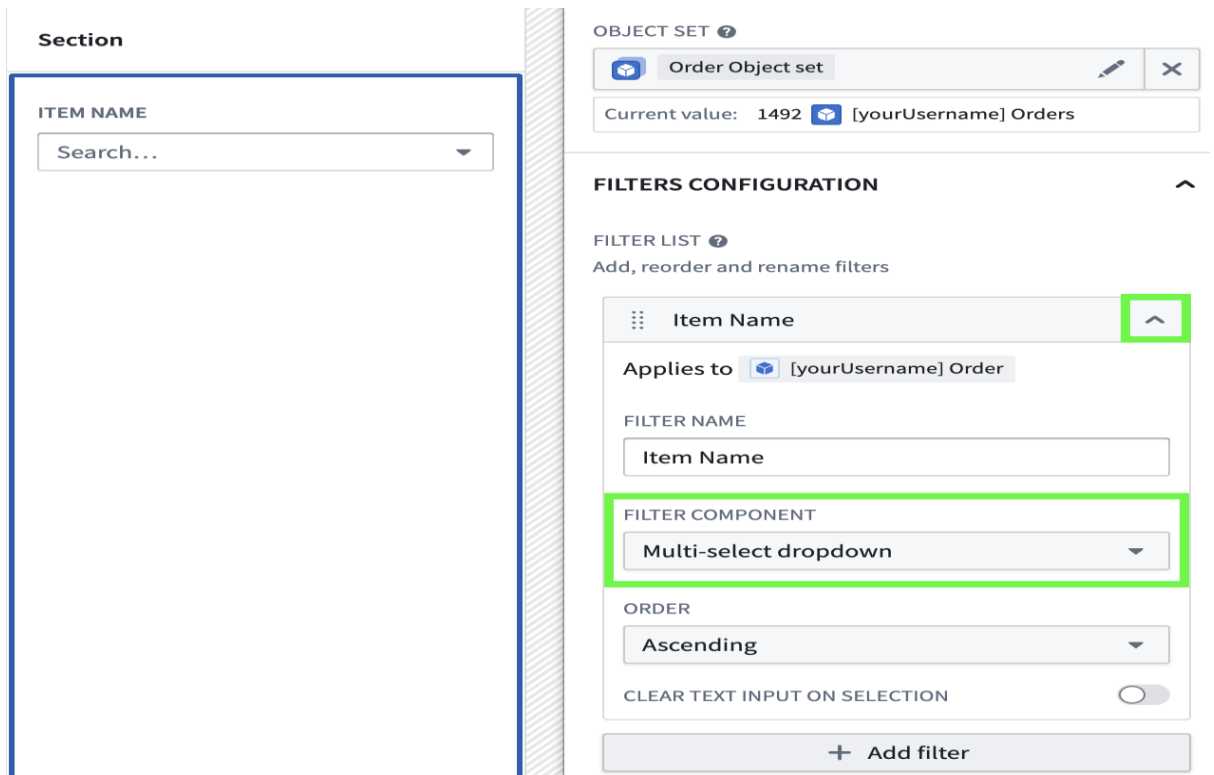
Step 3: Add filters

1. In the *Filter Configuration* section scroll down to the *Add filter* option under the *Filter List*

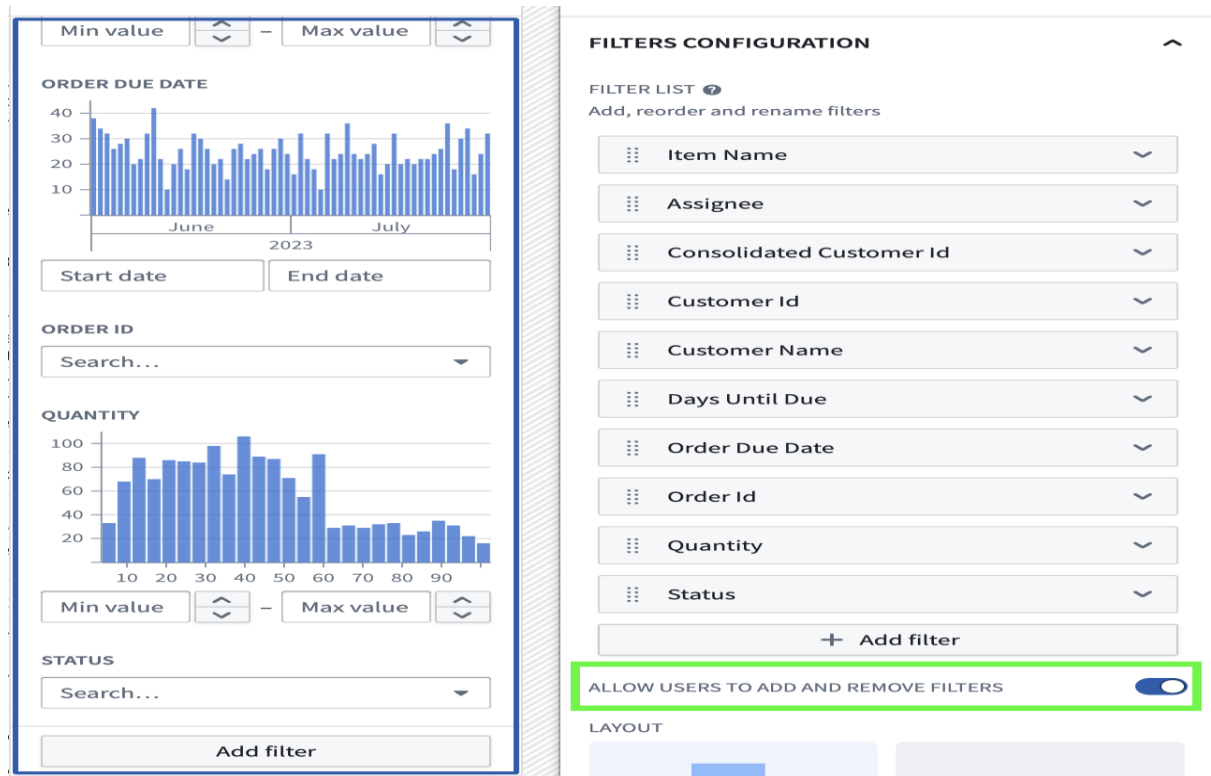
2. Search for and add *Item Name*



3. Once the filter has been added, click the arrow to the right of the filter name, and change the *filter component* to a *Multi-select dropdown*



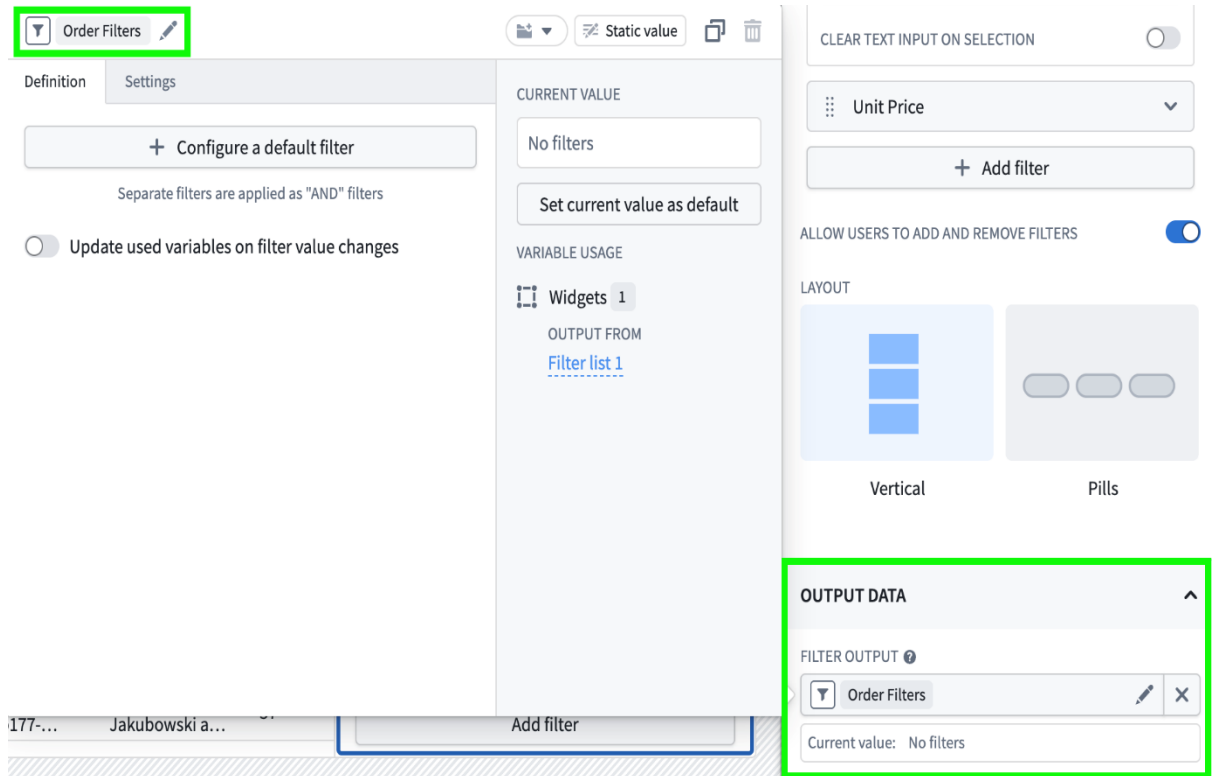
4. Repeat steps 2-3 for each other property name. Some properties will not have the option of a *Multi-select dropdown* in the *filter component*, in this case, just leave them as default.
5. Toggle on *Allow users to add and remove filters*



Step 4: Apply filters to object table

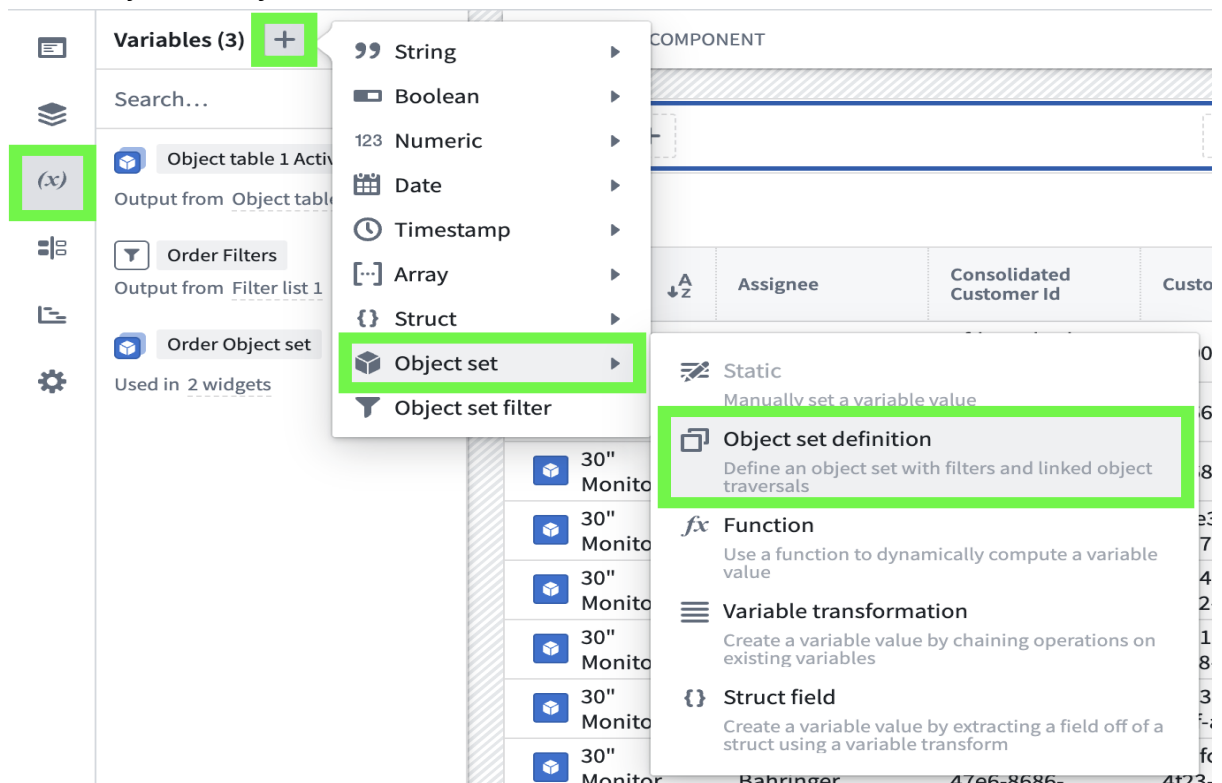
We currently have a Filter, but it isn't connected to the Object Table, so adding filters won't make any difference. We will need to create a new object set, add the filter to that, and then replace the current object set in the Object Table widget with the new one.

1. Rename the *Filter Output* to *Order Filters* (your *Current Value* might look different here if you have tried using the filters already, but it shouldn't make a difference)



2. Click on the *Variables* icon on the far left of the screen
3. Click on the *Plus* icon next to the *Variables*
4. Select *Object set*

5. Select *Object set definition*

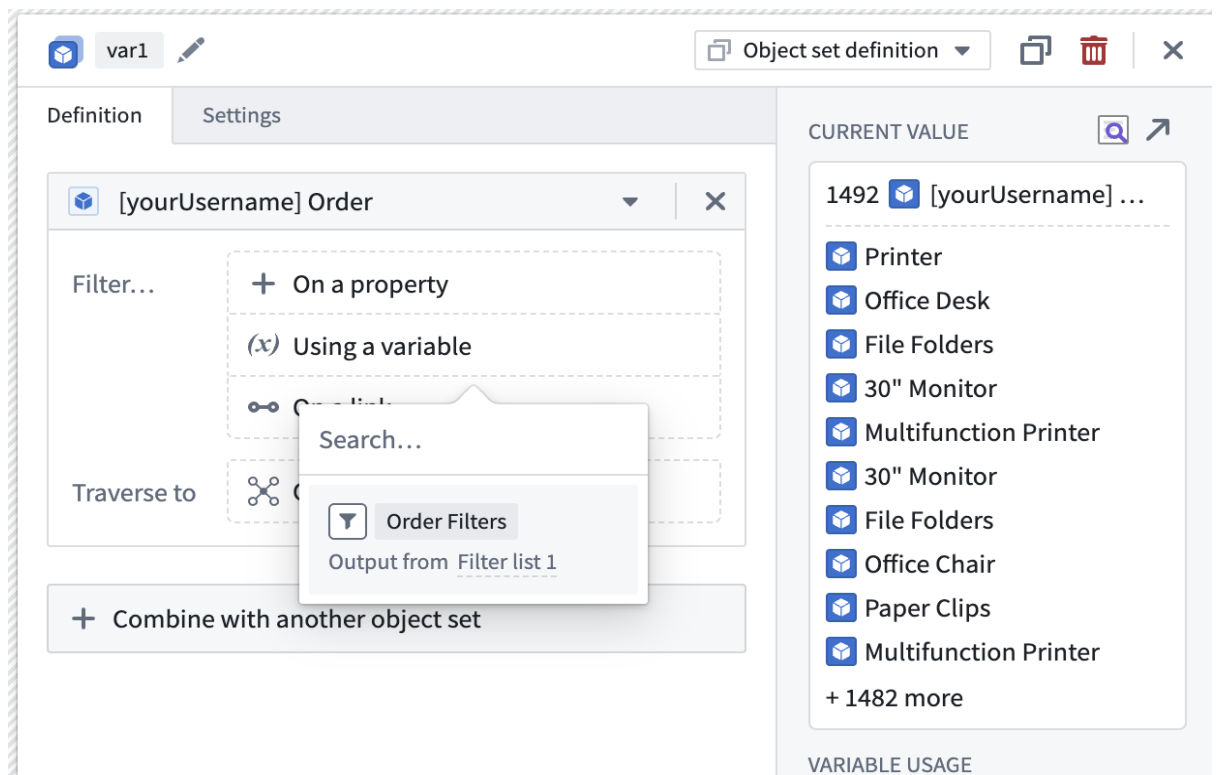


6. As before, search for and select your *Order* object set as the starting object set

7. Then click *Using a variable* next to *Filter*

8. Select *Order*

Filter



9. Rename the new Object Set from *var1* to *Filtered [username] Orders*
10. Close the *Filtered [username] Orders* pop-out by clicking the X in the top right
11. Click anywhere on your object table
12. In the edit window on the far right, under *Input Data*, click the X next to the *Object set*
13. Add the new *Filtered [username] Orders* variable

INPUT DATA

OBJECT SET 

 Filtered [username] Orders 



Current value: 1492  [yourUsername] Orders

ENABLE OBJECT TABLE FILTERING 

☐

Step 5: Configure filter display

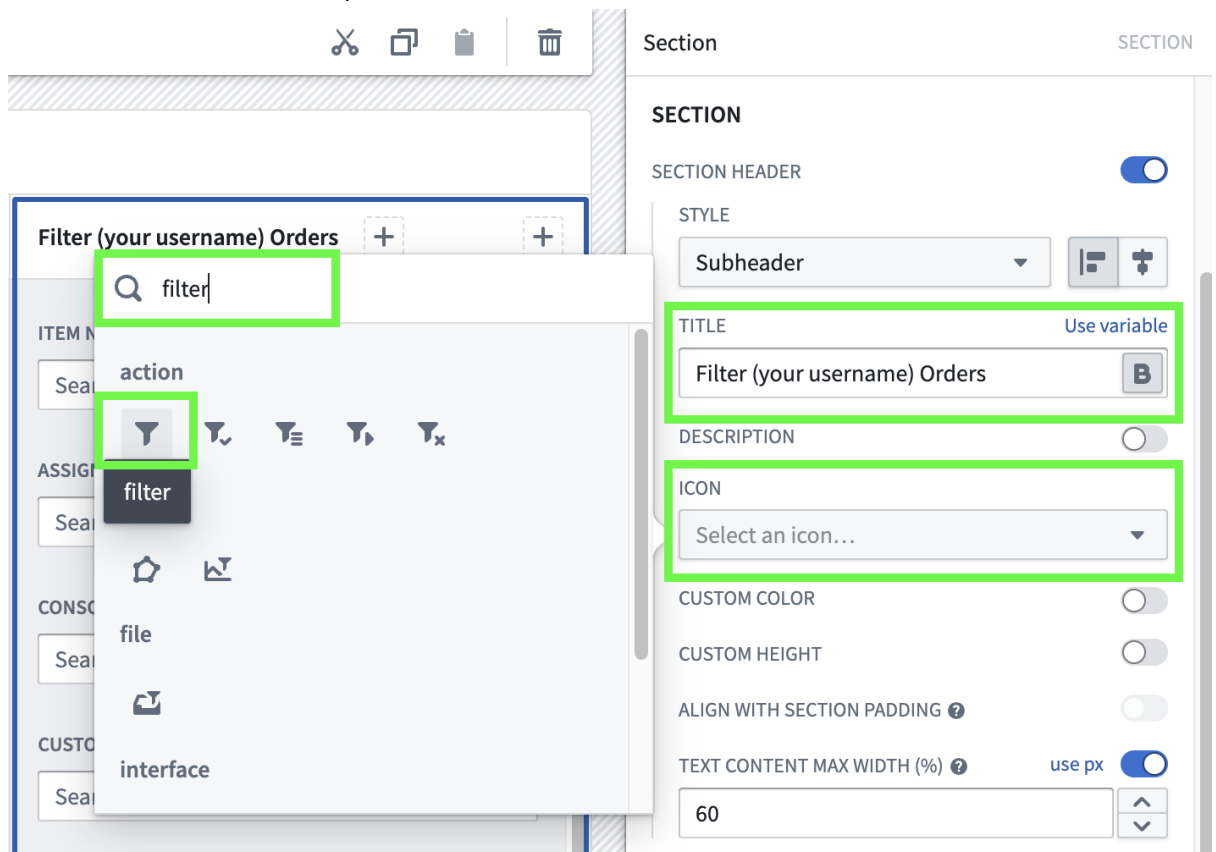
1. At the top of the *Filter list* edit window, change to the *Display* tab

2. Change the *Background Color* to *Light gray 4*

The screenshot shows a configuration panel for a widget titled 'Filter list 1'. At the top right, the text 'FILTER LIST' is visible. Below the title, there are three tabs: 'Widget setup', 'Metadata', and 'Display'. The 'Display' tab is selected and highlighted with a green border. To the right of the tabs is a book icon. Below the tabs, the section 'DIMENSIONS' is visible, containing 'ROW HEIGHT' settings with buttons for 'Auto (max)', 'Absolute', and 'Flex', and a numeric input field set to '1'. Below this is an 'OVERRIDE SECTION WIDTH' toggle switch. The 'FORMATTING' section follows, with a 'BACKGROUND COLOR' label. Under this label, there is a color palette with several color swatches. The first swatch is selected and highlighted with a green border, and a tooltip labeled 'Light gray 4' is shown next to it. To the right of the color swatches is a pencil icon for editing the color.

3. Select the section containing the *Filter list* widget (either by clicking the Section header above the *Filter list* widget or via the *Layout* tab on the far left)
4. Under *Section*, rename the *Title* from *Section* to *Filter (your username) Orders*

- Under *Icon*, select the dropdown and search for the *Filter* icon



- Toggle on *Collapsible*
- Under the *Collapsible* options, change the *Icon to expand* to be *Menu closed*
- Under the *Collapsible* options, change the *Icon to collapse* to be *Menu open*
- Change the *Header format* to *Contained*

10. Change the *Background Colour* to *Light Gray 4*

The screenshot displays a web application interface. On the left is a table titled 'Filter (your username) Orders' with columns: ITEM NAME, ASSIGNEE, CONSOLIDATED CUSTOMER ID, CUSTOMER ID, CUSTOMER NAME, and DAYS UNTIL DUE. Each column has a search input field. On the right is a 'Section' settings panel. The 'COLLAPSIBLE' toggle is turned on. The 'SECTION IS' dropdown is set to 'open'. The 'VARIABLE-BASED COLLAPSE STATE (OPTIONAL)' dropdown is set to 'Select boolean variable...'. The 'ICON TO EXPAND' dropdown is set to 'Menu closed'. The 'ICON TO COLLAPSE' dropdown is set to 'Menu open'. The 'ENABLE TABS' toggle is turned off. Under the 'FORMATTING' section, the 'HEADER FORMAT' dropdown is set to 'Title'. The 'BACKGROUND COLOR' dropdown is set to 'Light Gray 4', which is highlighted with a green box. The 'Section' settings panel is also highlighted with a green box.

Add Object Details

Step 1: Add a New Section

1. Click on the Object Table
2. Along the bar at the top of the screen, you should see the option to *Split section*, click this button
3. You will see some options on where to place the new section. If you mouse over each option, a green highlighted area will show you where the new section will go.

Select *New section on*

right

4. Click the newly created section, and in the edit window on the right, under *Section*, toggle on the *Section header*
5. Delete the *Title*
6. Under *Formatting*, change the *Header Format* to *Contained*

Section

EMPTY SECTION

DIMENSIONS

COLUMN WIDTH

Absolute Flex 300 px

RESIZABLE

SECTION

SECTION HEADER

STYLE

Subheader

TITLE Use variable

DESCRIPTION

ICON

Select an icon...

CUSTOM COLOR

CUSTOM HEIGHT

ALIGN WITH SECTION PADDING

TEXT CONTENT MAX WIDTH (%) use px

60

CONDITIONAL VISIBILITY

COLLAPSIBLE

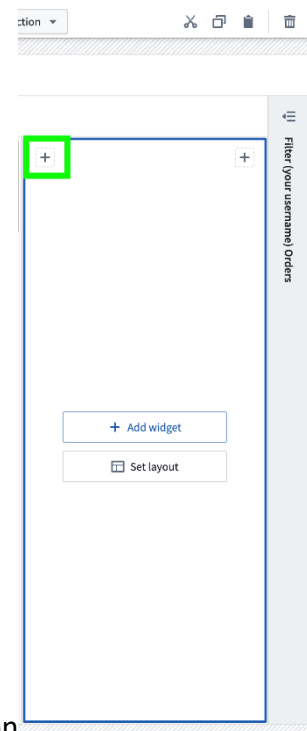
ENABLE TABS

FORMATTING

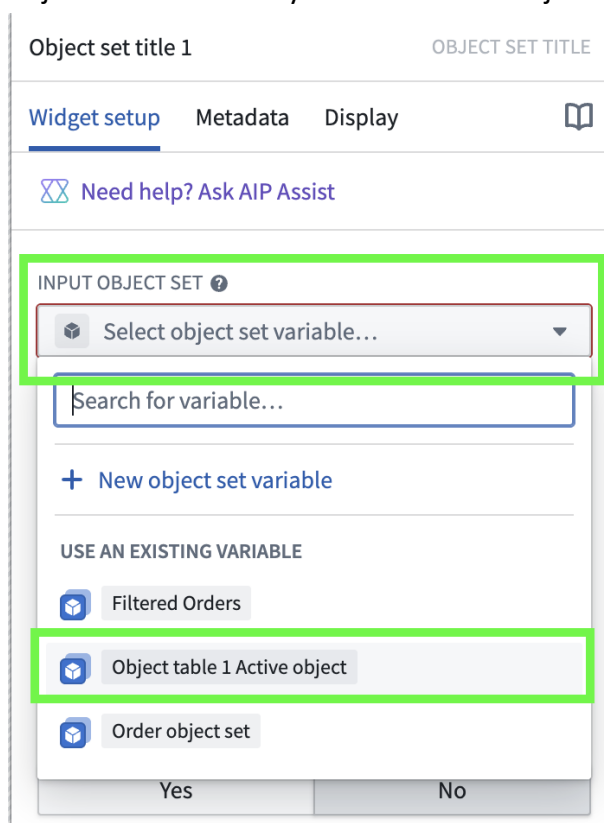
HEADER FORMAT

Title Title Title

Step 2: Add Object Set Title Widget

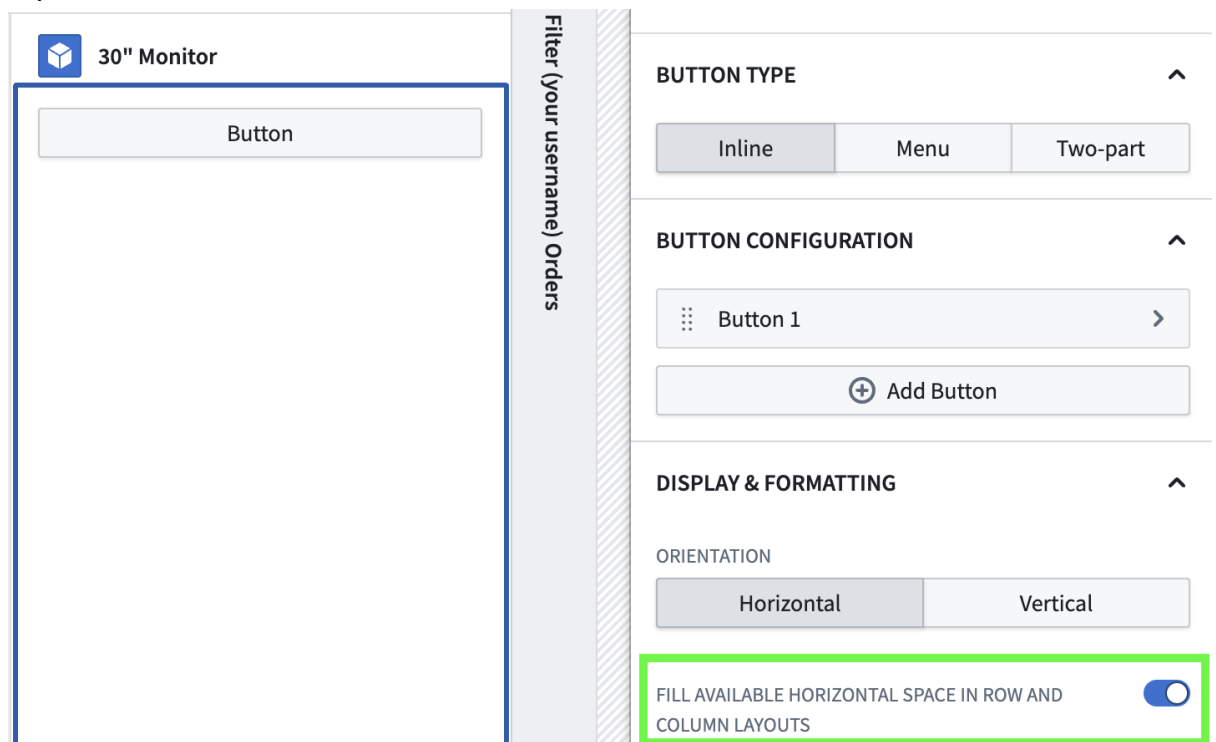


1. Click on the *Plus* icon at the top left of the Section
2. Search for and add *Object Set Title*
3. Set the *Input Object Set* to *Object table 1 Active object* (this variable contains the object that is currently selected in the Object Table)

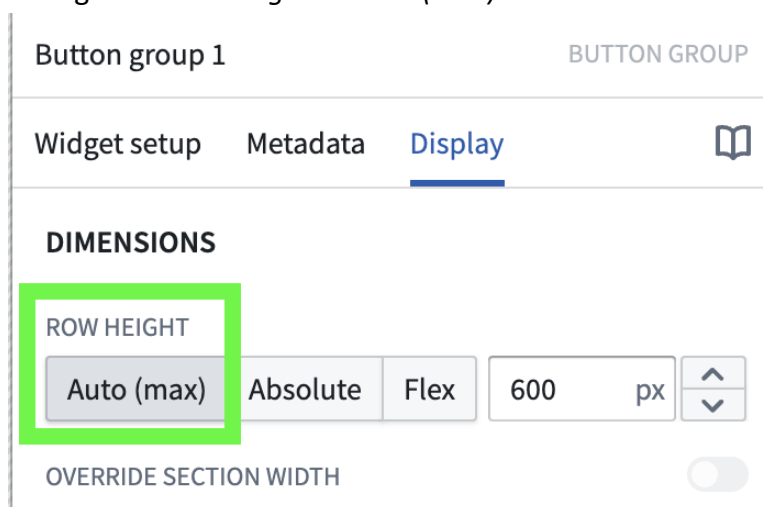


Step 3: Add Button

1. Click *Add widget* under the *Object Set Title*
2. Search for and add the *Button group* widget
3. In the edit window, under *Display & Formatting*, toggle on *Fill Available Horizontal Space and Column Layouts*



4. Under the Widget name, click on *Display*
5. Change the *Row Height* to *Auto (max)*

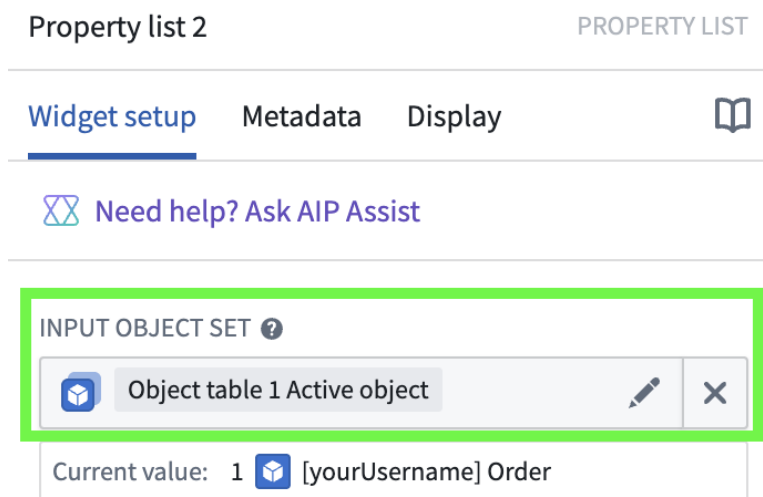


this button further later in the course

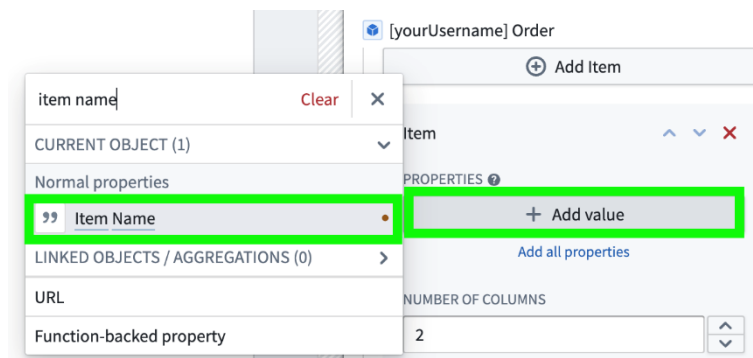
Note: We will configure

Step 4: Add Property List

1. Under the *Button Group*, click *Add Section*
2. Click *Add Widget*
3. Search for and add the *Property List* widget
4. In the edit window, set the *Input Object Set* to *Object table 1 Active object*



5. Under *[yourUsername] Order* there is a square box with *Item* at the top. In this box, under *Properties*, is the option to *Add value*. Click this.
6. Search for and add *Item Name*



7. You can add all properties in an object by clicking *Add all properties* under the *Add value* button
8. You can also delete properties from the view by clicking on the trashcan that appears when you hover over a *Property*

9. Set the *Number of Columns* to 1

The screenshot displays a software interface with two main sidebars. The left sidebar, titled 'Filter (your username) Orders', contains a section for '30" Monitor' with a 'Button' and a table of order details. The right sidebar, titled '[yourUsername] Order', contains a list of properties for an order item.

Filter (your username) Orders

30" Monitor

Button

Item Name	30" Monitor
Quantity	40
Order Due Date	Jul 16, 2023
Days Until Due	45
Consolidated Custo...	6fd6e8ab-3b15-497...
Customer Name	Bartell, Farrell and ...
Assignee	Gail Weber
Unit Price	90
Customer Id	A60051695
Order Id	A70587176-fb778d8...
Status	assigned

+ Add section

[yourUsername] Order

+ Add Item

Item

PROPERTIES ?

- Item Name
- Quantity
- Order Due Date
- Days Until Due
- Consolidated Customer Id
- Customer Name
- Assignee
- Unit Price
- Customer Id
- Order Id
- Status

+ Add value

Add all properties Remove all properties

NUMBER OF COLUMNS

1

ENABLE VALUE WRAPPING

10. Click the 'Display' tab in the Property list right sidebar

11. Set to 'Flex' and '1'

Property list 1 PROPERTY LIST

Widget setup Metadata **Display**

DIMENSIONS

ROW HEIGHT

Auto (max) Absolute **Flex** 1

Rename the Object Table

Step 1: Rename the Object Table section

1. Open the *Layout* tab and select the *Section* above *Object table 1*
2. In the edit window, locate the settings under *Section*. Toggle on *Section Header*
3. Change the *Style* to *Header*
4. Under *Title*, rename to All Orders
5. Change the *Icon* to *Th*

The screenshot shows the application interface with the 'All Orders' section selected. The configuration panel on the right is open, showing the 'Section' settings. The 'Section Header' toggle is turned on. The 'Style' is set to 'Header'. The 'Title' is set to 'All Orders'. The 'Icon' is set to 'Th'. The 'Row Height' is set to 'Flex' with a value of '2'. The 'Override Section Width' is set to 'Auto (max)'.

Item Name	Assignee	Consolidated Customer Id	Customer Id	Customer Name	Days Until
30" Monitor	Gail Weber	6f6e8ab-3b15-4570-9ba1-...	A60051695	Bartell, Farrell and Breitenberg	45
30" Monitor	Latoya Gulowski	6bf2af9-1cda-4b4f-8dc8-...	A56613925	Watsica, Turner and Toy	58
30" Monitor	No value	942baf52-5a85-4110-a887-...	A46828545	Fell Group	17
30" Monitor	Gail Weber	b3832f4-2e78-4530-a94f-...	44fe3e28-109a-4827-9860-...	King, Feeney and Kutch	53
30" Monitor	John Dooley	94df62c0-8f92-4bb2-a97b-...	73ac1b7-8550-4ca2-9f22-...	Boyer LLC	7
30" Monitor	No value	6667680a-7cbe-11ee-b962-...	6781013d-a846-40c5-9413-...	ACME Consolidated ...	41
30" Monitor	Kristen Mohr	3c474d5a-e256-465a-a2f0-...	4293a973-2f51-4f5f-ae1c-...	Schamberger - Medhurst	2
30" Monitor	Lorraine	abb464b4-5b6e-4b4b-8b4b-...	e82fc04c-a3fe-4b4b-8b4b-...	Brekke, Hahn and Breitenberg	8

Step 2: Remove section header

1. Click the *Section* title above the *All Orders* title
2. Go into the edit window and toggle off the *Section header*, so only the *All Orders* title should be seen

All Orders						30" Monitor	
Item Name	Assignee	Consolidated Customer Id	Customer Id	Customer Name	Days Until Due	Button	
30" Monitor	Gail Weber	6fd6e8ab-3b15-4970-98af-...	A60051695	Bartell, Farrell and Breitenberg	45	Item Name30" Monitor Quantity40 Order Due DateJul 16, 2023 Days Until Due45 Consolidated Cust...6fd6e8ab-3b15-497... Customer NameBartell, Farrell and ... AssigneeGail Weber Unit Price90 Customer IdA60051695 Order IdA70587176-fb778d8... Statusassigned + Add section	
30" Monitor	Latoya Gulgowski	6bf2afa9-1cda-484f-8dc8-...	A56613925	Watsica, Turner and Toy	58		
30" Monitor	No value	942bef52-5a85-4110-b887-...	A46828545	Feil Group	17		
30" Monitor	Gail Weber	b38352f4-2e78-4530-a94f-...	44fe3e28-105a-4827-9860-...	King, Feeney and Kutch	52		
30" Monitor	John Dooley	94dfd2c0-8f92-4bb2-a97b-...	73e4c1b7-b550-4ca2-9f22-...	Boyer LLC	7		
30" Monitor	No value	6667680a-7c9e-11ee-b962-...	6781013d-a846-40c8-9413-...	ACME Consolidated ...	41		
30" Monitor	Kristen Mohr	3c474b5a-e256-4f65-a2f0-...	4293c972-2f51-4f5f-ae1c-...	Schamberger - Medhurst	2		
30" Monitor	Lorraine Bahringer	abb464b4-5b6e-47e6-8686-...	e82fc04c-a3fe-4f23-9352-...	Brekke, Hahn and Stroman	8		
30" Monitor	Kristen Mohr	b38352f4-2e78-4530-a94f-...	A41637514	King, Feeney and Kutch	46		
30" Monitor	Alfredo Bins	9079871a-e79f-49df-a6ec-...	208184c5-d8a4-4b7b-9de8-...	Howe Group	38		
30" Monitor	Laverne Koss	6a0f29f5-5c51-479f-a1de-...	A16737082	ACME Consolidated ...	24		
30" Monitor	No value	b38352f4-2e78-4530-a94f-...	44fe3e28-105a-4827-9860-...	King, Feeney and Kutch	34		
30" Monitor	Lela Kemmer	f9d48f25-46b6-4c8f-bc0a-...	fd5a7d6b-5bab-4bb3-a293-...	Buckridge Inc	24		
30" Monitor	Taylor Hilll	c412823e-a899-4a2a-8f27-...	755b4e9f-a377-4f06-916a-...	Hegmann and Sons	38		
30" Monitor	No value	c412823e-a899-4a2a-8f27-...	A16929361	Hegmann and Sons	7		
30" Monitor	Laverne Koss	9c5e0213-1b3f-43d7-9e73-...	e3c8cfbf-6104-44ee-b177-...	Purdy, Jakubowski a...	57		

Configure Sections

Step 1: Split Section

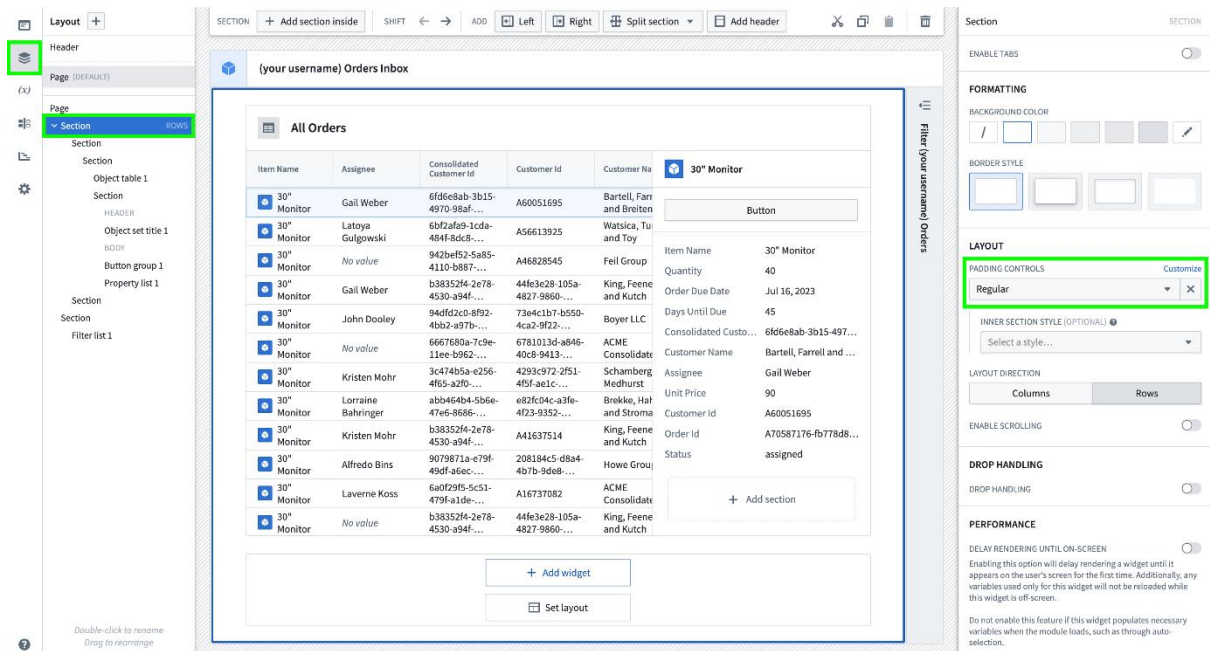
1. Open the *Layout* tab and select the first *Section* under *Page* (This is the section that contains both your *Object table* and Object details)
2. From the bar at the top of the screen, choose to *Split section*, and then select *New section below*

The screenshot shows a software interface with a sidebar on the left and a main content area. The sidebar has a 'Layout' tab selected, and a 'Section' tab is highlighted. The main content area displays a table of orders with columns: Item Name, Assignee, Consolidated Customer Id, Customer Id, Customer Name, Days Until Due, and a '30" Monitor' section. The '30" Monitor' section contains a table with columns: Item Name, Quantity, Order Due Date, Days Until Due, Consolidated Customer Id, Customer Name, Assignee, Unit Price, Customer Id, Order Id, and Status. A dropdown menu is open, showing options: 'New section above', 'New section below', 'New section on left', and 'New section on right'.

Item Name	Assignee	Consolidated Customer Id	Customer Id	Customer Name	Days Until Due
30" Monitor	Gail Weber	6fd6e8ab-3b15-4970-98af-...	A60051695	Bartell, Farrell and Breitenberg	45
30" Monitor	Latoya Gulgowski	6bf2afa9-1cda-484f-8dc8-...	A56613925	Watsica, Turner and Toy	58
30" Monitor	No value	942bef52-5a85-4110-b887-...	A46828545	Feil Group	17
30" Monitor	Gail Weber	b38352f4-2e78-4530-a94f-...	44fe3e28-105a-4827-9860-...	King, Feeney and Kutch	52
30" Monitor	John Dooley	94dfd2c0-8f92-4bb2-a97b-...	73e4c1b7-b550-4ca2-9f22-...	Boyer LLC	7
30" Monitor	No value	6667680a-7c9e-11ee-b962-...	6781013d-a846-40c8-9413-...	ACME Consolidated ...	41
30" Monitor	Kristen Mohr	3c474b5a-e256-4f65-a2f0-...	4293c972-2f51-4f6f-a01c-...	Schamberger - Modhurst	2
30" Monitor	Lorraine Bahringer	abb464b4-5b6e-47e6-8686-...	e82fc04c-a3fe-4f23-9352-...	Brekke, Hahn and Stroman	8
30" Monitor	Kristen Mohr	b38352f4-2e78-4530-a94f-...	A41637514	King, Feeney and Kutch	46
30" Monitor	Alfredo Bins	9079871a-e79f-49df-a6ec-...	208184c5-d8a4-4b7b-9de8-...	Howe Group	38
30" Monitor	Laverne Koss	6a0f29f5-5c51-479f-a1de-...	A16737082	ACME Consolidated ...	24
30" Monitor	No value	b38352f4-2e78-4530-a94f-...	44fe3e28-105a-4827-9860-...	King, Feeney and Kutch	34
30" Monitor	Lela Kemmer	f9d48f25-46b6-4c8f-bc0a-...	fd5a7d6b-5bab-4bb3-a293-...	Buckridge Inc	24
30" Monitor	Taylor Hilll	c412823e-a899-4a2a-8f27-...	755b4e9f-a377-4f06-916a-...	Hegmann and Sons	38
30" Monitor	No value	c412823e-a899-4a2a-8f27-...	A16929361	Hegmann and Sons	7
30" Monitor	Laverne Koss	9c5e0213-1b3f-43d7-9e73-...	e3c8cfbf-6104-44ee-b177-...	Purdy, Jakubowski a...	57

Step 2: Format Section

1. In the *Layout* tab, reselect the first *Section* under *Page* again (This is the section that contains both your *Object table*, Object details panel and the newly create section)
2. In the edit-window, under *Layout*, change the *Padding Controls* from *No Padding* to *Regular*



Step 3: Configure Sections

1. Click anywhere in the newly created section (except the *Add widget* or *Set layout* parts)
2. In the edit-window for the Section, set the *Row Height* to *Flex 1*
3. Add a *Section Header* and rename the *Title* to *Charts*
4. Click on *Set Layout*
5. Click on *Columns*

Tip: If you make a mistake, you can press Ctrl+Z to undo a change

(your username) Orders Inbox

All Orders

Item Name	Assignee	Consolidated Customer Id	Customer Id	Customer Na
30" Monitor	Gail Weber	6fd6e8ab-3b15-4970-98af-...	A60051695	Bartell, Farr and Breiten
30" Monitor	Latoya Gulowski	6bf2afa9-1cda-484f-8dc8-...	A56613925	Watsica, Tui and Toy
30" Monitor	No value	942bef52-5a85-4110-b887-...	A46828545	Feil Group
30" Monitor	Gail Weber	b38352f4-2e78-4530-a94f-...	44fe3e28-105a-4827-9860-...	King, Feene and Kutch
30" Monitor	John Dooley	94dfd2c0-8f92-4bb2-a97b-...	73e4c1b7-b550-4ca2-9f22-...	Boyer LLC
30" Monitor	No value	6667680a-7c9e-11aa-b96f-...	6781013d-a846-47e9-9d13-...	ACME Cronenlat

30" Monitor

Item Name

30" Monitor

Quantity

40

Order Due Date

Jul 16, 2023

Order Detail Desc

AC

+ Add section

Charts +

Layout

Determines how components will be arranged in this section

Columns

Rows

Tabs

Flow

Toolbar

Loop

Set layout

SECTION NAME

Section

DIMENSIONS

ROW HEIGHT

Auto (max)

Absolute

Flex

1

Override Section Width

SECTION

SECTION HEADER

STYLE

Subheader

TITLE

Charts

DESCRIPTION

ICON

Select an icon...

CUSTOM COLOR

CUSTOM HEIGHT

ALIGN WITH SECTION PADDING

TEXT CONTENT MAX WIDTH (%)

60

CONDITIONAL VISIBILITY

COLLAPSIBLE

ENABLE TABS

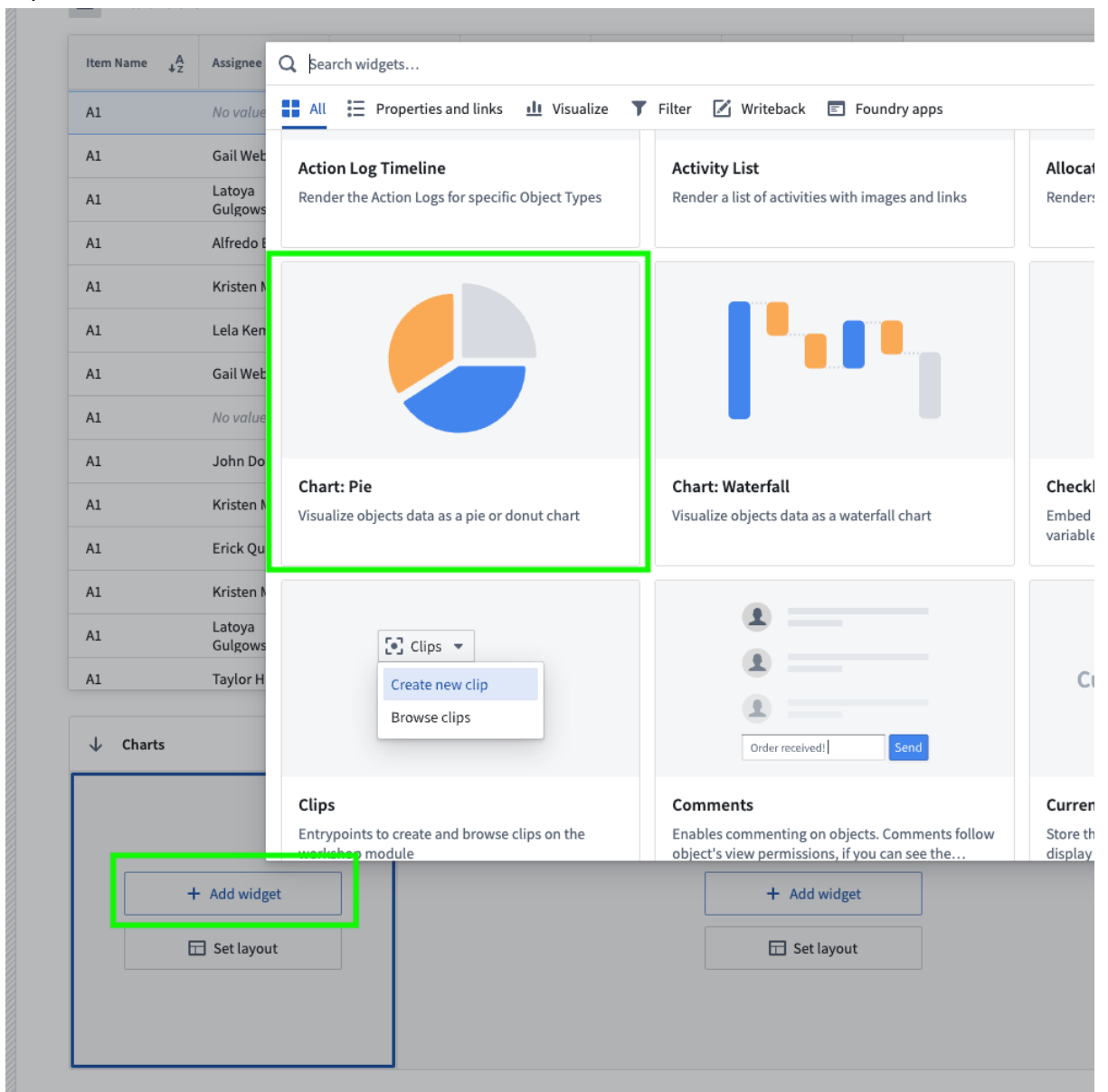
Create Pie Chart

Step 1: Add Pie Chart

In this section, we will add a Pie Chart to our application to display orders based on status.

1. Click on *Add Widget* in the left Section
2. Click on *Chart: Pie*. You can search for it if needed using the "Search widgets..." search bar at the

top.



3. Navigate to the right side panel where settings are configurable for the chart
4. Under *Input Object Set*, select *Filtered [username] Orders*
 - Note the *Filtered [username] Orders* is an Object Set that represents the output of the filter section on the right-hand side of the application. By selecting this as the input to the Pie Chart, the Orders that are visualized in the chart will be determined by the filter settings.
5. Set *Group By* to *Status*
6. Click on *Show Legend*

INPUT OBJECT SET

 Filtered [username] Orders  

Current value: 1492  [username] Orders

GROUP BY

 Status 

ENABLE ONTOLOGY COLORS 

☒

AGGREGATION

METRIC

 Count 

ENABLE NUMERIC FORMATTING

☐

RADIUS



0% 99%

PADDING

None Compact Normal **Large**

LEGEND

SHOW LEGEND

☒

Inside chart

Next to chart

Left

Right

Top

Bottom

Create Bar Chart

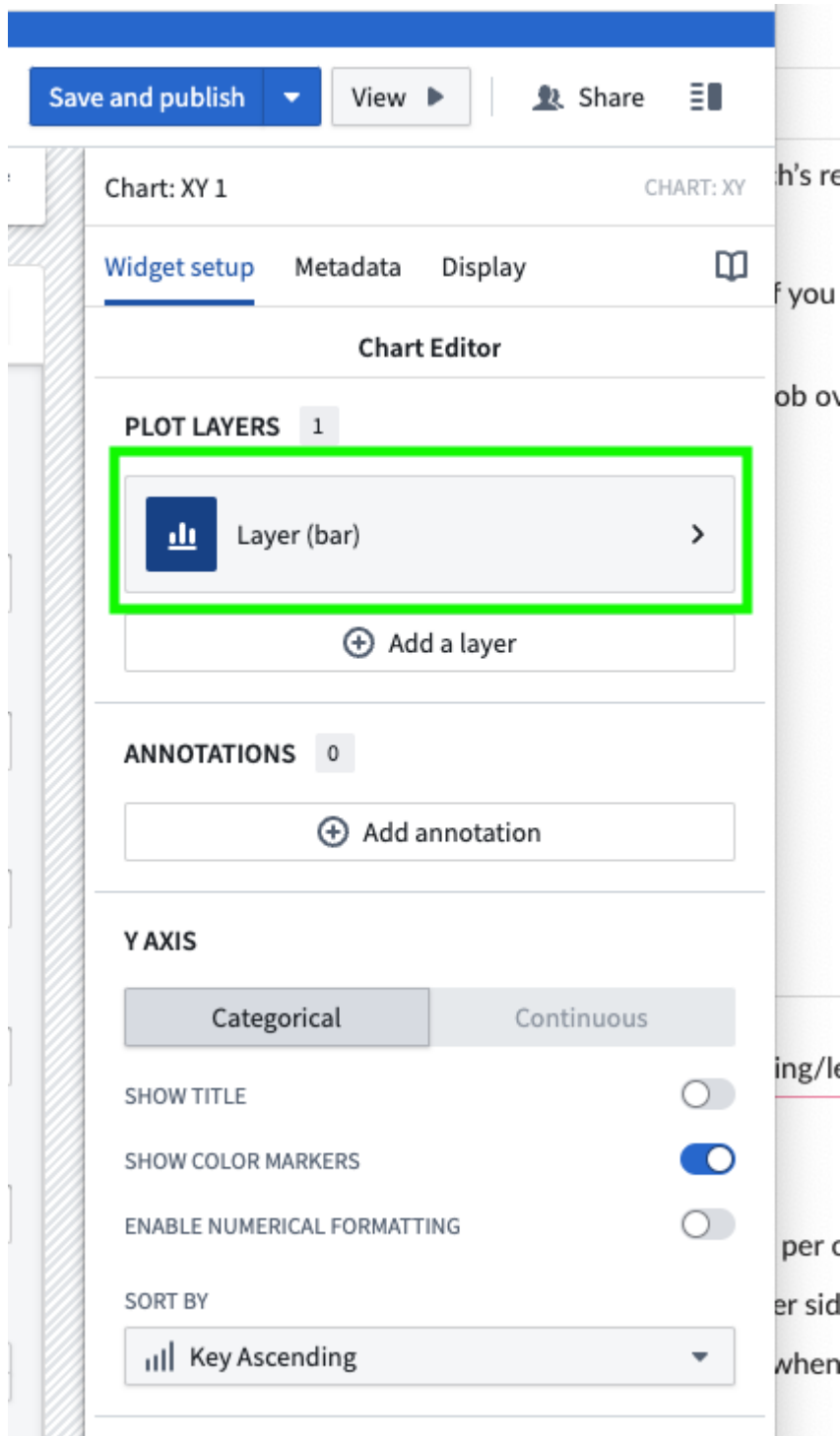
Step 1: Create Bar Chart

In this lesson we will add a Bar Chart to visualize orders by the number of days until they are due as well as their status.

1. Navigate to the empty section next to your new Pie Chart
2. Click on *Add Widget*
3. Click on *Chart: XY*

The screenshot shows a dashboard editor interface. On the left, there is a table with columns for status (A1) and names (Latoya Gulgows, Alfredo, Kristen M, Lela Ken, Gail Web, No value, John Do, Kristen M, Erick Qu, Kristen M, Latoya Gulgows, Taylor H). Below the table is a pie chart titled 'Charts' showing the distribution of orders by status: assigned (43%), closed (42%), and open (15%). The legend indicates: assigned (blue), closed (red), and open (green). In the center, there is a widget selection panel with several options: 'Origin' (a bar chart), 'Filter list' (a text box), 'Object list' (a list of objects), and 'Object view' (a text box). The 'Chart: XY' widget is highlighted with a green box. On the right, there is a sidebar with buttons for 'Map', 'Visuali', 'interac', 'Farr', 'Farr', 'Farr', 'Farr', 'Grai', 'Pivot', 'Enable', and 'object'. At the bottom right, there is a button labeled '+ Add widget' which is also highlighted with a green box.

4. On the right side of the screen, click on *Plot Layers*



5. Click on *Layer (bar)*
 6. Under *Data Input*, select Object set variable *Filtered [username] Orders*
 7. Under *X Axis property*, select *Days Until Due*
 8. Under *Bar Series* and *Segment By*, select *Status*
 9. Scroll to the top, click *Chart Editor* to take you out of the editing mode
- Hint: Don't worry if your chart looks strange -- we'll fix it!*

Save and publish

View

Share



Chart: XY 1

CHART: XY

Widget setup

Metadata

Display



< Chart Editor

Layer (bar)

TITLE

Layer (bar)

DATA INPUT

Object set

Function

Time series set

Filtered (your username) Orders



Current value: 1492 (your username) Orders

LAYER TYPE



Bar Chart



Line Chart



Scatter Chart

SHOW LABELS



X AXIS PROPERTY

PROPERTY

123 Days Until Due



BUCKETING

Exact Value



LIMIT

Set category limit...



BAR SERIES

Use multiple series



Count

+ Add step

CUMULATIVE SUM



SEGMENT BY (OPTIONAL)

Status



SEGMENT DISPLAY OVERRIDES

+ Add Segment

10. Scroll all the way to bottom and click on *Bar Orientation*

11. Click *Customize*

12. Select *Vertical*

Save and publish

View

Share

Chart: XY 1

CHART: XY

Widget setup

Metadata

Display

+ Add a layer

ANNOTATIONS

0

+ Add annotation

Y AXIS

Categorical

Continuous

SHOW TITLE

SHOW COLOR MARKERS

ENABLE NUMERICAL FORMATTING

SORT BY

||| Key Ascending

X AXIS

Value

Axes display settings are automatically generated by series data.

Customize

ONTOLOGY FORMATTING

ENABLE ONTOLOGY COLORS

LEGEND

SHOW LEGEND

TOOLTIPS

SHOW TOOLTIPS

EXPORTS

ALLOW EXPORTS

BAR ORIENTATION

Bar Orientation settings are automatically generated by series data.

Customize

Move Charts to the Top

Step 1: Adjust Charts

1. Click on *Charts*
2. Click on the up arrow next to *Move*

NEW SECTION + Add section inside MOVE **↑** ↓ ADD Above Below Split section ▼

(your username) Orders Inbox

All Orders

Item Name	Assignee	Consolidated Customer Id
A1	No value	6a0f29f5-5c51-479f-a1de-
A1	Gail Weber	eabbbf77-3914a66-94cf-
A1	Latoya Gulgowski	b38352f4-2e74530-a94f-
A1	Alfredo Bins	672d3a1d-d74fca-bf22-
A1	Kristen Mohr	ca79e915-18a43da-824f-
A1	Lela Kemmer	891dffa3-3e614ad5-b5b5-
A1	Gail Weber	ca79e915-18a43da-824f-
A1	No value	942bef52-5a84110-b887-
A1	John Dooley	f9d48f25-46b4c8f-bc0a-

A1

View full details

Item Name A1

Assignee No value

Consolidated Custo... 6a0f29f5-5c51-479f-a1

Customer Id A16737082

Customer Name ACME Consolidated E

Days Until Due 53

Order Due Date Jul 24, 2023, 2:00 AM

Order Id A37897985-2876d968-

Quantity 24

Status open

Charts +

assigned: 43%

closed: 42%

open

Count

Days Until Due

Filter (your user)

ITEM NAME Search...

ASSIGNEE Search...

CONSOLIDATED CUSTOMER ID Search...

CUSTOMER ID Search...

CUSTOMER NAME Search...

DAYS UNTIL DUE

ORDER DUE DATE Start date

Ad

+ Ac

Configure Object Table

Step 1: Configure the Object Table

1. Click on the table of Order Objects

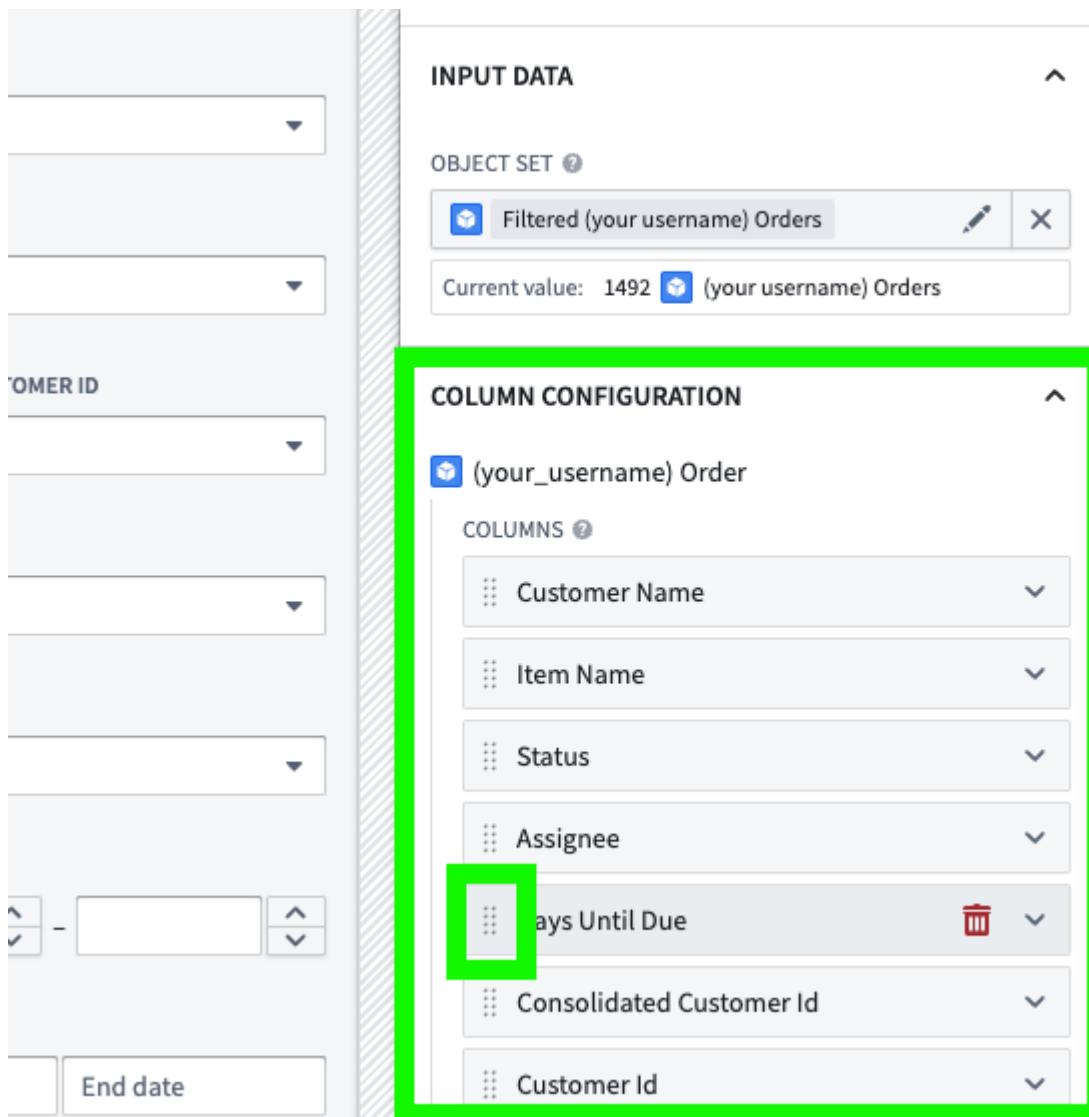
- Under *Column Configuration* on the far right, ensure *Status* is one of the selected properties to be displayed

The screenshot shows a dashboard interface for '(your_username) Orders Inbox'. It includes a 'Charts' section with a pie chart showing order status distribution (assigned: 43%, closed: 41%, open) and a bar chart showing 'Count' vs 'Days Until Due'. Below the charts is a table titled 'All Orders' with columns 'Customer Name', 'Item Name', and 'Status'. The table lists several orders, including 'Bartell, Farrell and ...', 'Watsica, Turner and Toy', 'Feil Group', 'King, Feeney and Kutch', and 'Boyer LLC'. A '30" Monitor' button is also visible. On the right, the 'COLUMN CONFIGURATION' panel is open, showing a list of columns. The 'Status' column is highlighted in green, indicating it is selected for display.

Customer Name	Item Name	Status
Bartell, Farrell and ...	30" Monitor	assigned
Watsica, Turner and Toy	30" Monitor	closed
Feil Group	30" Monitor	open
King, Feeney and Kutch	30" Monitor	closed
Boyer LLC	30" Monitor	closed

- Drag the property names to reorder them so that the first five are:

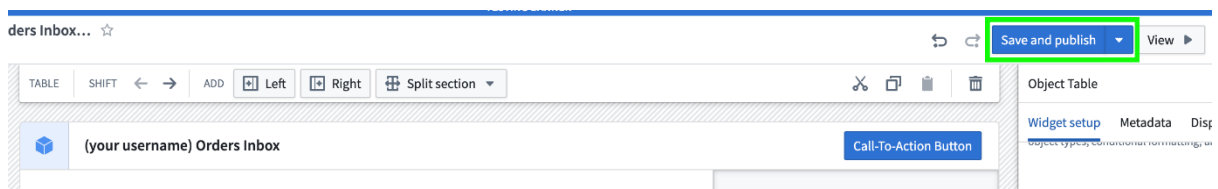
- Customer Name, Item Name, Status, Assignee, and Days Until Due*



Saving and Viewing your Workshop Module

Step 1: Save Your Work

1. Click on Save and publish
2. Click on View (*Tip: Open it in a new tab*)



Conclusion

At this point we have a read-only dashboard with basic filtering capabilities. Let's next make it interactive and operational by allowing people to assign and close orders. Effectively, updating the fields of the Objects in the Ontology.

Note: Your workshop may look slightly different than the below. Once you've configured and setup the action in the next two sections, feel free to experiment with layouts, formatting, and displayed properties to get your desired look.

Configuring Actions

Introduction

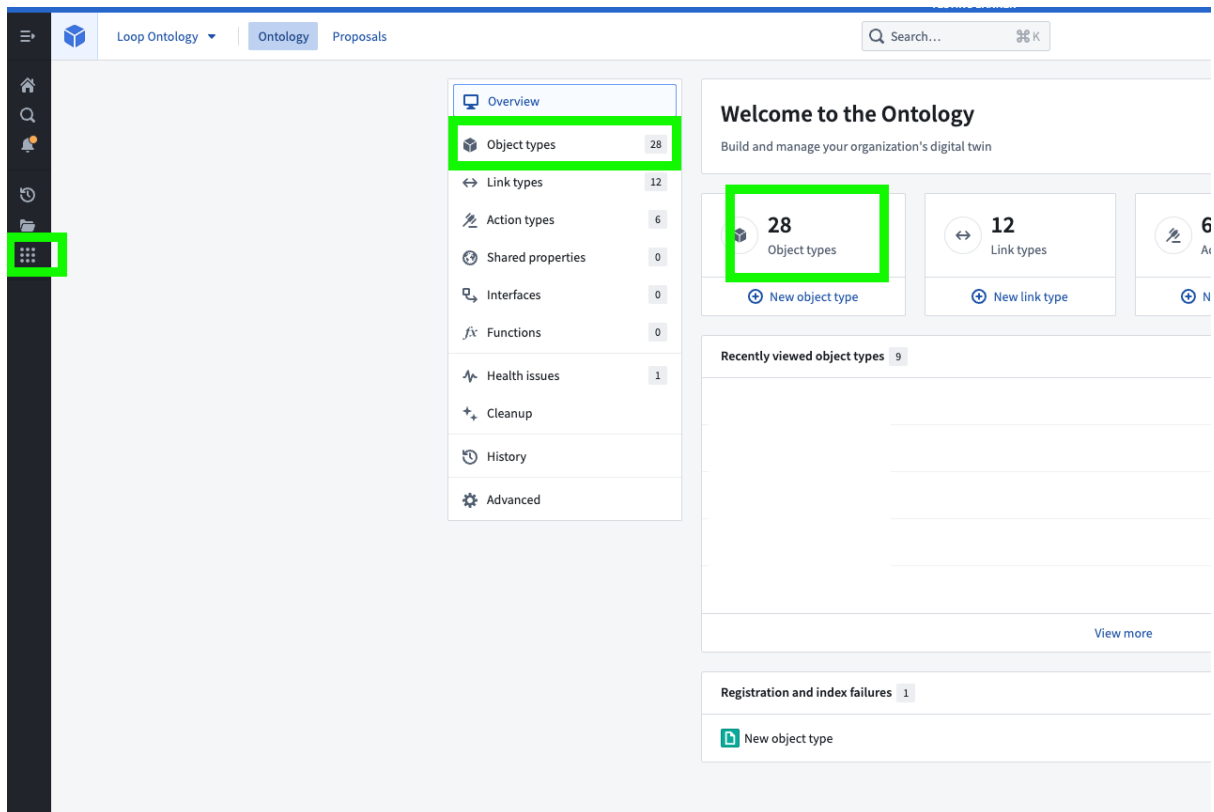
We now want to make it possible to interact and edit the Ontology. We will do this by creating **Actions**. Actions are events and edits that are performed on an object or set of objects in the Ontology. For example, you might have an Action to set a status as closed, or to create a new Order, or delete an incorrect Order. You can also define API bindings for Actions so that external applications may also interact with the Ontology!

By the end of this section you will have created an "Assign Order" Action that performs two actions at once: It sets the assignee to whatever name you provide, and it automatically sets the Status to "assigned."

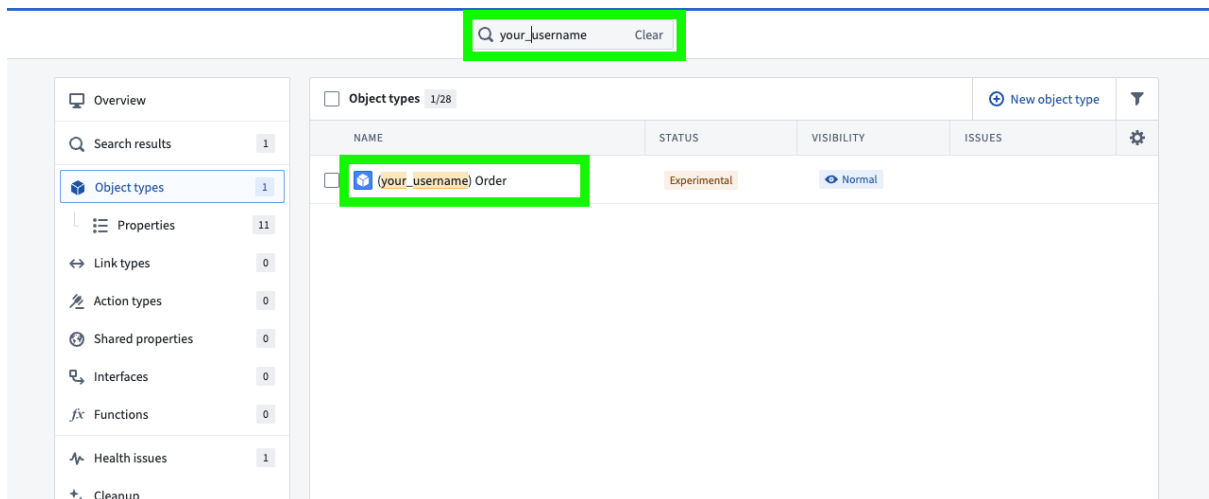
Enabling edits to your object

Step 1: Enable Edits on Order Object

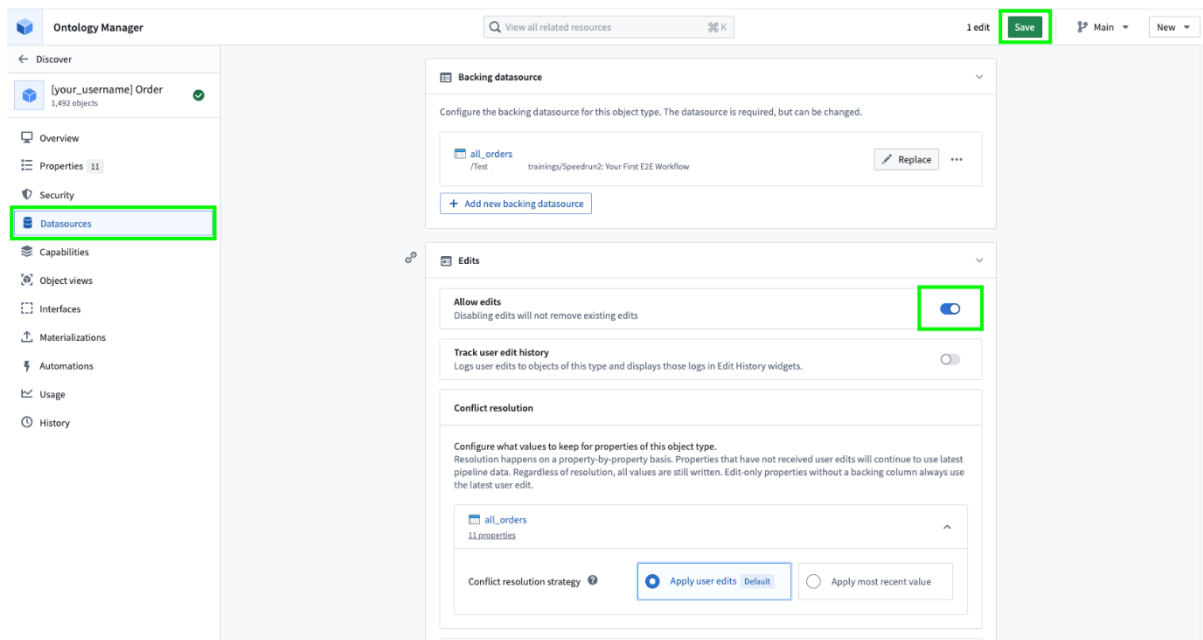
1. Navigate to the Ontology Manager application
2. Click on Object Types



3. Search [username] Order (Tip: Be careful, there may be many similar objects in the Ontology. You want to select your Object you created in this course)
4. Click on your object



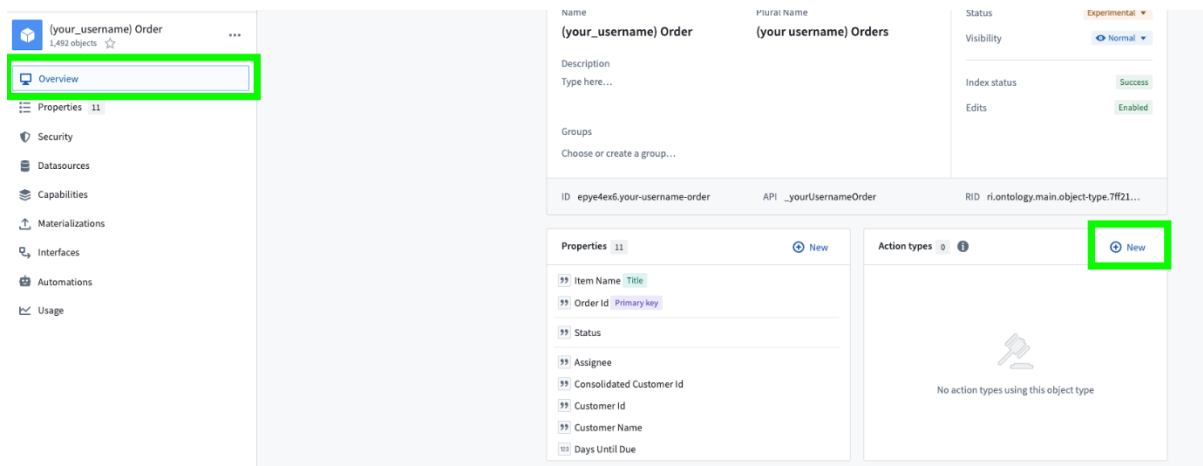
5. Click on *Datasources* on the left
6. If not already enabled, toggle on *Allow edits*. You can ignore any other toggles or settings.
7. Click *Save*
8. Click *Save to Ontology*



Create the "Assign Order" Action

Step 1: Create Action

1. Click on *Overview*
2. Click on *New* under *Action types*



3. Click on *Modify objects* under *Object actions*
 - If your Order object type is not selected in the dropdown, select it now.
4. Click *Next*

Create a new action type

1 Action type

2 Mapping

3 Metadata

4 Submission criteria

STEP 1

Select an action type you want to configure

Enable users to make changes to the ontology by configuring actions they can execute.

ObjectLinkFunctionWebhookInterfaceNotification

Object type

[your_username] OrderExperimental

Object actions

Create object

Configure an action type that adds a new object instance

Modify object(s)

Configure an action type that edits existing object instances.

Modify or create object

Modify an existing instance, otherwise create a new instance.

Delete object(s)

Configure an action type that deletes object instances.

SkipNext

5. Click on *Add Property*
6. Select *Assignee*
7. Click on *Add Property*
8. Select *Status*

STEP 2

Map action parameters

Map the action parameters that will be used as inputs to this action

(your_username) Order

PROPERTY	MAP TO
Assignee	Assignee

+ Add property

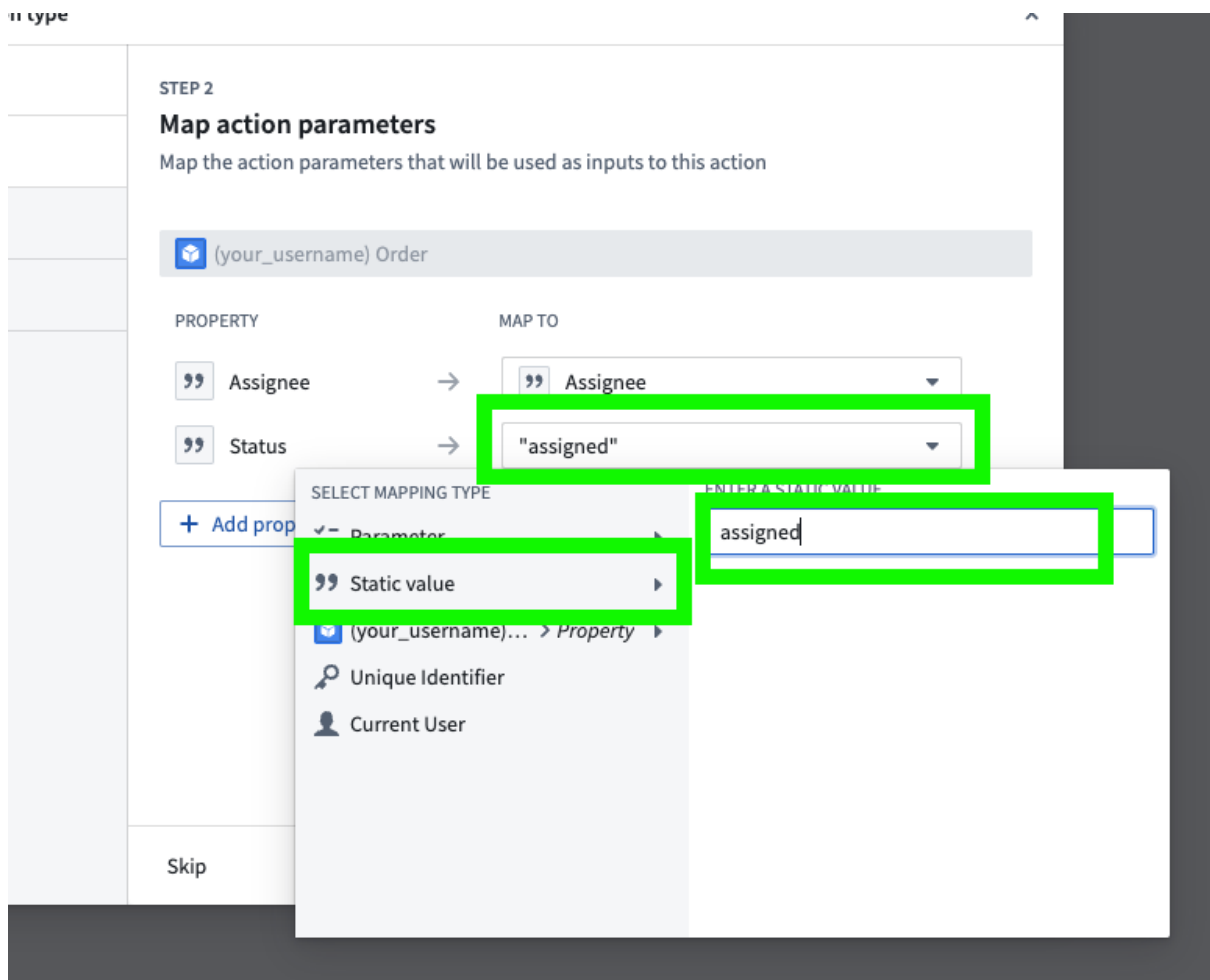
Search...

- Consolidated Customer Id
- Customer Id
- Customer Name
- Days Until Due
- Item Name Title
- Order Due Date
- Order Id Primary key
- Status

Back Next

9. On the right hand side under *Map To*, select *Status*
10. From the drop down, select *Static value*
11. Type in assigned in the text box
12. Click *Next*

Updating the Status rule to a static value in the Action rules will always set the Status value as "assigned." Leaving the Assignee rule as-is will allow a user to input text to update the Assignee value - we'll see this later in the Action form settings.



13. Set *Action Type Name* to [username] Assign Order. Prefixing the action with your username is a useful way to avoid confusing it with others

14. Click *Next*

STEP 3

Configure action type metadata

Use a familiar name or helpful description to enhance this action type's discoverability.

Action type name

Assign Order



Description • optional

Enter description...

Icon • Autogenerated, cannot be modified



Skip

Back

Next

For this Action, we want to set yourself as the only person who can run this Action. If you were creating a workflow used by a team, you'd likely want to select a user group here or select *Allow all users in organization to execute this action*.

15. Click on *User*
16. Select your username
17. Click on *Next*

Create a new action type

1 Action type

2 Mapping

3 Metadata

4 Submission criteria

STEP 4

Configure who can execute this action type

Choose who can submit this action type

☐ Organization

☐ Group

☒ User

Only this user can submit

your_username

You

Submitting actions

Skip

Back

Create

18. Make sure you are going to be saving in the correct folder.

19. Click on *Create*

STEP 5

Save location

Select a location to save this action type to



Speedrun: Your First E2E Workflow



Browse



Permissions



Skip

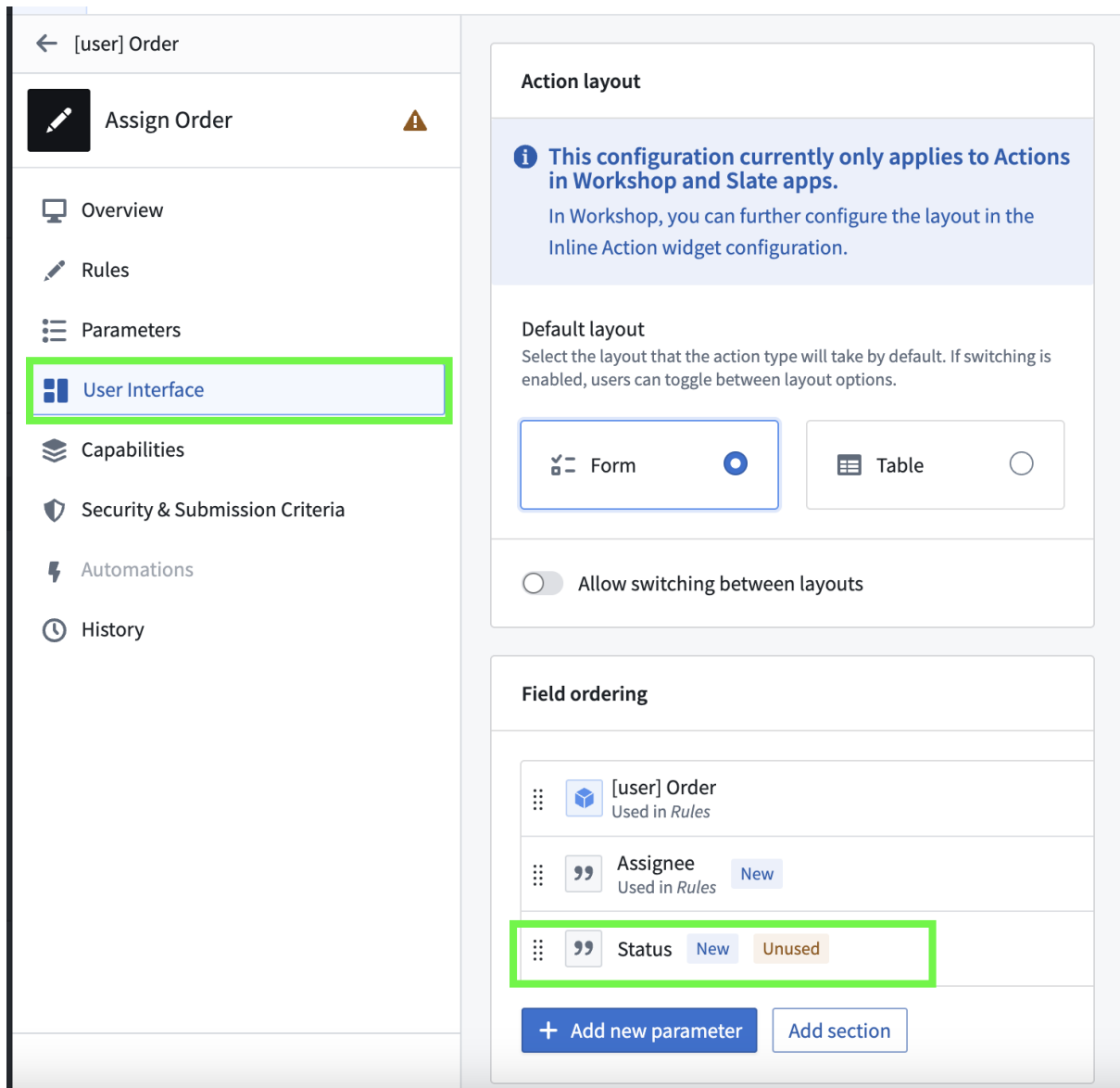
Back

Create

Finish configuring the Action

Step 1: Configure Action Fields

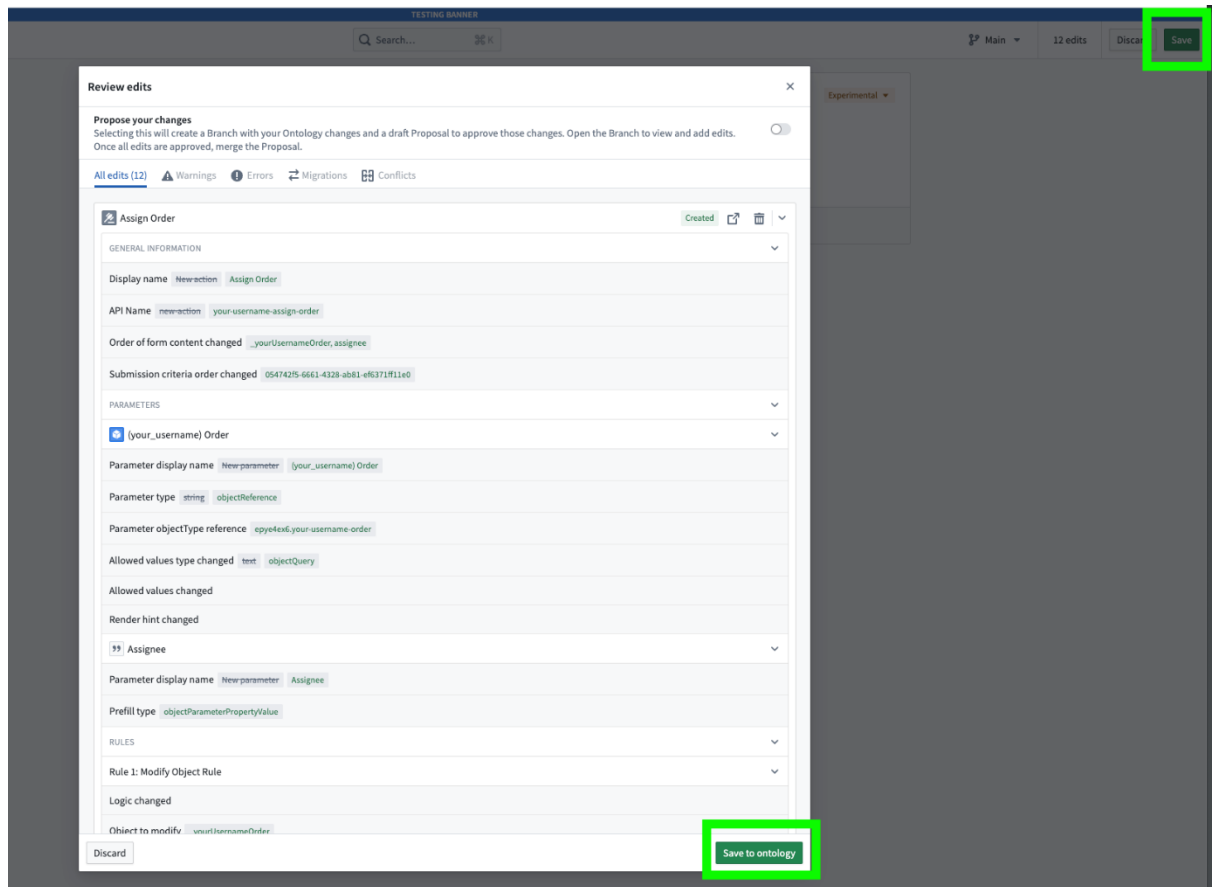
1. Select *User Interface* on the left sidebar
2. Select *Status* under *Field Ordering* in the middle (*If Status is not one of the parameters shown, skip to 4*)



Tip: Hide Form Preview pane on the right (to avoid clutter on next screen)

3. Delete the Status parameter
4. Click **Save** (on top right)

5. Click *Save to ontology*



Conclusion

We have now created a basic “Assign Order” action that will allow us to modify Orders. In a real-world setting we would configure more robust permissions, perhaps add data validations to make sure that incorrect values don’t make their way in, and define API bindings so that external systems and applications may interact with the Ontology.

Making your Workshop Application Operational

Introduction

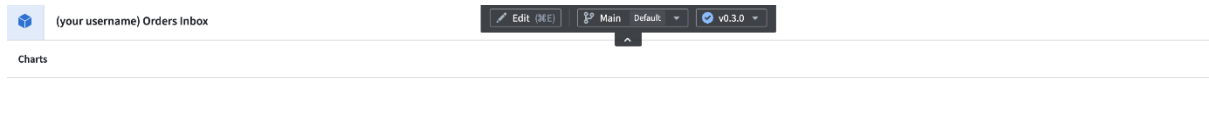
While read-only dashboards can be very useful, it’s important that multiple people and teams be able to collaborate, organize themselves, and update state. Otherwise we won’t have solved the Fulfillment Department’s problem! You’ll see how easy this is by adding the “Assign Order” action to the Workshop application you’ve built.

Open your Workshop Module

Option 1

If you still have the *View Mode* Workshop open on a tab, let's go back to *Edit Mode*.

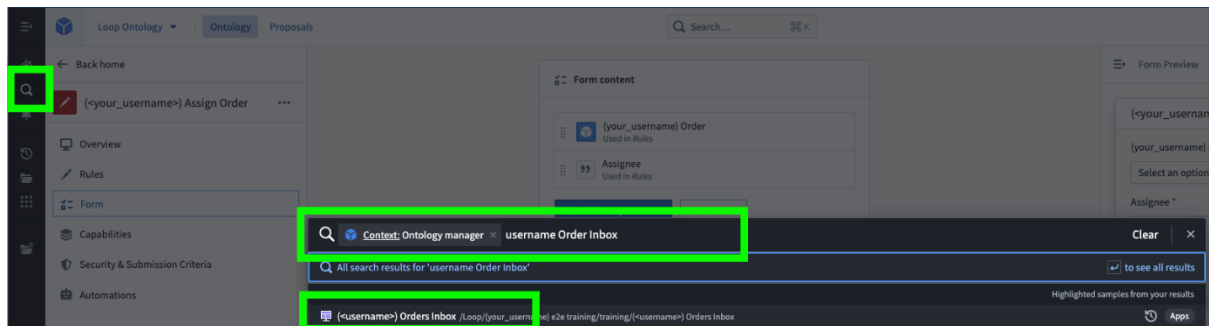
1. Click on the three grey dots at the top of the screen to reveal the dropdown
2. Click on *Edit*



Option 2

If you need to find your Workshop module by searching

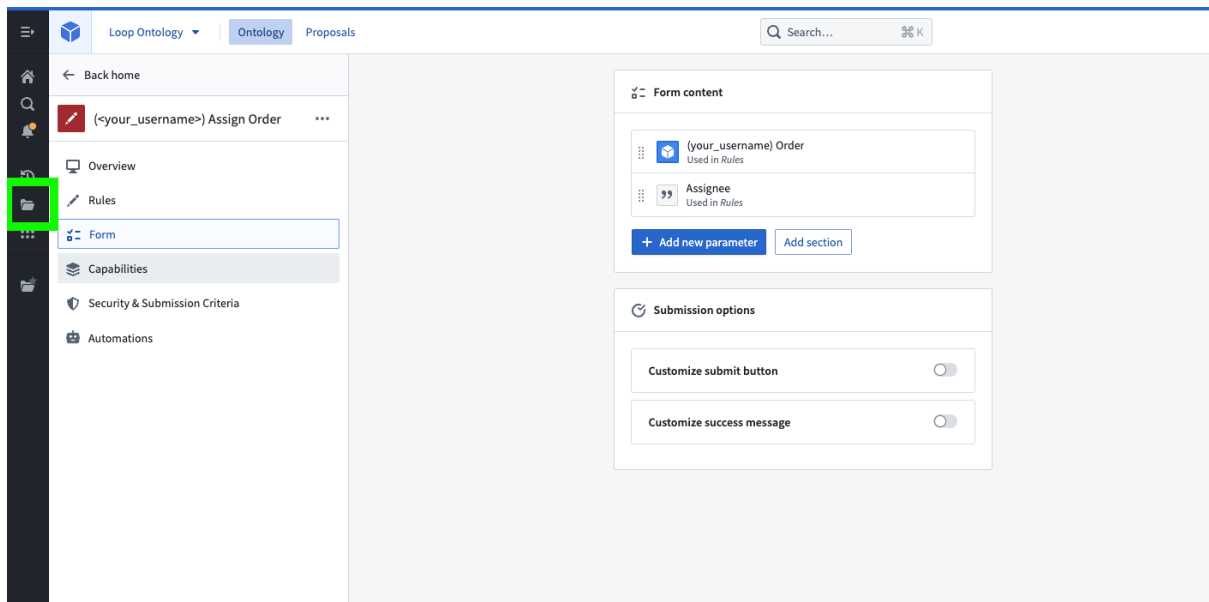
1. Click on *Search* on the sidebar on the left
2. Type in the name of your Workshop module
3. Click on the correct result (*Tip: There might be several Workshop modules with similar names. Make sure you click on the right one!*)



Option 3

If you need to find your Workshop module by navigating to your folder:

1. Click on *Files* on the sidebar on the left. This will take you back to the home directory for your files
2. Navigate to your training folder and select your Workshop module



Add Actions to your Workshop

Step 1: Configure Buttons for Action

1. Earlier, we added a button to our Workshop. It should be near lower middle of the Workshop.
2. Select the button so that you see the 'Button group' right side bar
3. Click into the 'Button Configuration' (it should be called "Button 1")

The screenshot shows the 'Button Configuration' panel on the right side of the interface. The panel is titled 'Button group 1' and has tabs for 'Widget setup', 'Metadata', and 'Display'. The 'Widget setup' tab is active. Under 'BUTTON TYPE', 'Inline' is selected. Under 'BUTTON CONFIGURATION', 'Button 1' is selected and highlighted with a green box. The 'DISPLAY & FORMATTING' section shows 'ORIENTATION' set to 'Horizontal', 'FILL AVAILABLE HORIZONTAL SPACE IN ROW AND COLUMN LAYOUTS' checked, 'LABEL ALIGNMENT' set to 'Center', 'STYLE' set to 'Standard', 'SIZE' set to 'Regular', and 'GROUP BUTTONS' checked.

4. Set the text to Assign
5. Set *On Click* to *Action*
6. Search and select the [username] Assign Order Action you created earlier. You should see it is associated with your Order object
7. Under *Parameter Defaults*, select the *Parameter to configure* equal to [username] Order
8. Set the *Local Default* value: Object table 1 Active object
 1. This will automatically pass the Order selected in the Object table into the Action form rather than the user having to manually select the Order to update.
9. Scroll to the top, click *Go back* or the back arrow

Button group 1 BUTTON GROUP

Widget setup Metadata Display

< Assign

TEXT Use variable

Assign

scroll down if needed

Button group 1

BUTTON GROUP

Widget setup

Metadata

Display

CONDITIONAL VISIBILITY

ON CLICK

Action

[username] Assign Order

on  [your_username] Order



Default layout

 Form

 Table

End-user features

SWITCH LAYOUT

PARAMETER DEFAULTS

Local default values for parameters

Select parameter to configure

 [your_username] Order





LOCAL DEFAULT VALUE

Object table 1 Active object



Local override applied

VISIBILITY IN FORM

Visible

Disabled

Hidden

- Click on Save and Publish
- Click on View *(Tip: Open it in a new tab)*
- We'll test your new Action out in the next lesson

Test Your New Application

Step 1: Locate Order

1. Ensure your application is in View mode rather than Edit. If you're in Edit mode you'll see a *View* button in the top right corner and can click it to enter View mode.
2. Navigate to the Filters tool on the far right
3. Under Order ID, paste the following Order ID
 1. 8d64e941-799c-4e35-b913-d0e3475c307ec206b101-f8b3-4bbe-a1f4-c674dc3124c8
4. Select the option from the drop down. This should update your table and charts to only display a singular order

ORDER DUE DATE

Start date End date

ASSIGNEE

Search...

CONSOLIDATED CUSTOMER ID

Search...

CUSTOMER ID

Search...

CUSTOMER NAME

Search...

DAYS UNTIL DUE

-

ITEM NAME

Search...

ORDER ID

206b101-f8b3-4bbe-a1f4-c674dc3124c8

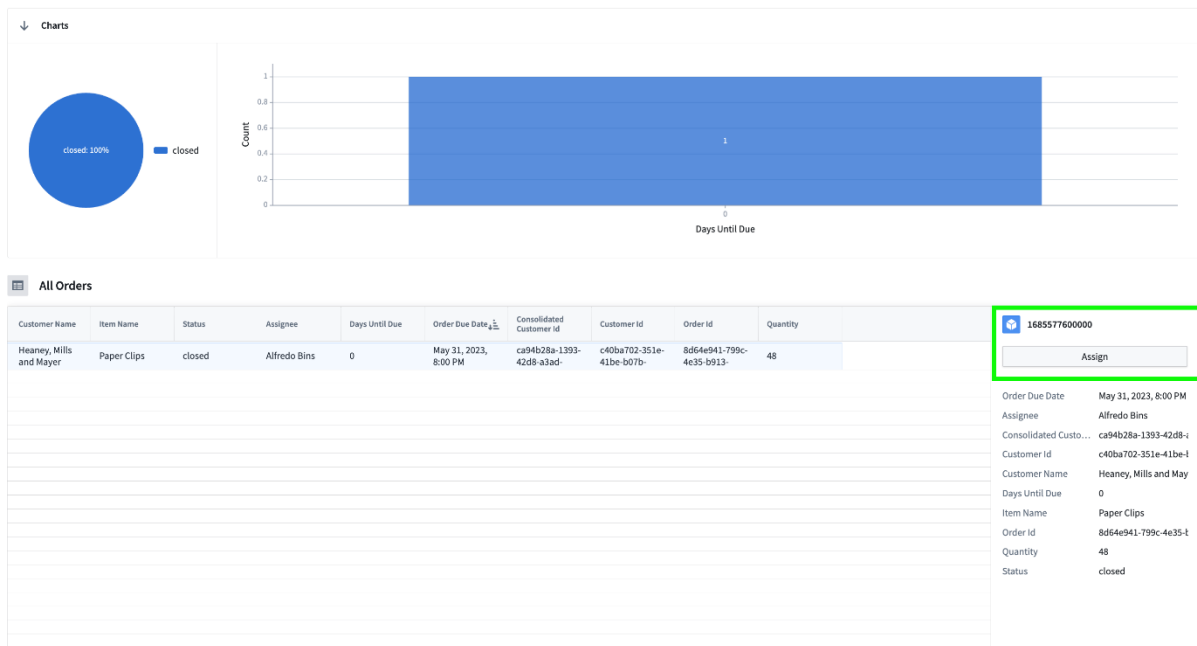
8d64e941-799c-4e35-b913-d0e3475c307ec206b101-f8b3-4bbe-a1f4-c674dc3124c8

STATUS

Search...

Add filter

5. Click on the Order in the Object table
6. Click on the *Assign* button



7. In the Action form, remove *Alfredo Birns* from the Assignee and add *Gail Weber*

8. Click *Submit*

[yourUsername] Assign Order

ORDER *

1685577600000

ASSIGNEE *

Gail Weber

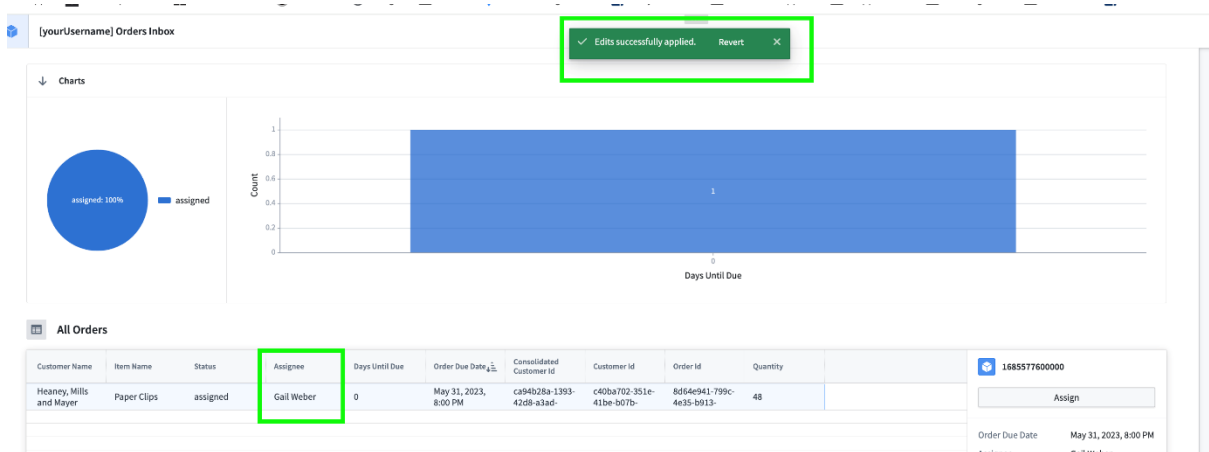
Edited

Cancel Submit

9. Back viewing your application, you should see a green pop-up at the top of your screen verifying that your Action was implemented.

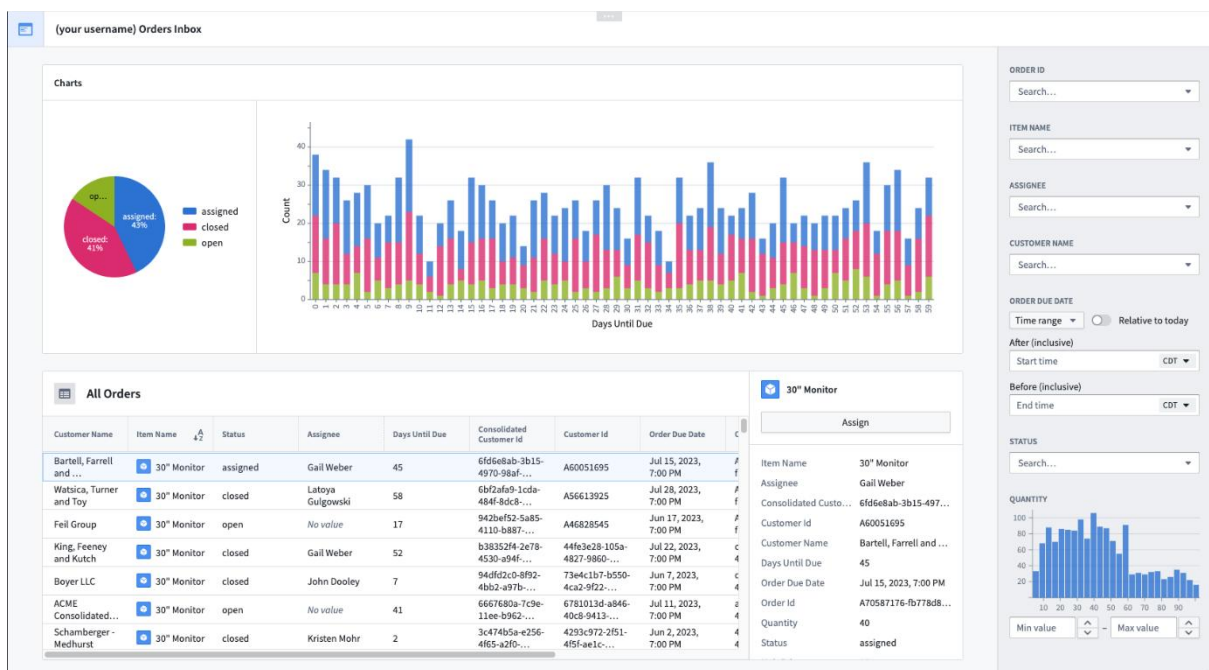
10. In the Object table, also note how the name in the Assignee column has been updated.

Congrats! You've successfully updated an Object via an Action.



Wrapping Up

Conclusion

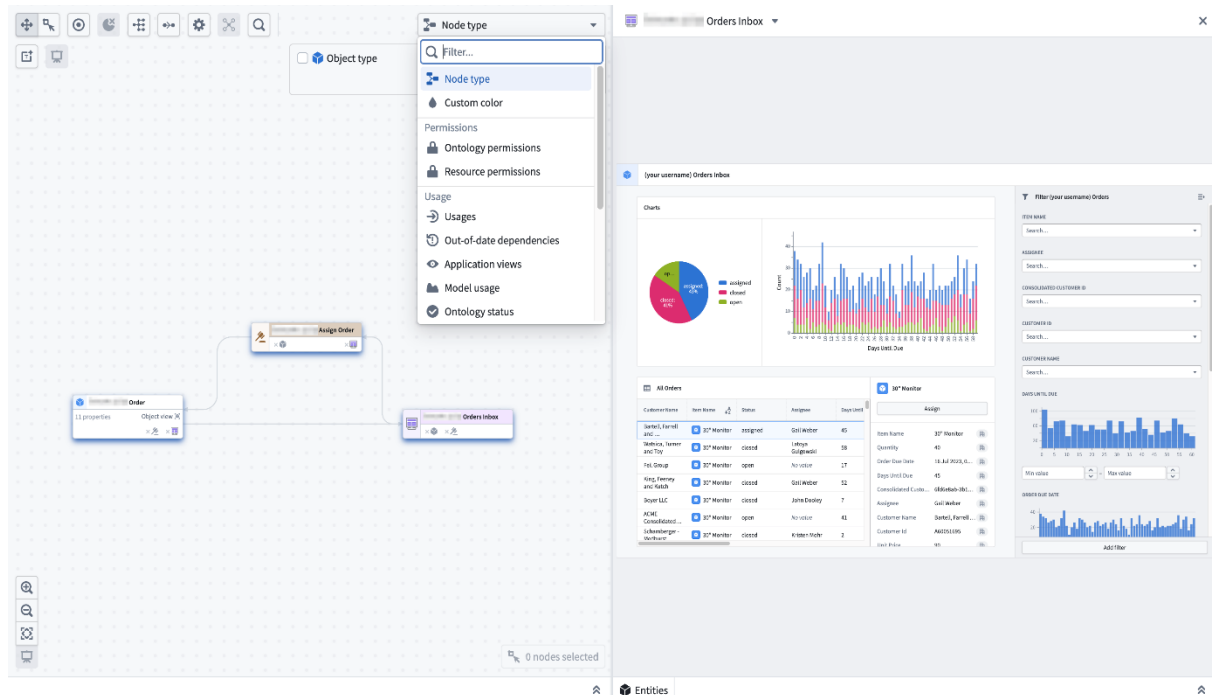


You've now created an operational dashboard for the Fulfillment Department that serves as the single source of truth and cockpit about allocations for upcoming orders.

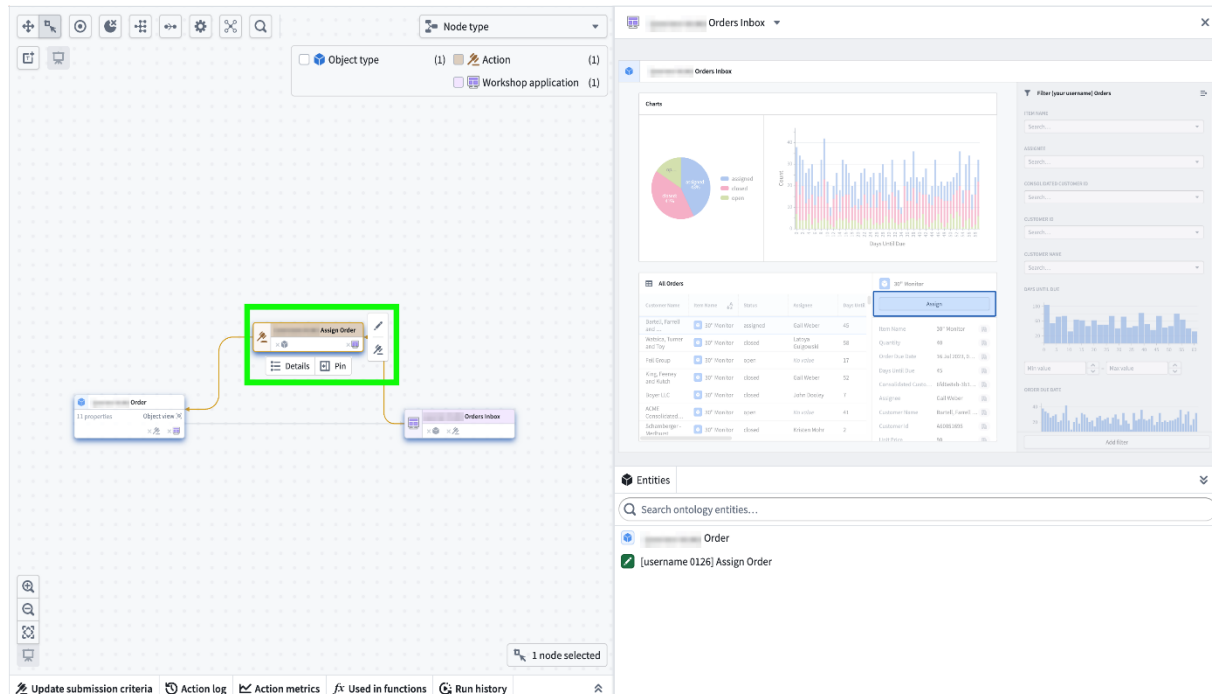
Visualizing and Managing Your Operational Dashboard in Workflow Lineage

1. Open your workshop in Workflow Lineage by pressing CMD + i on your workshop (or CTRL + i)

- This will take you to a pre-populated graph of everything backing your Workshop application

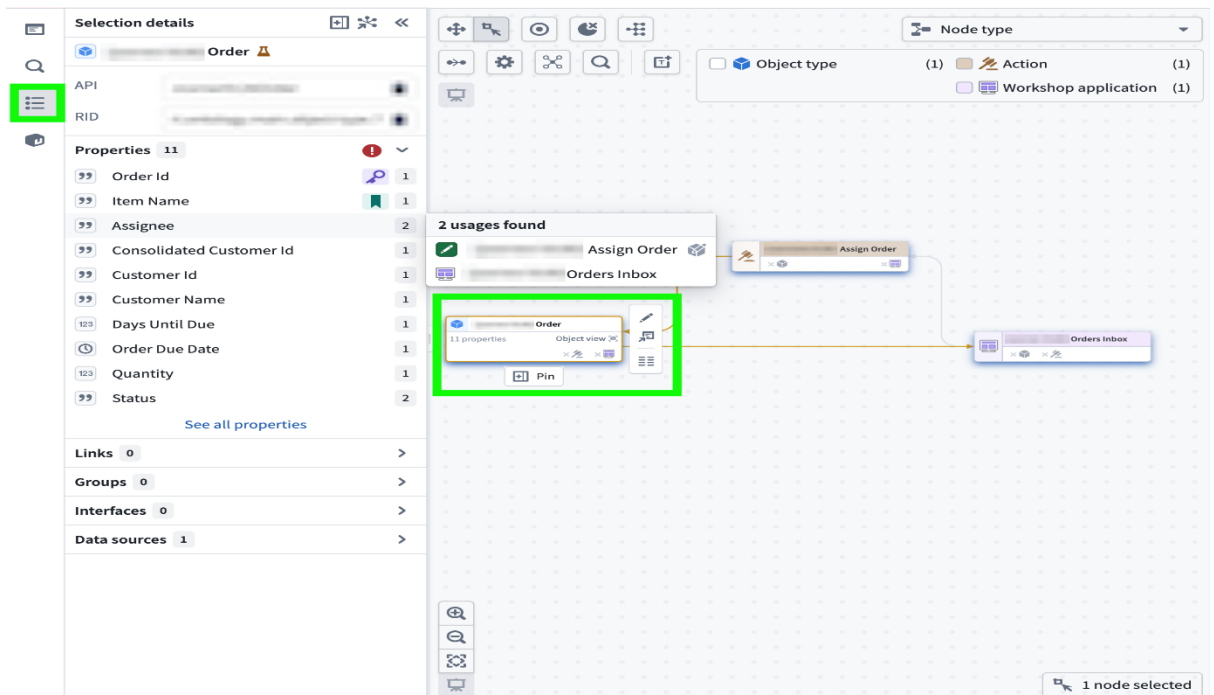


- If you select the action node you will see the corresponding button highlight on the right hand side



- If you select the Object node and open the selection details on the left hand side you can see all downstream usages of the

properties



As you add more and more complexity to your application, Workflow Lineage will automatically keep your graph up to date, and you can use the different color modes and bottom tab to save time and make informed decisions. You'll be able to:

- Bulk update [function backed actions](#), [workshops](#), logics
- Manage and bulk change [security submission criteria](#)
- [Publish](#) and [delete](#) multiple Workshops in one go
- [View AIP usage metrics](#) including successful/attempted/failed token usage counts
- ...and more!

[Learn more about all of Workflow Lineage's capabilities here](#)

It is relatively simple to make this even more useful and more secure:

- Enhance visualizations
- Send email notifications to users when assigned to an order
- Add comments section to provide status updates for each order
- Replace static data file imports with live feeds from the transactional systems
- Control who can assign and mark closed an order
- Automatically generate audit trails of who made a modification or status change

- Write back the actions (assignee, status) back to the transactional systems in real-time

You now have the basics for developing operational data applications!